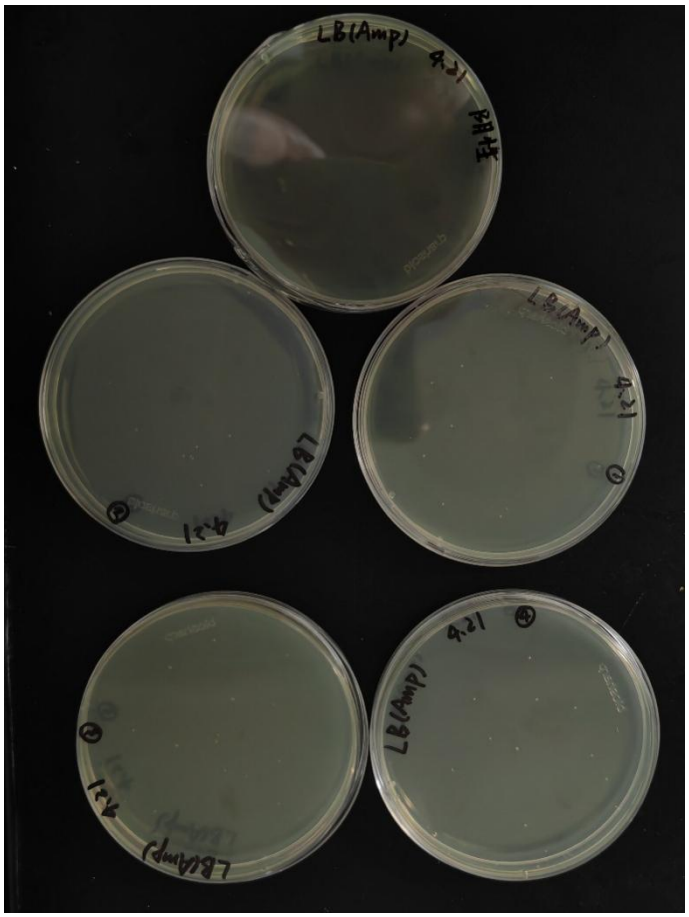
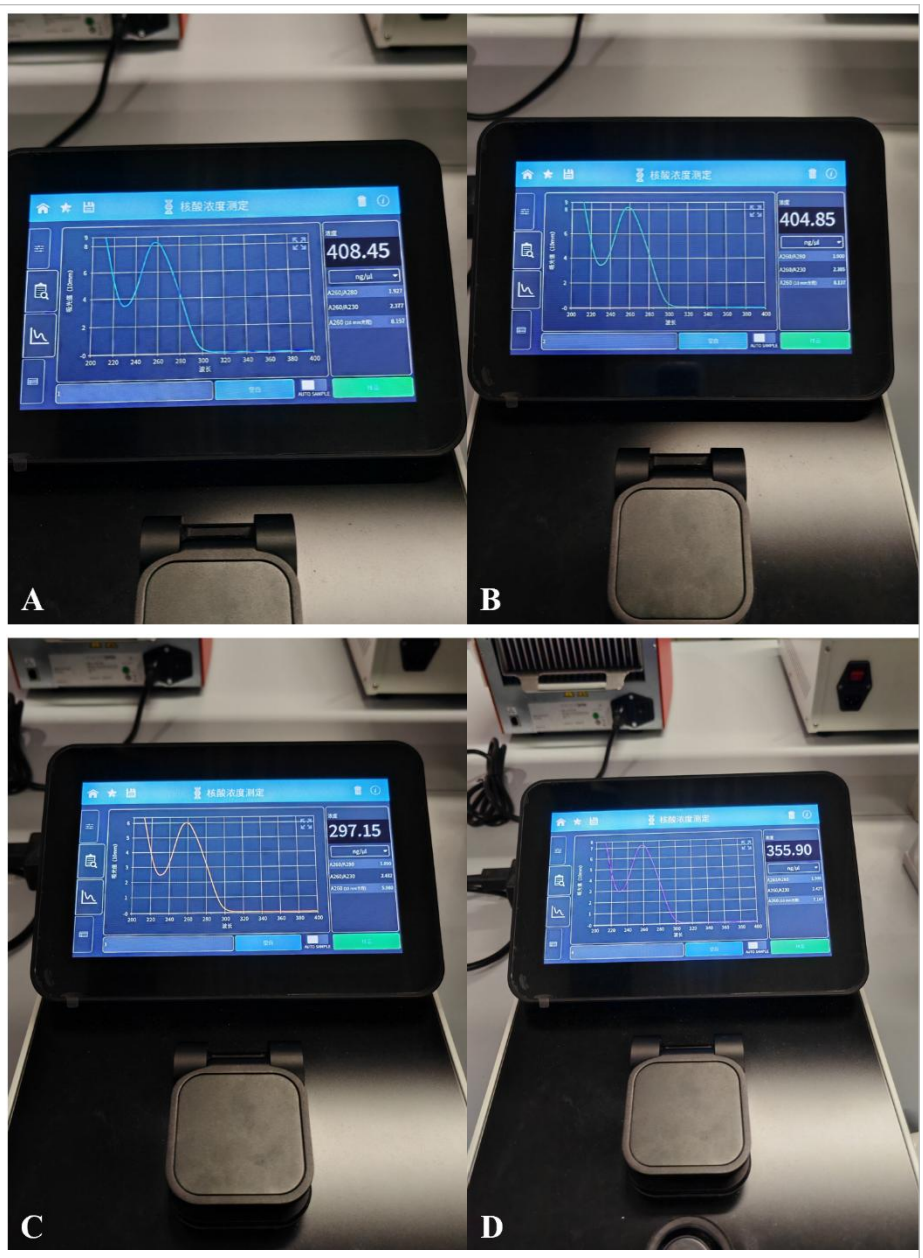
Notebook：Construction of the PAGER Antigen-Gated System for Influenza A Virus Detection

No 1 Date：April 21 to 24 Experimenters: Aishi Zeng Recorder: Aishi Zeng

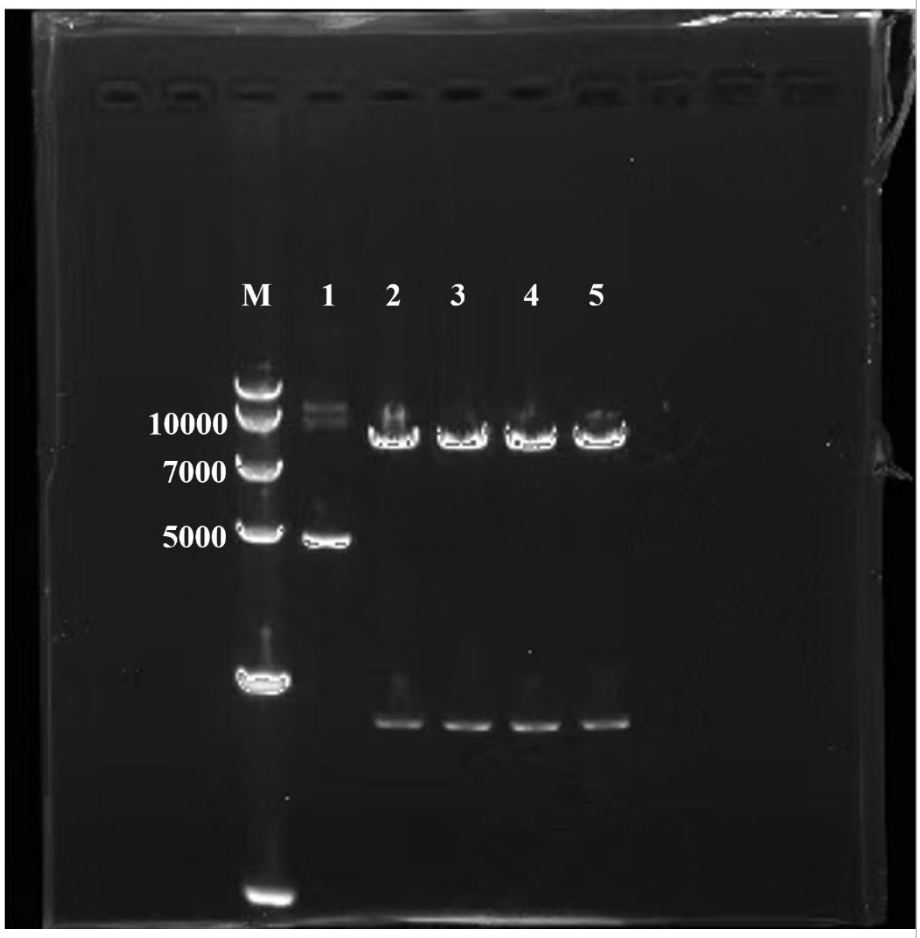
Experiment	PAGER plasmid transformation into E. coli, plasmid amplification, extraction and double- digest verification
Note	Protocol 1、Protocol 3、Protocol 4、Protocol 5
Result	April 22  Figure 1: E. coli transformation LB-Amp agar plates Brief overview: PAGER plasmid was transformed into TOP10 E. coli. The negative control plate (top) showed no colony growth, confirming no contamination. The four experimental LB-Amp plates produced visible transformant colonies, indicating successful transformation. April 24



Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	408.45 ng/μL	1.927
B	404.85 ng/μL	1.900
C	297.15 ng/μL	1.890
D	355.90 ng/μL	1.908

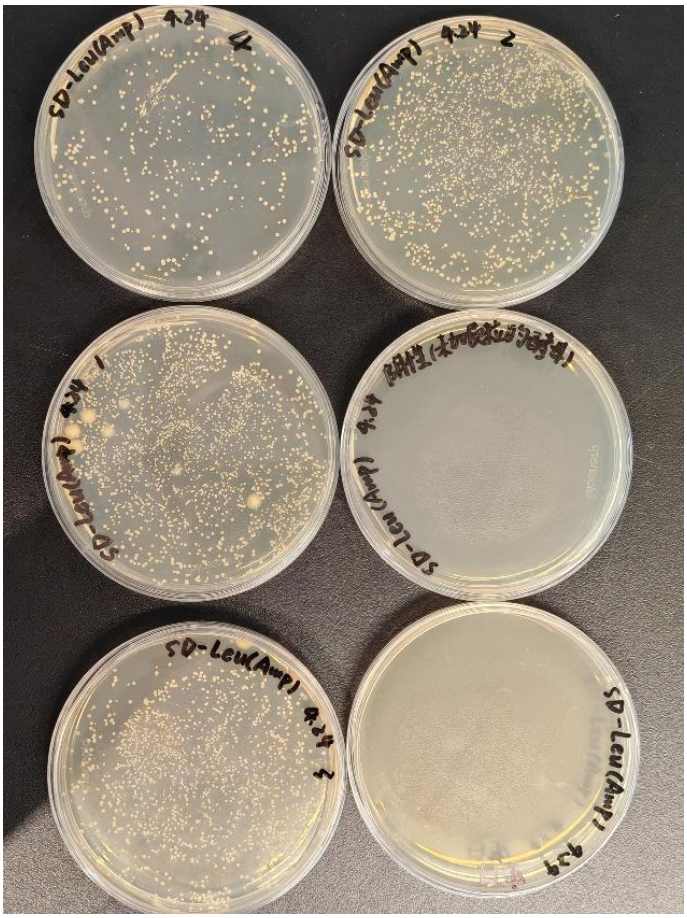
Figure 2. NanoDrop measurement results of four extracted PAGER plasmid samples. (A) Sample 1, concentration: 408.45 ng/μL; (B) Sample 2, concentration: 404.85 ng/μL; (C) Sample 3, concentration: 297.15 ng/μL; (D) Sample 4, concentration: 355.90 ng/μL.

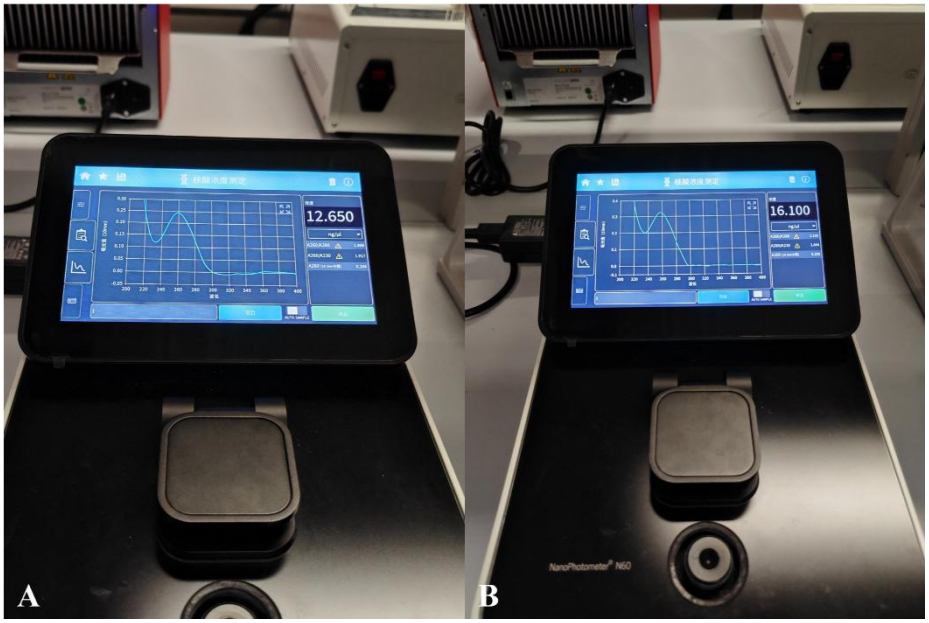
The absorbance spectrum and nucleic acid concentration of each plasmid

	sample were detected by NanoDrop spectrophotometer. All four samples exhibited acceptable concentration and A260/A280 ratio, demonstrating high plasmid quality for subsequent experiments. April 24  Figure 3. Agarose gel electrophoresis of PAGER plasmid double digestion. M: DNA Marker; Lane 1: undigested PAGER plasmid; Lanes 2–5: plasmids digested with NdeI and BamHI- HF. The double digestion is expected to produce two fragments of 8877 bp and 2145 bp. The gel result shows that the undigested plasmid presents three bands at the position of supercoiled plasmid. All four digested samples display two distinct bands consistent with the expected fragment sizes, confirming the identity of the PAGER plasmid.
Explanation	This set of experiments was performed to amplify and obtain sufficient, high- quality PAGER plasmid for downstream yeast- based sensor experiments. The PAGER plasmid was transformed into TOP10 *E. coli* for plasmid propagation. The blank negative- control plate ruled out contamination of the experimental system. After culture and plasmid extraction, all four samples reached acceptable concentration and A260/A280 purity. *Nde*I- *Bam*HI- HF double- digestion further validated plasmid integrity, ensuring no structural rearrangement occurred during bacterial amplification. Verified plasmids are essential starting material for subsequent yeast transformation and sensor protein characterization.

Next plan	Carry out yeast transformation, followed by colony PCR and sequencing validation for engineered strain construction.

No 2 Date：April 26 to June 9 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Transformation of PAGER plasmid into Saccharomyces cerevisiae and strain validation
Note	Protocol 4、Protocol 5、Protocol 12、Protocol 13、Protocol 23
Result	April 26  Figure 1: SD- Leu(Amp) selective plates after yeast transformation. The negative control plate and blank control plate showed no colony growth, demonstrating effective selection pressure. Plate 1 and Plate 3 were contaminated with large irregular colonies. Plate 2 and Plate 4 generated evenly distributed small yeast transformant colonies with no obvious contamination, available for colony selection. April 30

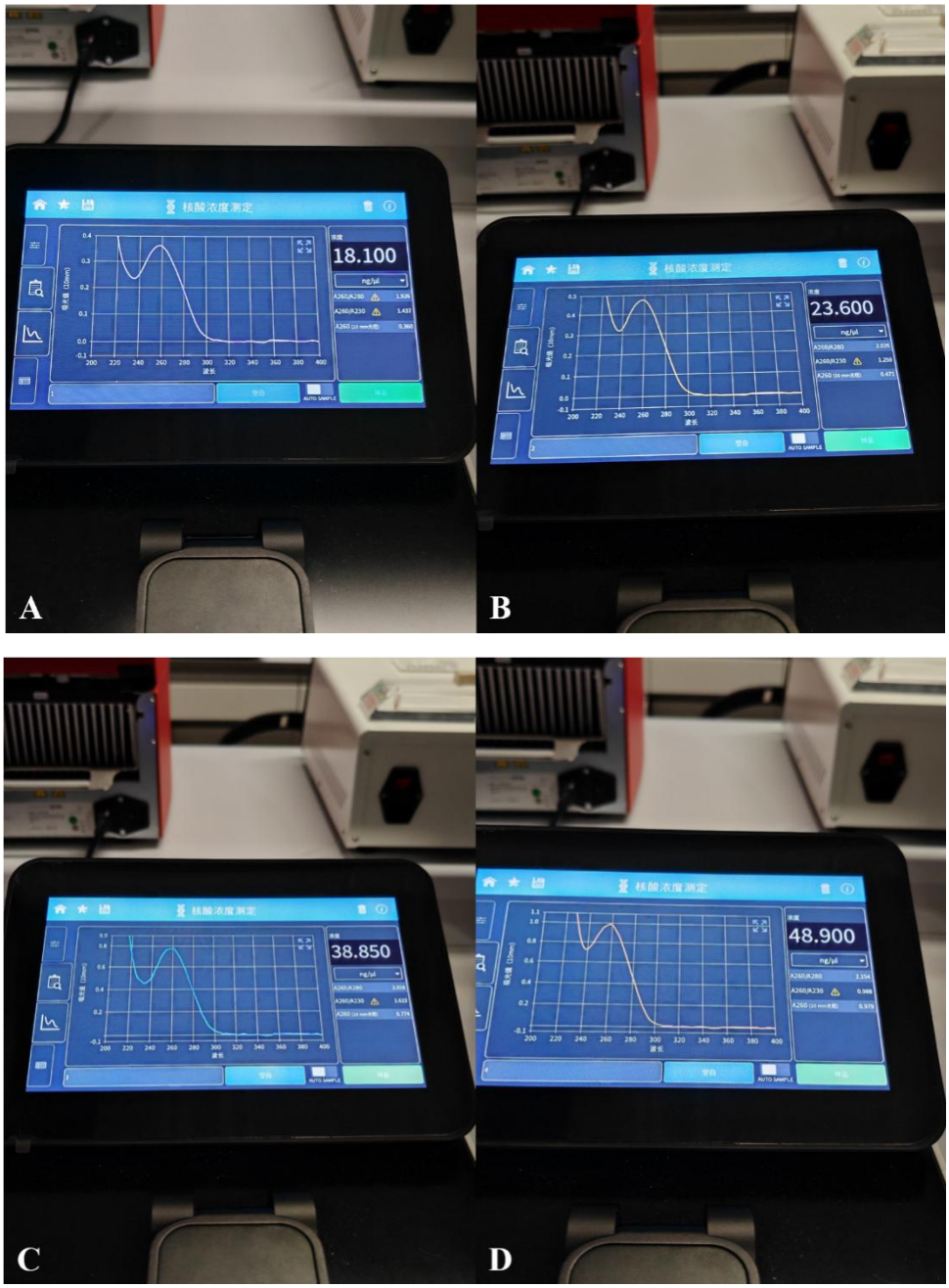


Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	12.650 ng/μL	1.888
B	16.100 ng/μL	2.190

Figure 2. NanoPhotometer N60 quantification of plasmids extracted from yeast transformants (first extraction). (A) Measurement result of sample A, with DNA concentration of 12.650 ng/μL. (B) Measurement result of sample B, with DNA concentration of 16.100 ng/μL.

Both samples show low DNA yield and suboptimal purity. The plasmid quality was inadequate to perform restriction digestion assays as used for E. coli plasmid validation.

May 3



Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	18.100 ng/μL	1.926
B	23.600 ng/μL	2.026
C	38.850 ng/μL	2.018
D	48.900 ng/μL	2.154

Figure 3. NanoPhotometer N60 quantification of plasmids extracted from yeast transformants (second extraction).(A) Sample A, concentration = 18.100 ng/μL. (B) Sample B, concentration = 23.600 ng/μL. (C) Sample C, concentration = 38.850

ng/μL. (D) Sample D, concentration = 48.900 ng/μL.

Higher plasmid yields were achieved in the second extraction, yet the A260/A230 ratios indicated unsatisfactory purity.

May 3

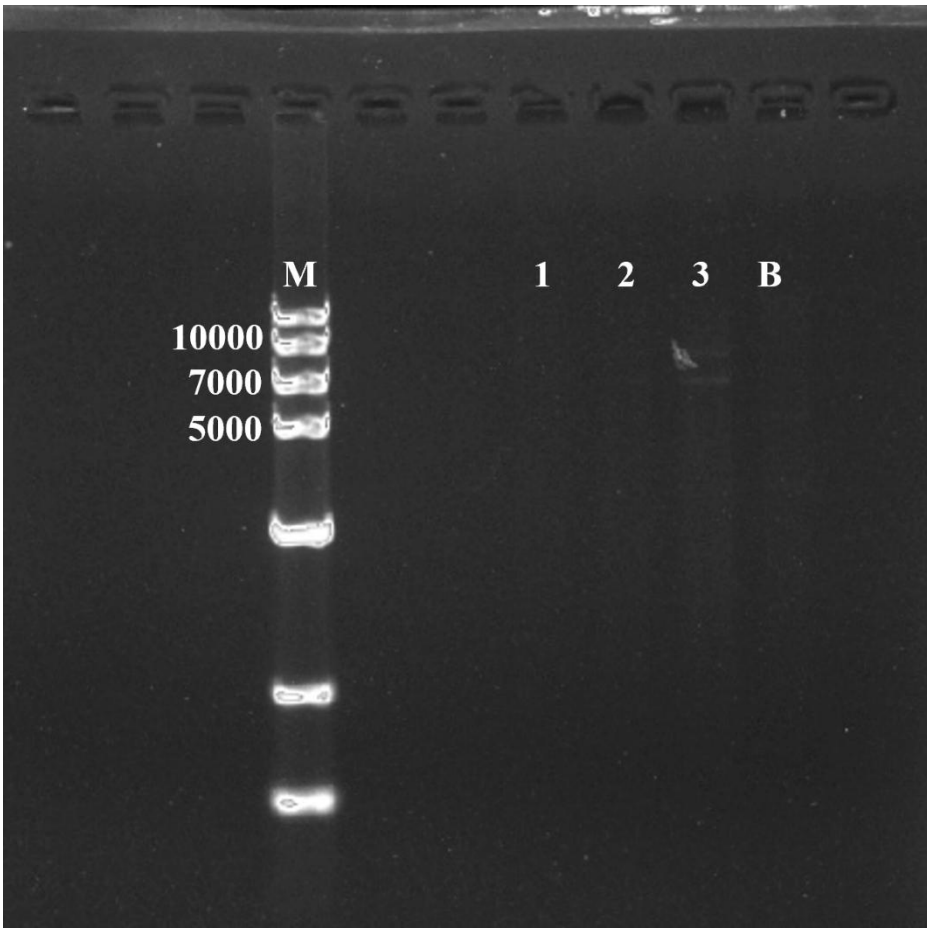
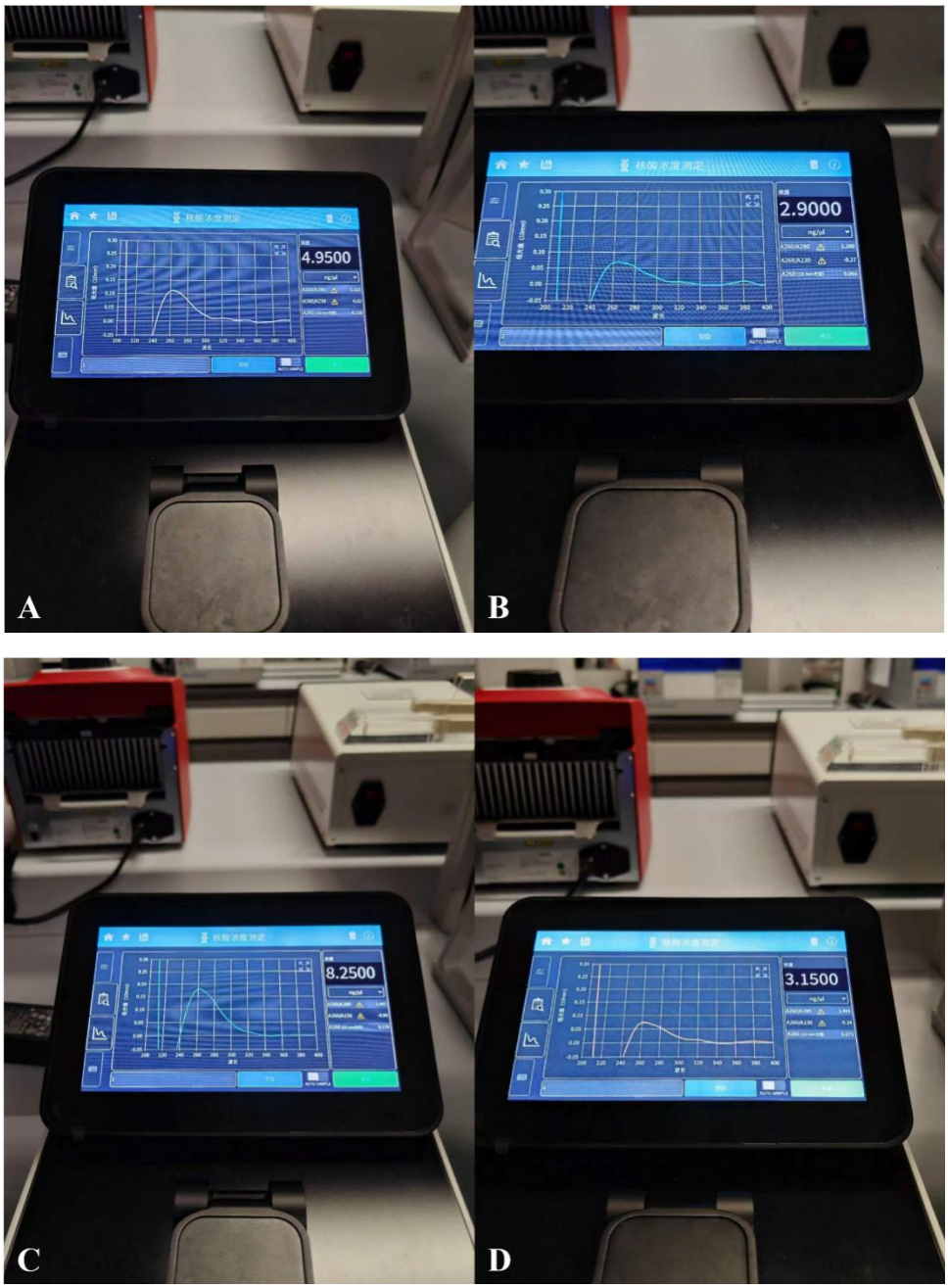


Figure 4. Agarose gel electrophoresis of double-digested yeast plasmids from the second extraction.M: DNA marker; Lane 1, 2, 3: double digested sample B, sample D and sample C from the second yeast plasmid extraction, respectively; Lane B: blank control with water.

No obvious bands were detected in all sample lanes. The low plasmid concentration of yeast extracts resulted in the absence of observable bands after restriction digestion.

May 9

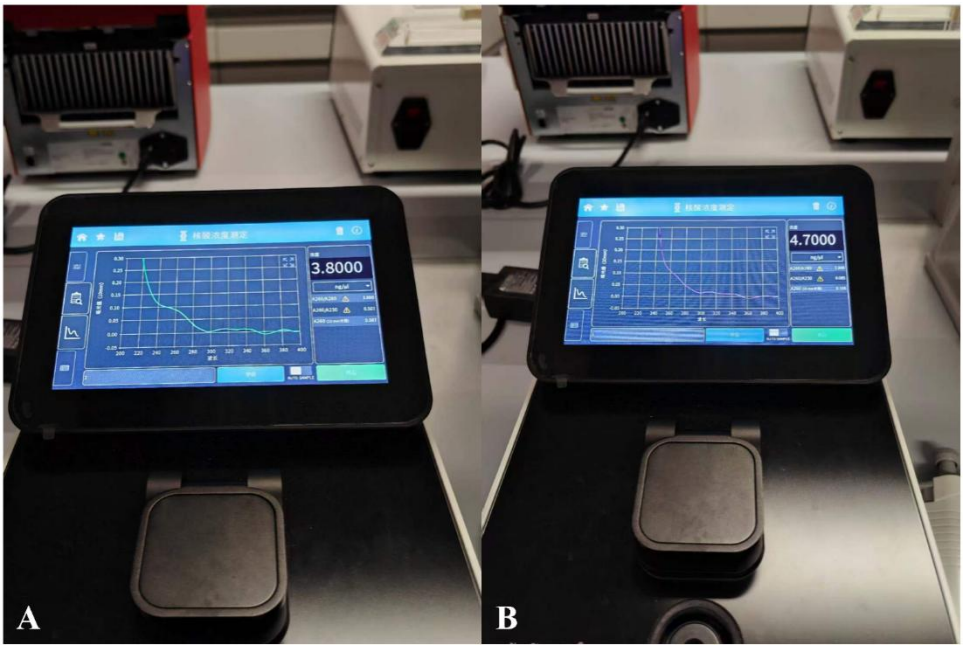


Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	4.9500ng/µl	1.523
B	2.900ng/µl	1.289
C	8.2500ng/µl	1.447
D	3.1500ng/µl	1.465

Figure 5. NanoPhotometer N60 quantification of plasmids extracted from yeast transformants (third extraction). (A) Sample A, concentration = 4.9500 ng/µL. (B) Sample B, concentration = 2.9000 ng/µL. (C) Sample C, concentration = 8.2500 ng/µL. (D) Sample D, concentration = 3.1500 ng/µL.

All samples showed low DNA concentration and suboptimal A260/A280 values.

May 11

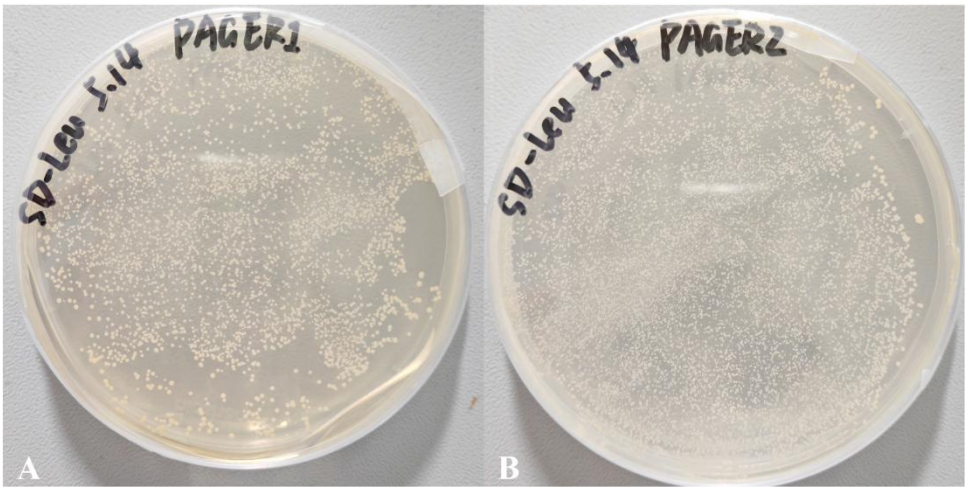


Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	4.7000ng/μl	2.848
B	3.8000ng/μl	3.800

Figure 6. NanoPhotometer N60 quantification of plasmids extracted from yeast transformants (fourth extraction). (A) Sample A, concentration = 3.8000 ng/μL. (B) Sample B, concentration = 4.7000 ng/μL.

Both samples show low DNA concentration and poor plasmid quality. The result remains unsatisfactory, and a new round of yeast transformation will be performed subsequently.

May 19



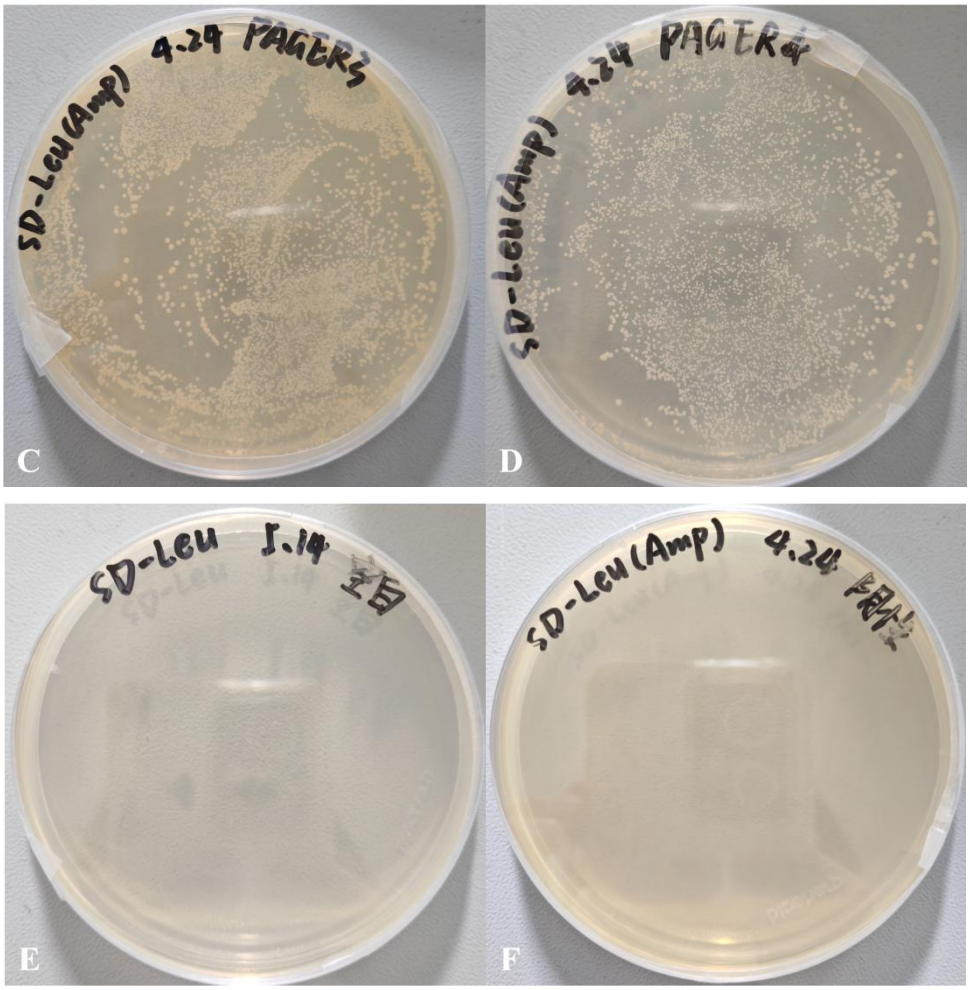


Figure 7. Plate results of the second yeast transformation. (A) PAGER1 transformed yeast, (B) PAGER2 transformed yeast, (C) PAGER3 transformed yeast, (D) PAGER4 transformed yeast: experimental groups. Abundant yeast colonies appeared on plates A–D, indicating successful transformation. (E) Blank control, sterile water was spread on the plate. (F) Negative control, sterile water replaced plasmid in the transformation reaction.

No colonies grew on E and F. All six plates were free of contamination.

May 26

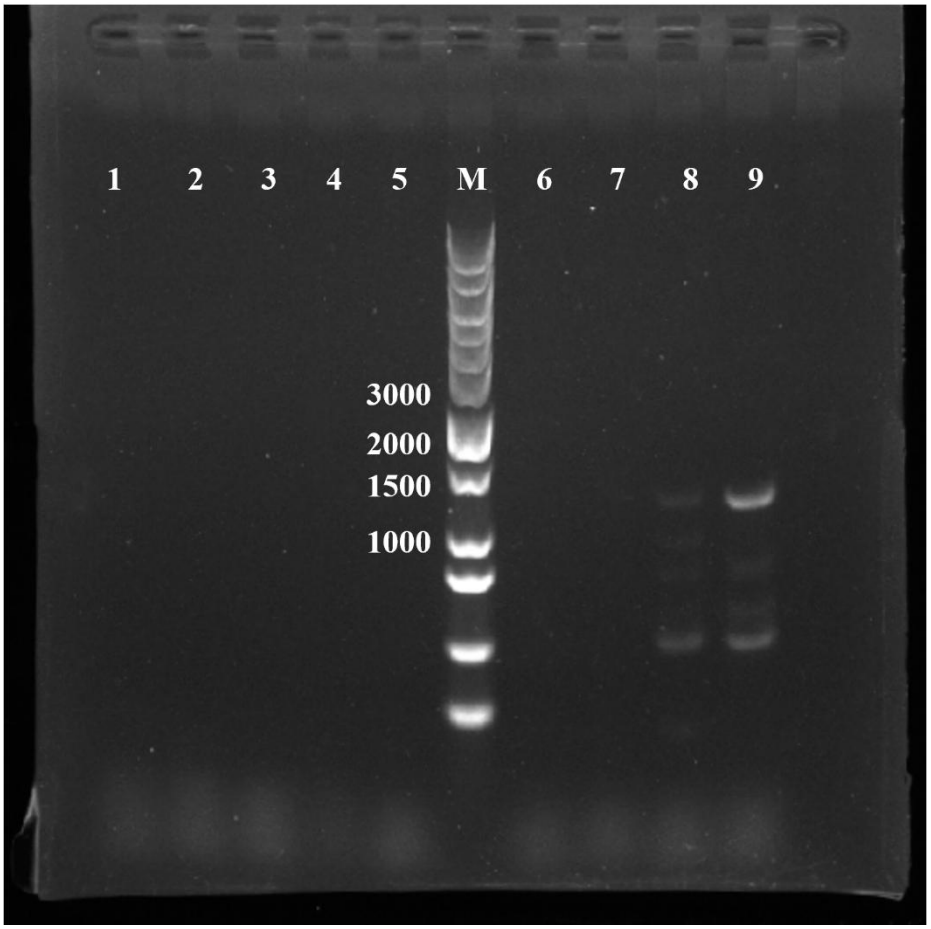


Figure 8. Agarose gel electrophoresis result of the first yeast colony PCR. Lane 1: BY4741 negative control; Lane 2–5: colony PCR products of colonies from the second yeast transformation; Lane M: DNA molecular weight marker; Lane 6–7: colony PCR products of colonies from the first yeast transformation; Lane 8–9: PCR products using yeast plasmids from the first plasmid extraction as templates. The expected amplicon size was 2086 bp. None of the lanes showed the target band of correct size.

May 28

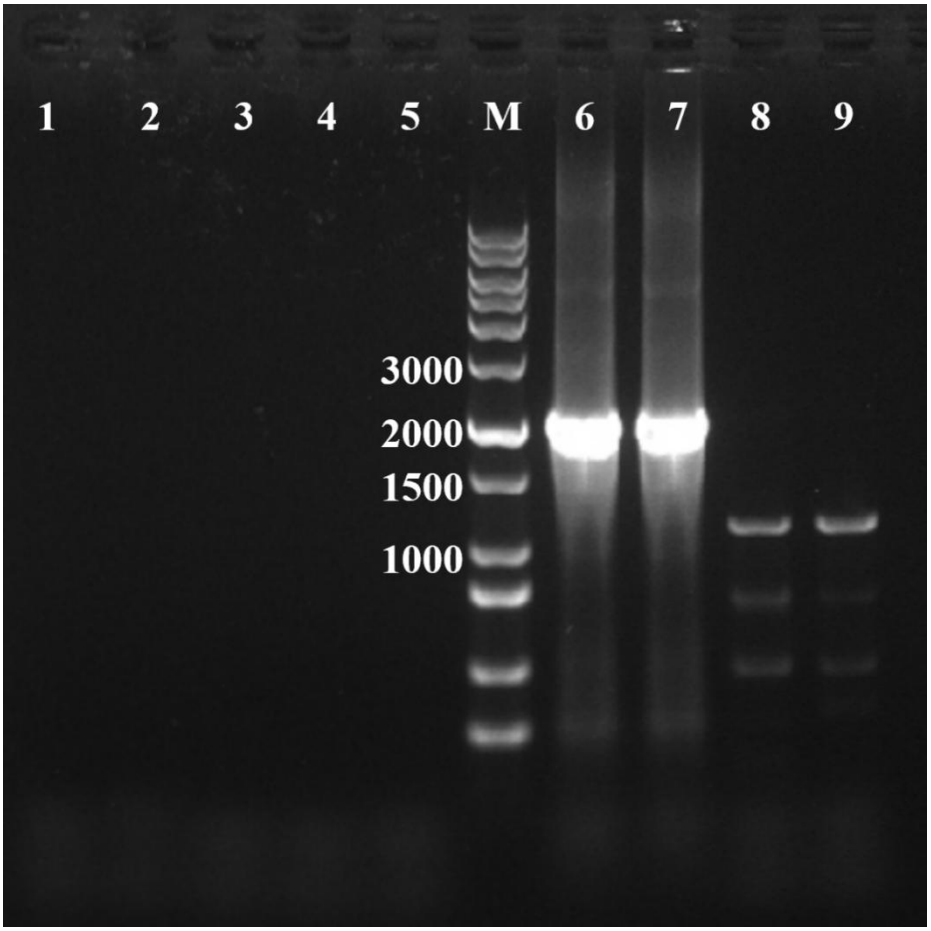


Figure 9. Agarose gel electrophoresis result of the second yeast colony PCR. Lane 1: BY4741 colony PCR; Lane 2–3: colony PCR from the second yeast transformation plate; Lane 4–5: colony PCR from the first yeast transformation plate; Lane M: 10000 DNA marker; Lane 6–7: PCR products using E. coli extracted plasmids as templates; Lane 8–9: PCR products using first-round yeast extracted plasmids as templates.

The expected amplicon size was 2086 bp. Only E. coli plasmid templates yielded correct target bands, while no valid amplification was observed for all yeast samples.

May 31

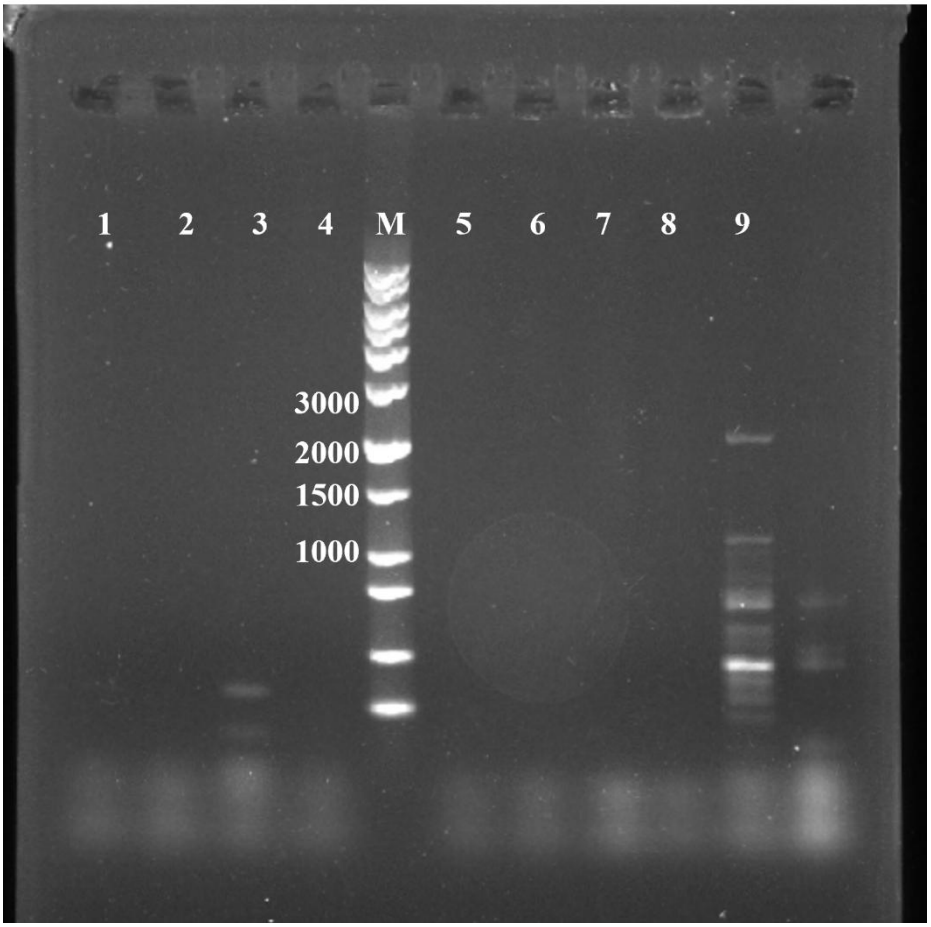


Figure 10.Agarose gel electrophoresis of yeast colony PCR under different cell lysis treatments. Lanes 1–8: colony PCR of colonies from the second yeast transformation under different lysis conditions. Lane 1 and 2: samples treated with 30 μL 20 mM NaOH at 98 °C for 10 min; Lane 3 and 4: samples treated with 10 μL 20 mM NaOH at 98 °C for 15 min; Lane 5 and 6: samples treated with 20 μL 0.2% SDS at 98 °C for 10 min; Lane7 and 8: samples treated with 0.2 M NaOH plus 0.2% SDS at 98 °C for 10 min. Lane M: DNA marker. Lane9: PCR using the third batch of yeast extracted plasmid as template.

Only the plasmid template from the third yeast plasmid extraction achieved target amplification, and no target band was obtained from all colony PCR groups.

June 5

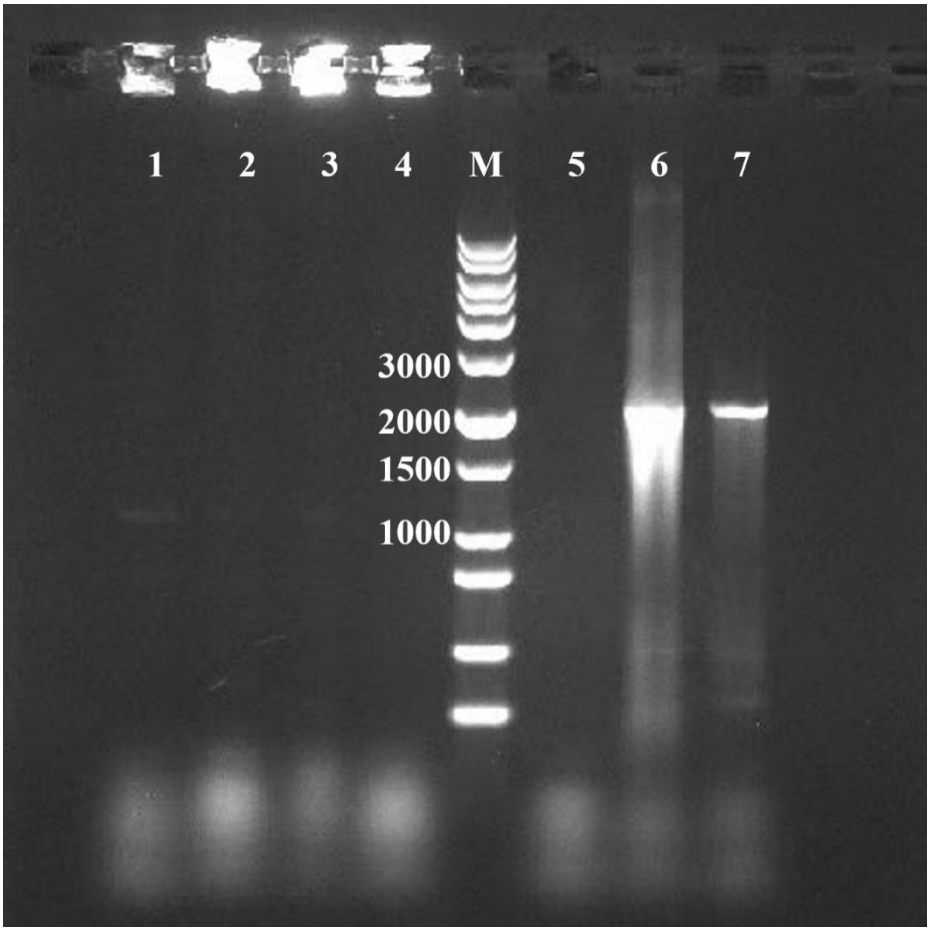


Figure 11. Agarose gel electrophoresis of yeast colony PCR. Lanes 1–4: yeast colony PCR samples; Lane M: DNA molecular weight marker; Lane 5: BY4741 yeast control; Lane 6: PCR using E. coli extracted plasmid as template; Lane 7: PCR using sample 1 of yeast extracted plasmid as template.

Only Lane 6 and Lane 7 produced the target band, and no target fragment was amplified from yeast colony PCR samples and BY4741 control.

June 6

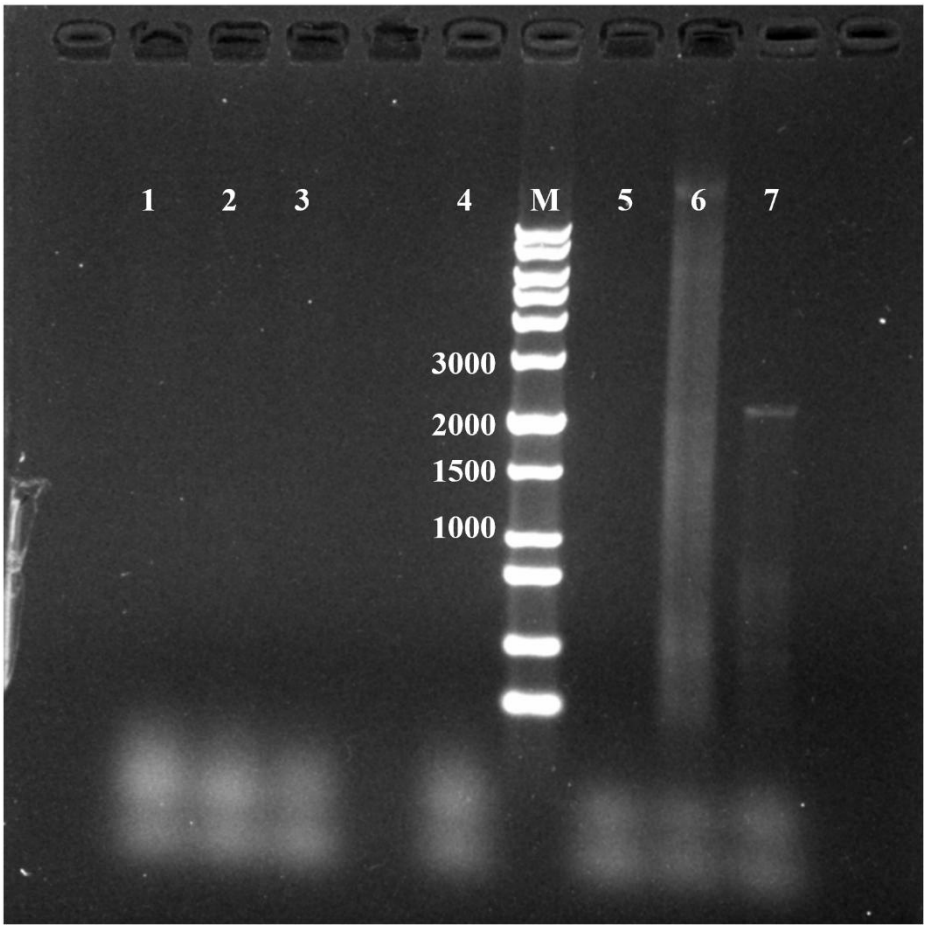


Figure 12. Agarose gel electrophoresis of yeast colony PCR. Lanes 1–4: yeast colony PCR samples; Lane M: DNA molecular weight marker; Lane 5: BY4741 yeast control; Lane 6: PCR using E. coli extracted plasmid as template; Lane 7: PCR using sample 1 of yeast extracted plasmid as template.

Only Lane 6 and Lane 7 produced the target band, and no target fragment was amplified from yeast colony PCR samples and BY4741 control.

June 6

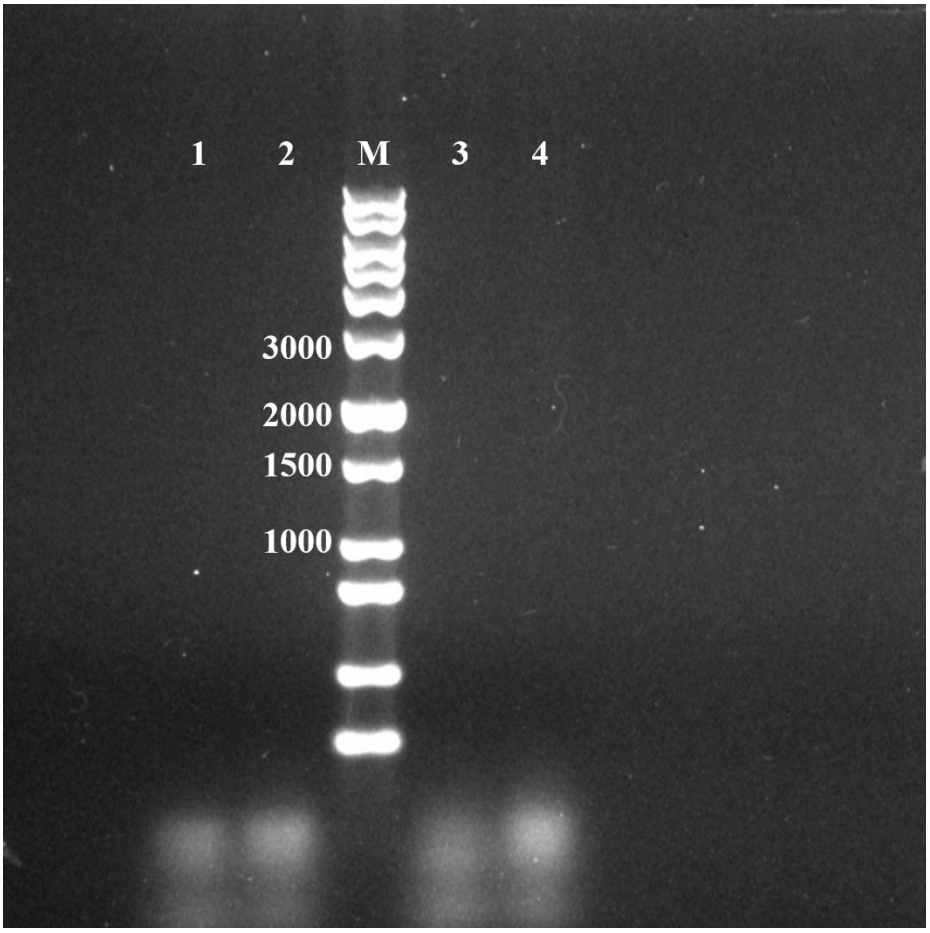


Figure 13. Agarose gel electrophoresis of yeast colony PCR with modified PCR template loading volume. Lanes 1–4: yeast colony PCR samples with template volumes of 0.5 μL, 1 μL, 1.5 μL and 2 μL, respectively. Lane M: DNA molecular weight marker. No target bands were detected in all samples.

June 9

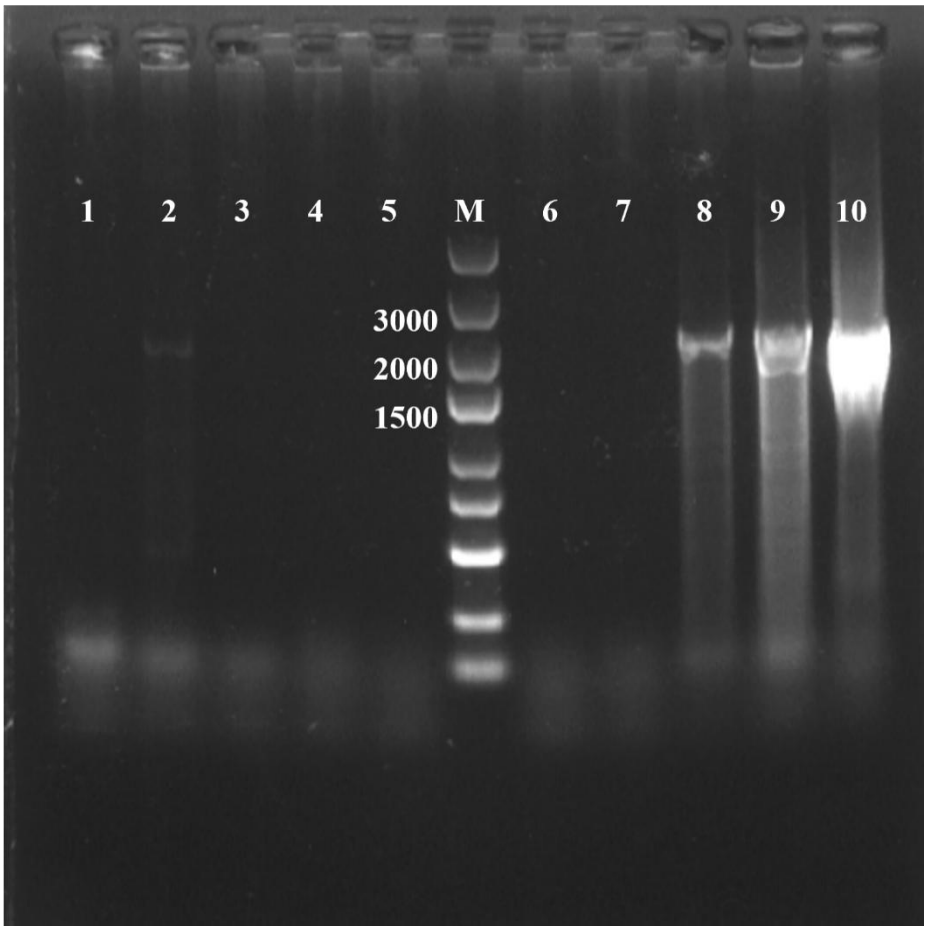
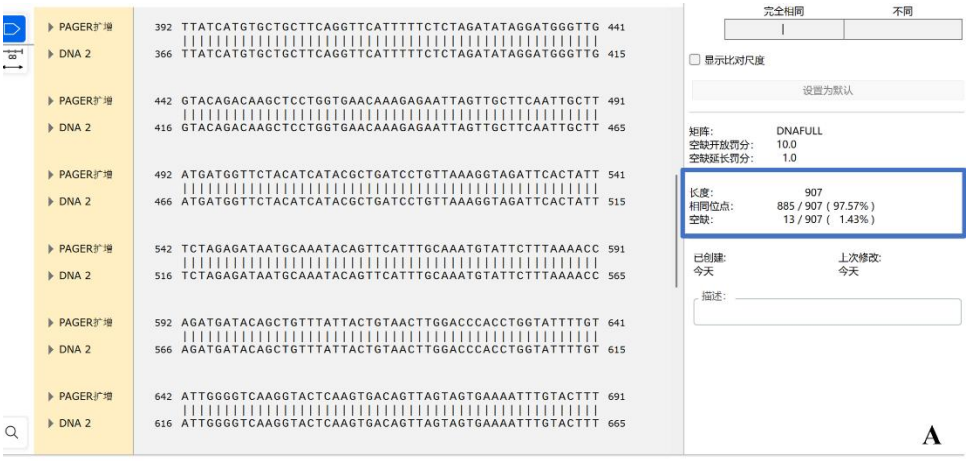


Figure 14. Agarose gel electrophoresis of yeast colony PCR under different thermal lysis temperatures. Lanes 1, 4, 7: yeast colony samples lysed at 90 °C; Lanes 2, 5, 8: yeast colony samples lysed at 80 °C; Lanes 3, 6, 9: yeast colony samples lysed at 70 °C; Lane M: DNA molecular weight marker; Lane 10: PCR using E. coli plasmid extracted on 24 April as template.

Target bands were successfully amplified in Lane 2, Lane 9 and Lane 10, indicating successful colony PCR validation for these samples.

June 16



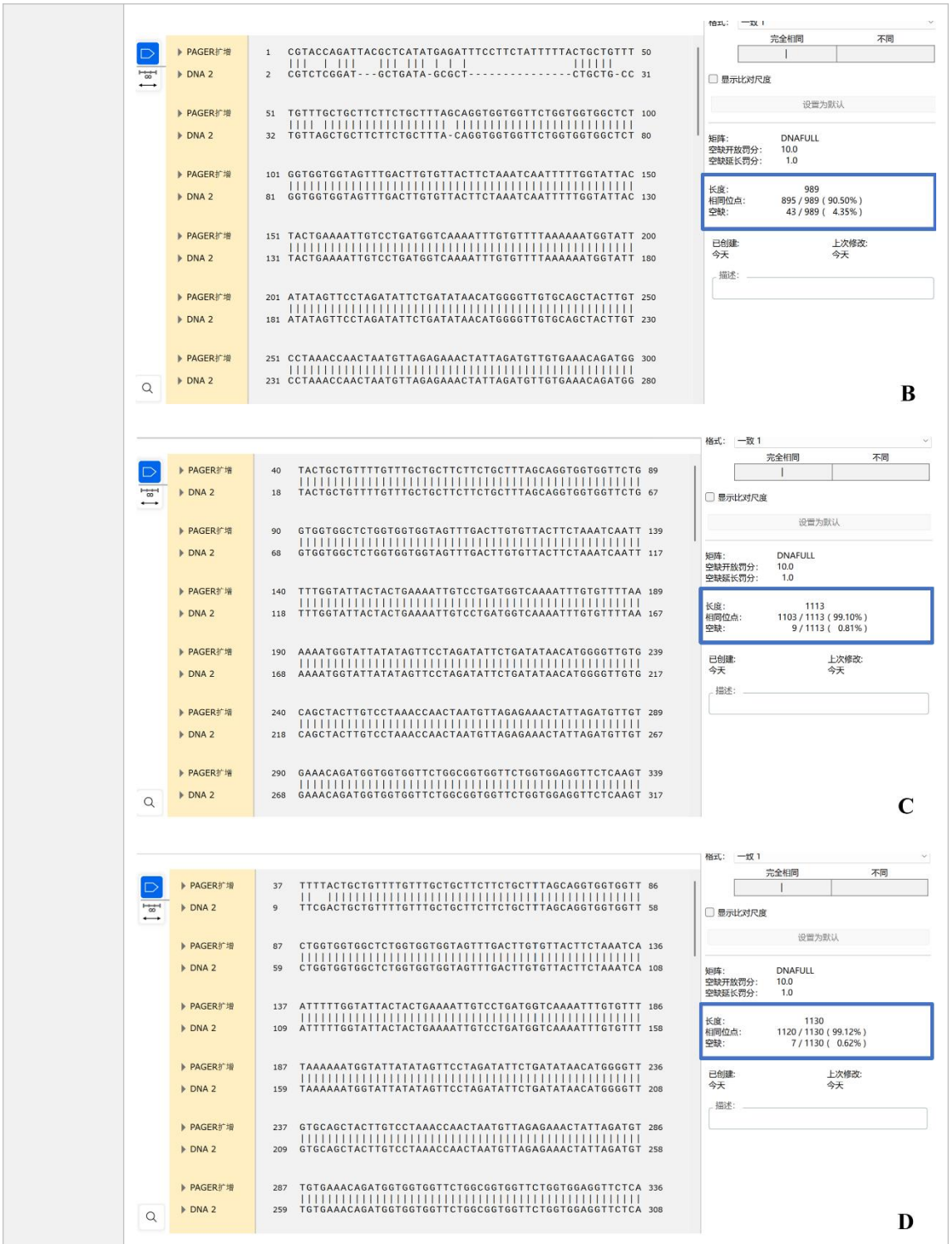
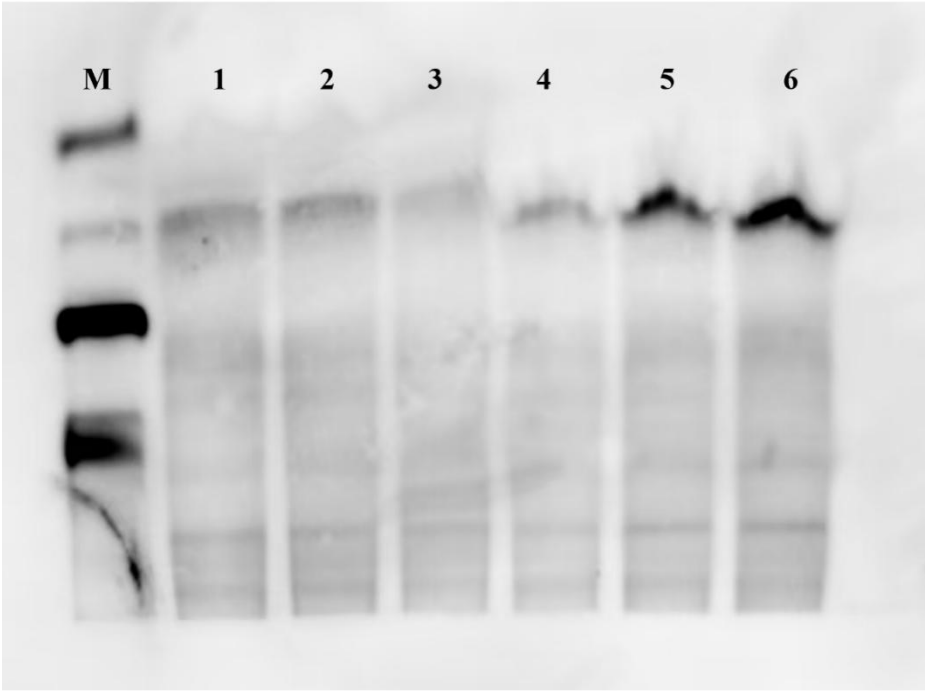


Figure 15. Sequence alignment of positive yeast transformants. Subgraphs A, B, C and D show sequence pairwise alignment between four sequenced yeast samples and the reference PAGER amplified sequence. A: length = 907 bp, 97.57% identity; B: length = 989 bp, 90.50% identity; C: length = 1113 bp, 99.10% identity; D: length = 1130 bp, 99.12% identity.

The majority of samples exhibited high consistency with the reference sequence, confirming the correct insertion of the PAGER target fragment. Minor gaps were regarded as sequencing artifacts, and the overall sequencing results were reliable.

Explanation	This series of experiments aims to construct and validate yeast strains carrying the PAGER plasmid for subsequent protein expression and functional assay. At the early stage, we lacked mature conditions for yeast colony PCR. Yeast cell wall is thick and hard to fully lyse, so many initial PCR attempts failed to produce target bands. It took substantial time to screen and optimize lysis temperature, template dosage and other parameters. After iterative optimization of thermal lysis conditions, we obtained positive clones by colony PCR. Sequencing alignment further confirmed that the target PAGER sequence was correctly inserted into yeast without critical mutation or rearrangement, which provides qualified positive strains for the following functional characterization.
Next plan	Use sequence-verified positive yeast strains to carry out protein expression detection and subsequent functional fluorescence assays.

No 3 Date：July 10 to 19 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	SDS- PAGE and Western Blot protein expression detection of yeast PAGER strain
Note	Protocol 18
Result	July 10 to 11  Figure 1. Western blot result from the first trial for target protein detection. Lane M: protein molecular weight marker with poor definition, which cannot be used for precise band size estimation. Lanes 1–4: total protein samples extracted from PAGER-modified yeast strains. Lanes 5–6: total protein samples from wild-type BY4741 yeast. No obvious differential bands were observed between modified yeast and wild-type BY4741 samples. July 13 to 14

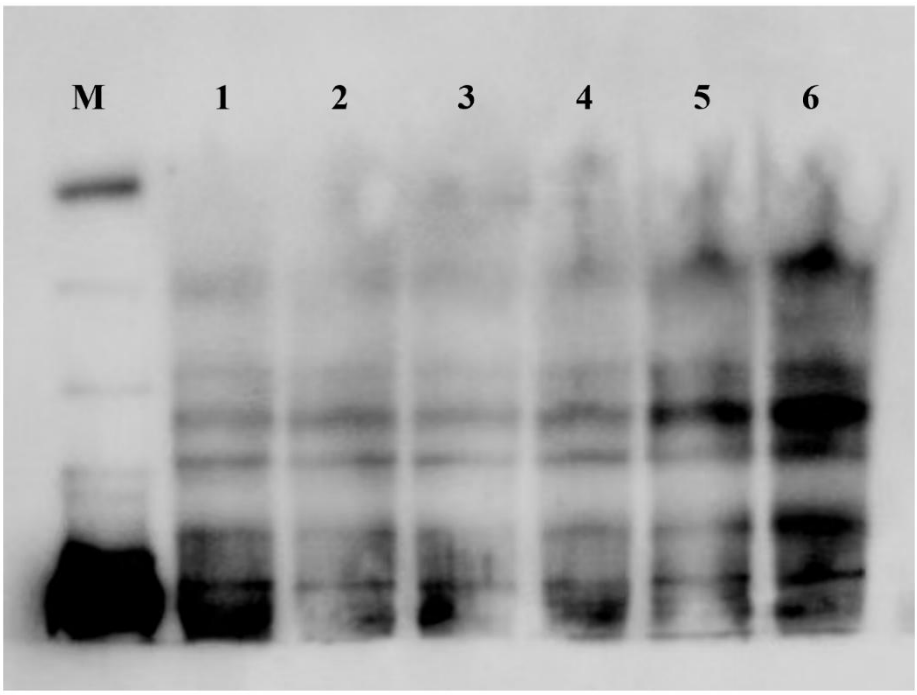


Figure 2. Western blot result from the second trial for target protein detection. Lane M: protein molecular weight marker. Lanes 1–4: total protein samples extracted from PAGER-modified yeast strains. Lanes 5–6: total protein samples from wild-type BY4741 yeast.

Consistent with the first trial, no obvious differential bands were observed between modified yeast and wild-type BY4741 samples.

July 16

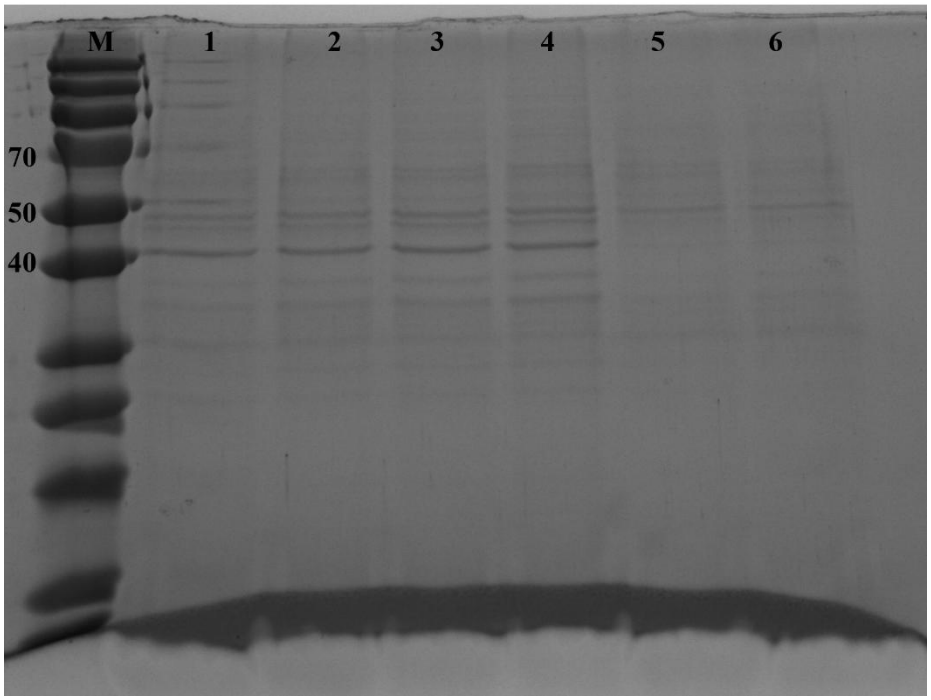
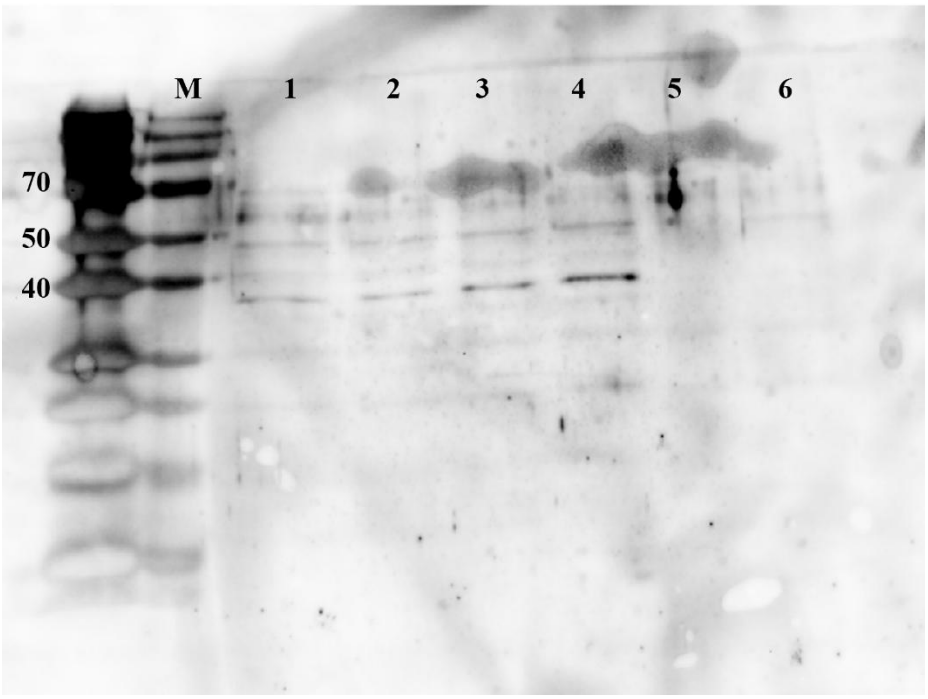
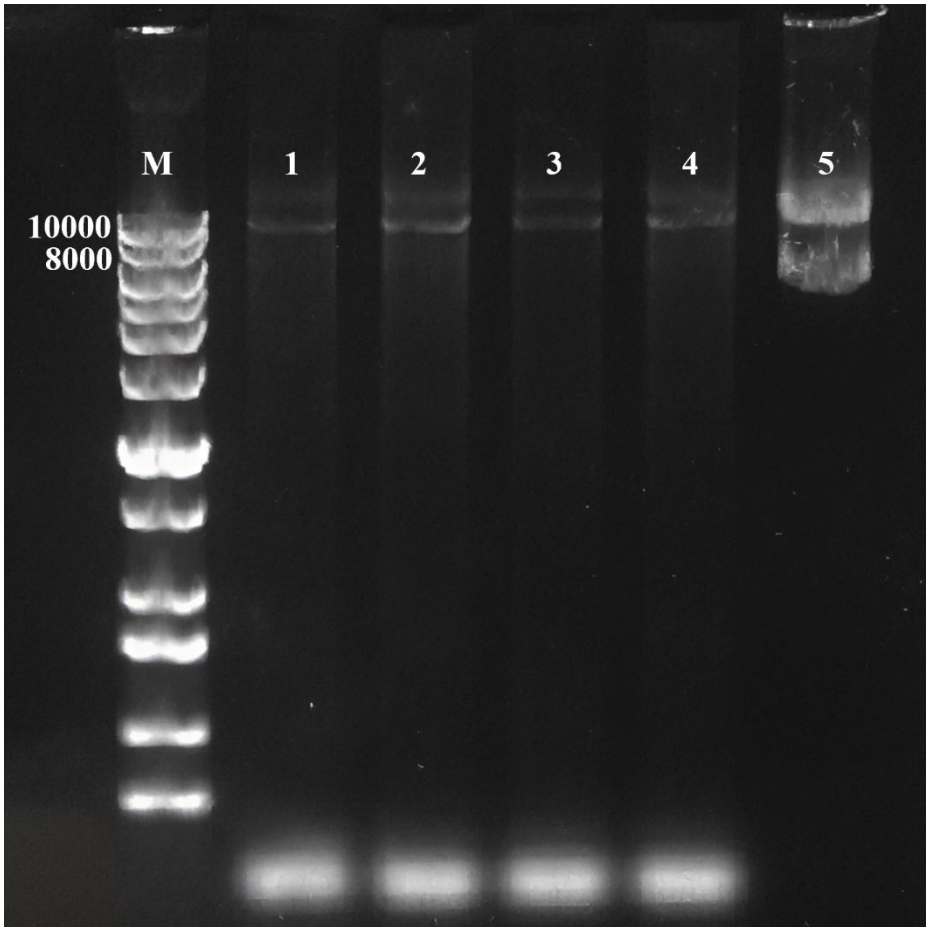


Figure 3. SDS-PAGE analysis of total protein extracted from yeast strains. Lane M: protein molecular weight marker, with 40 kDa and 70 kDa positions labeled. Lanes 1–4: total protein from PAGER-modified yeast strains. Lanes 5–6:

	total protein from wild-type BY4741 yeast. A differential band near 40 kDa was detected in lanes 1–4, which was not present in wild-type samples. The theoretical molecular weight of target protein is 70 kDa, so the observed band shows an abnormal migration position. July 16 to 17  Figure 4. Western blot detection using muscarinic Acetylcholine Receptor 1 Rabbit Polyclonal Antibody.Lane M: protein molecular weight marker, with 40 kDa, 50 kDa and 70 kDa positions labeled. Lanes 1–4: total protein from PAGER-modified yeast strains. Lanes 5–6: total protein from wild-type BY4741 yeast. Specific bands near 40 kDa were detected in modified yeast samples, which were not present in wild-type controls. The theoretical molecular weight of the target protein is 70 kDa, hence the observed bands show an unexpected migration position.
Explanation	This set of SDS-PAGE and Western blot experiments aimed to validate target PAGER protein expression in sequence-verified yeast transformants. The first two WB attempts showed no obvious differential bands. Later SDS-PAGE and Western blot detected specific bands near 40 kDa in modified yeast, which deviated from the expected 70 kDa of full-length target protein. Since we did not acquire the expected protein band, we suspect there are problems with target protein expression, such as protein truncation or degradation inside yeast cells. Therefore, new gene constructs will be designed and built to improve protein expression for follow-up functional tests.
Next plan	Remove redundant sequences between promoter and target gene, construct new plasmids, and repeat protein expression detection.

No 4 Date：July 20 to 24 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Second plasmid reconstruction: removal of redundant sequences between promoter and target gene, preparation of linear fragments with homology arms for in vivo circularization in yeast
Note	Protocol 7、Protocol 12
Result	July 20  Figure 1. Agarose gel electrophoresis of PCR-generated linear fragments for the second plasmid construction.Lane M: DNA molecular weight marker, with 8000 bp and 10000 bp labeled. Lanes 1–4: PCR-amplified linear fragments. Lane 5: circular plasmid as negative control. The expected size of target fragment is 9522 bp. Target bands at the expected size were detected in lanes 1–4, while extra non-specific miscellaneous bands from non-specific PCR amplification were also observed. July 21

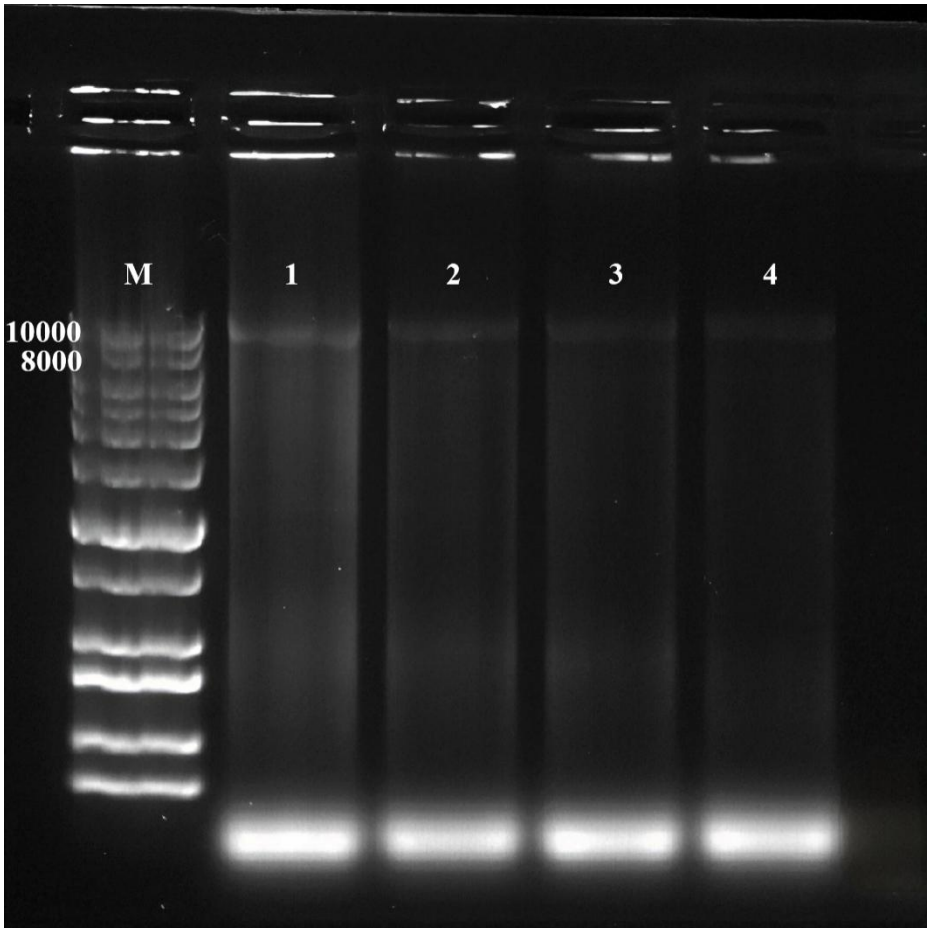
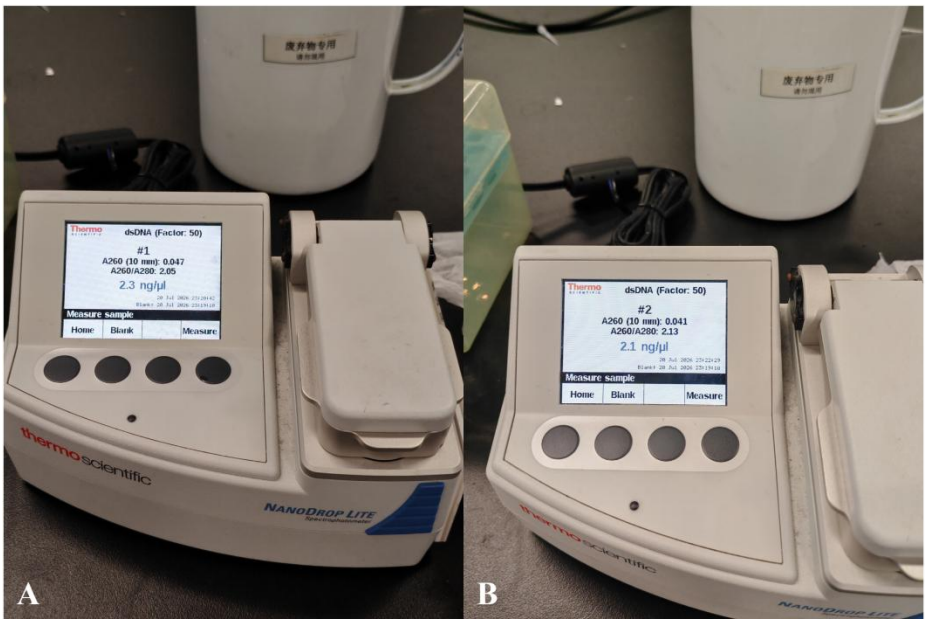


Figure2. Agarose gel electrophoresis of the second linearization PCR product. M: DNA molecular weight marker; Lane 1–4: four parallel PCR replicates. Target bands were successfully amplified, yet the bands were faint, indicating low product yield. The target bands will be excised and purified by gel extraction.

July 21



Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	2.3ng/μl	2.05
B	2.1ng/μl	2.13

Figure 3: Nanodrop measurement results of gel-extracted linear fragments. Panel A: Sample 1, concentration = 2.3 ng/μL, A260/A280 = 2.05. Panel B: Sample 2, concentration = 2.1 ng/μL, A260/A280 = 2.13.
The A260/A280 values are within the acceptable range for DNA, but the DNA concentration of both samples is relatively low.

July 22

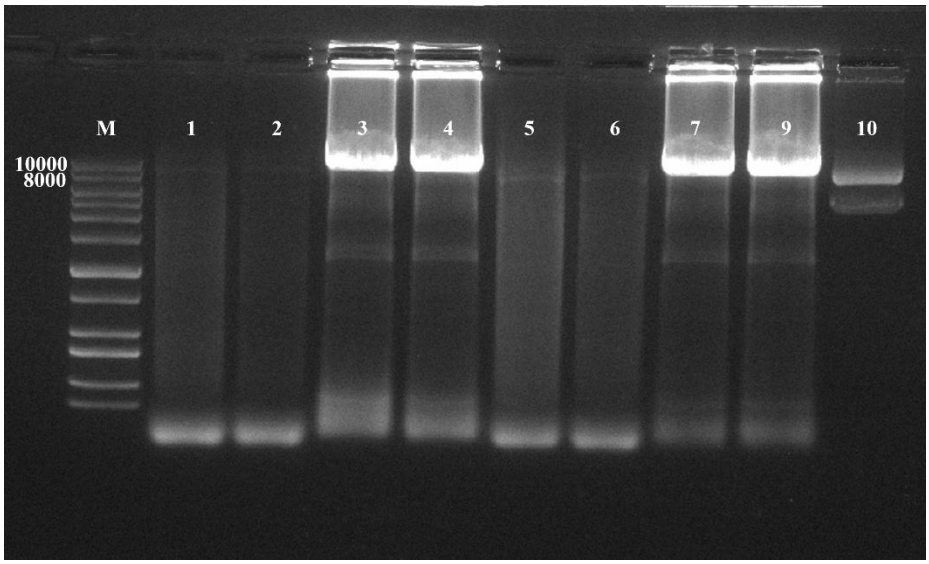
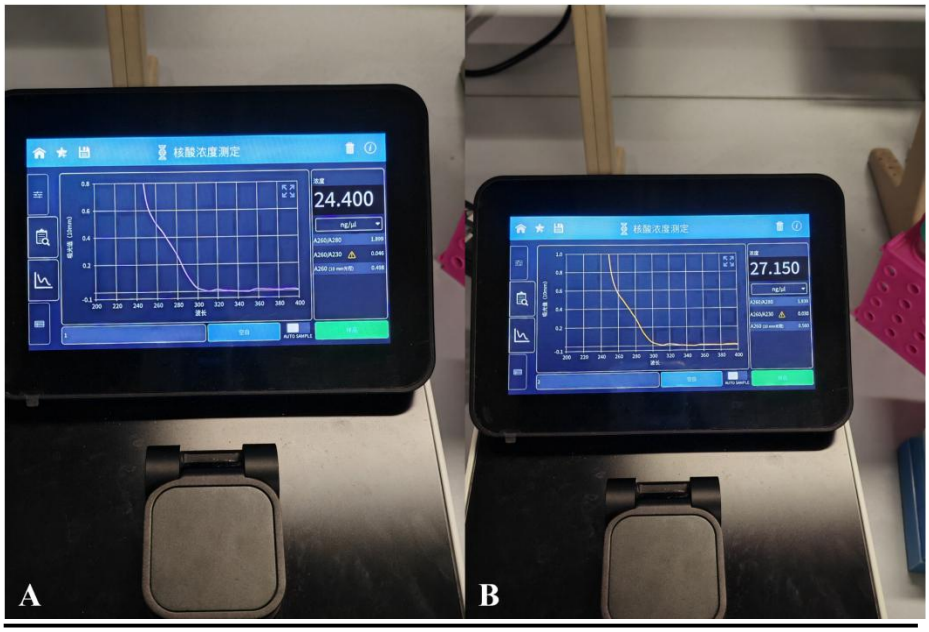


Figure 4. Agarose gel electrophoresis of linearization PCR products with different DNA polymerases.Lane M: DNA molecular weight marker, with 8000 bp and 10000 bp labeled. Lanes 1,2,5,6: PCR products amplified by ordinary Taq polymerase. Lanes 3,4,7,8: PCR products amplified by high-fidelity DNA polymerase. Lane 10: circular blank plasmid as control.
Target bands were obtained in all PCR samples. The bands from high-fidelity polymerase groups (lanes 3,4,7,8) were markedly brighter than those amplified by ordinary Taq polymerase, showing improved amplification yield.

July 22



Extracted plasmid Sample	Concentration	Purity (A260/A280)
A	24.400ng/μl	1.899
B	2.7.150ng/μl	1.939

Figure 5. Nucleic acid quantification of gel-purified linear fragments amplified by high-fidelity polymerase.(A) Sample A, DNA concentration = 24.400 ng/μL. (B) Sample B, DNA concentration = 27.150 ng/μL.

The use of high-fidelity polymerase greatly improved the yield of recovered linear DNA fragments, and the concentration meets the requirement for yeast transformation.

July 23

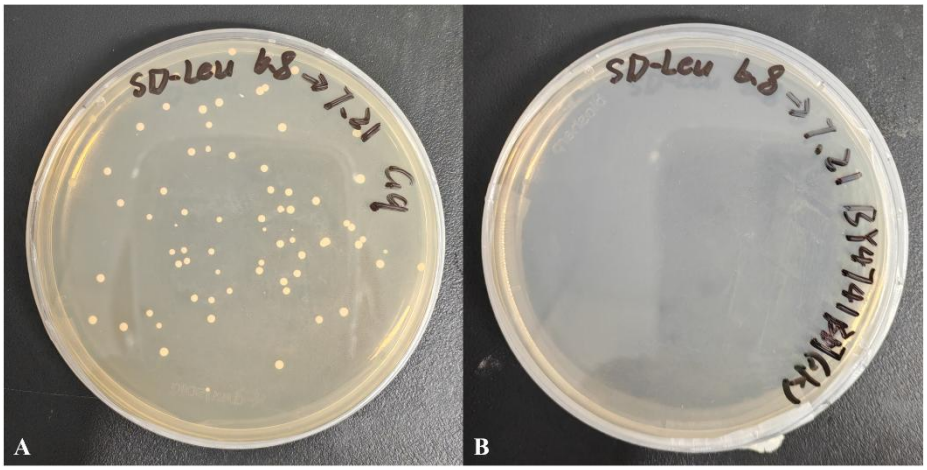


Figure 6. Yeast transformation screening on SD-Leu selective plates.(A) Experimental plate transformed with purified linear DNA fragments, numerous yeast colonies were obtained. (B) Negative control plate (BY4741 without DNA transformation), no colonies grew.

The plate results confirm successful yeast transformation and effective selection.

July 24 to 25

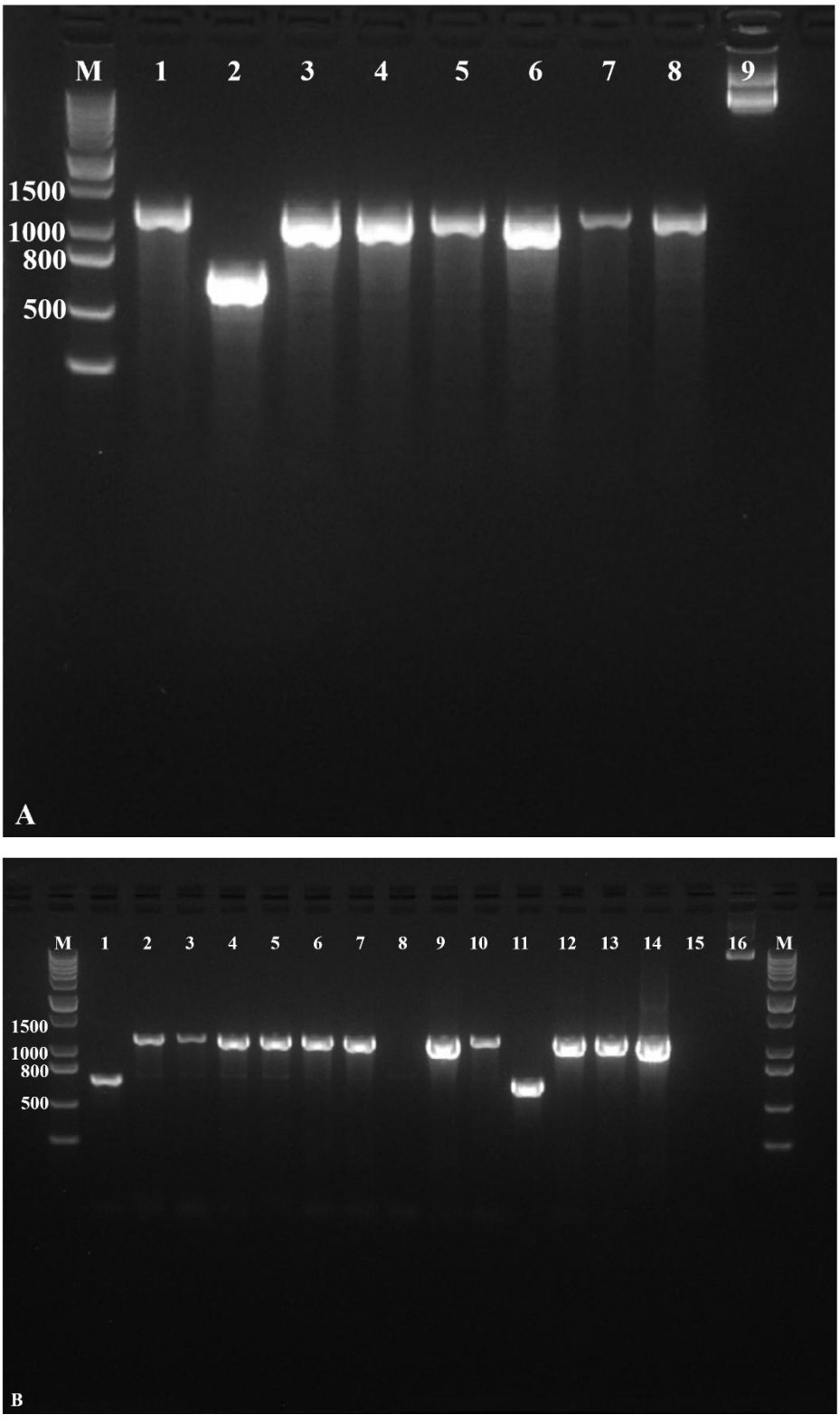


Figure 7. Agarose gel electrophoresis of colony PCR products for yeast transformant screening.(A) Eight picked yeast colonies (Lanes1–8). Only Lane 2 exhibited the target band.(B) Lane1: previously verified positive colony from panel A. Lanes2–11: ten newly picked colonies; only Lane11 produced the target band. Lane14: PCR product amplified from original plasmid template.

	Lane15: PCR with wild-type BY4741 as template. Lane16: unamplified original plasmid loaded directly as control. Two colonies with target-sized PCR amplicons were regarded as successfully transformed yeast strains carrying the reconstructed plasmid.
Explanation	This experiment focused on second-round plasmid reconstruction. Based on previous SDS-PAGE and Western blot results, we suspected that redundant sequences between the promoter and target gene caused abnormal transcription or premature translation termination, leading to truncated protein products. To eliminate this interference, these redundant sequences were removed. Linear DNA fragments carrying homology arms were then generated by PCR. Instead of assembling plasmids in vitro, these linear fragments rely on yeast endogenous homologous recombination machinery to complete self-circularization inside yeast cells, so as to obtain the corrected recombinant plasmid for subsequent protein-expression experiments.
Next plan	Cultivate verified positive yeast strains and wild-type BY4741 control group, complete functional validation experiments and record fluorescence signal outputs.

No 5 Date：August 1 to 5 Experimenters: Aishi Zeng Recorder: Aishi Zeng

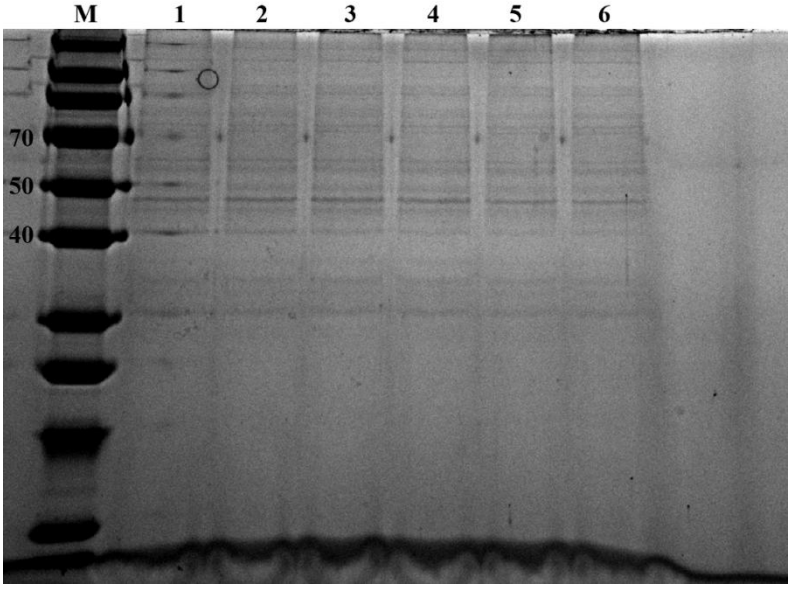
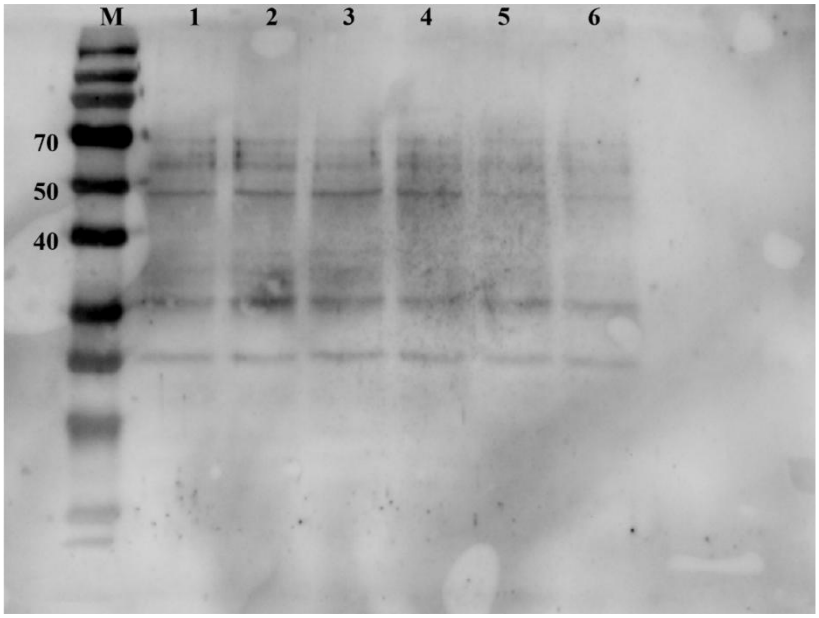
Experiment	SDS- PAGE and Western-Blot detection of target protein from second-generation reconstructed yeast strains
Note	Protocol 18
Result	August 1  Figure 1. Self-cast SDS-PAGE analysis of total protein extracted from engineered yeast strains.Lane M: protein molecular weight marker; Lanes 1–2: protein of first-generation modified yeast; Lanes 3–4: protein of second-generation modified yeast (redundant sequences between promoter and target gene removed); Lanes 5–6: wild-type BY4741 negative control. No specific band different from negative control was found around the expected molecular weight of 70 kDa in lanes 1–4. August 1 to 2 

Figure 2. Western blot detection of total protein extracted from engineered yeast strains.Sample loading order is identical to the SDS-PAGE gel. Lane M: protein marker; Lane 1–2: first-generation modified yeast; Lane3–4: second-generation modified yeast; Lane5–6: wild-type BY4741 negative control.
No obvious specific band distinct from the negative control was observed near the expected 70 kDa position.

August 4

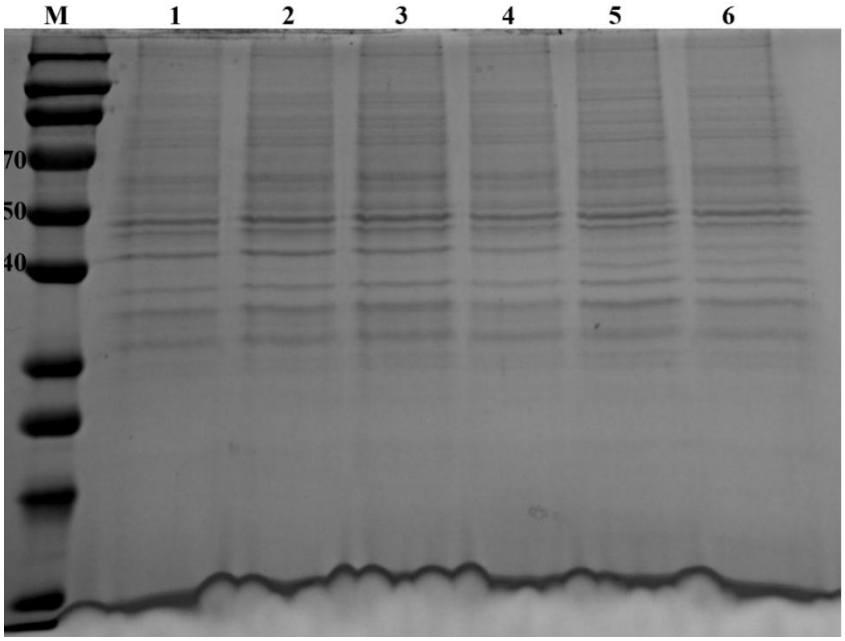
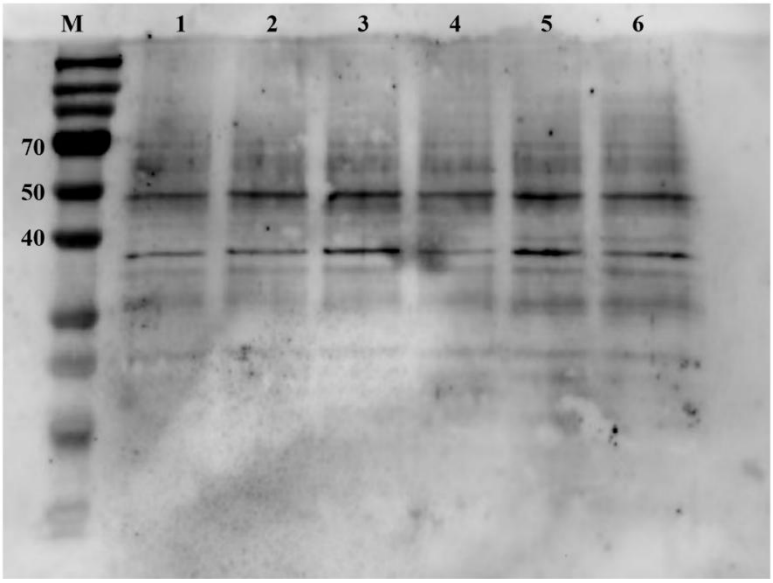
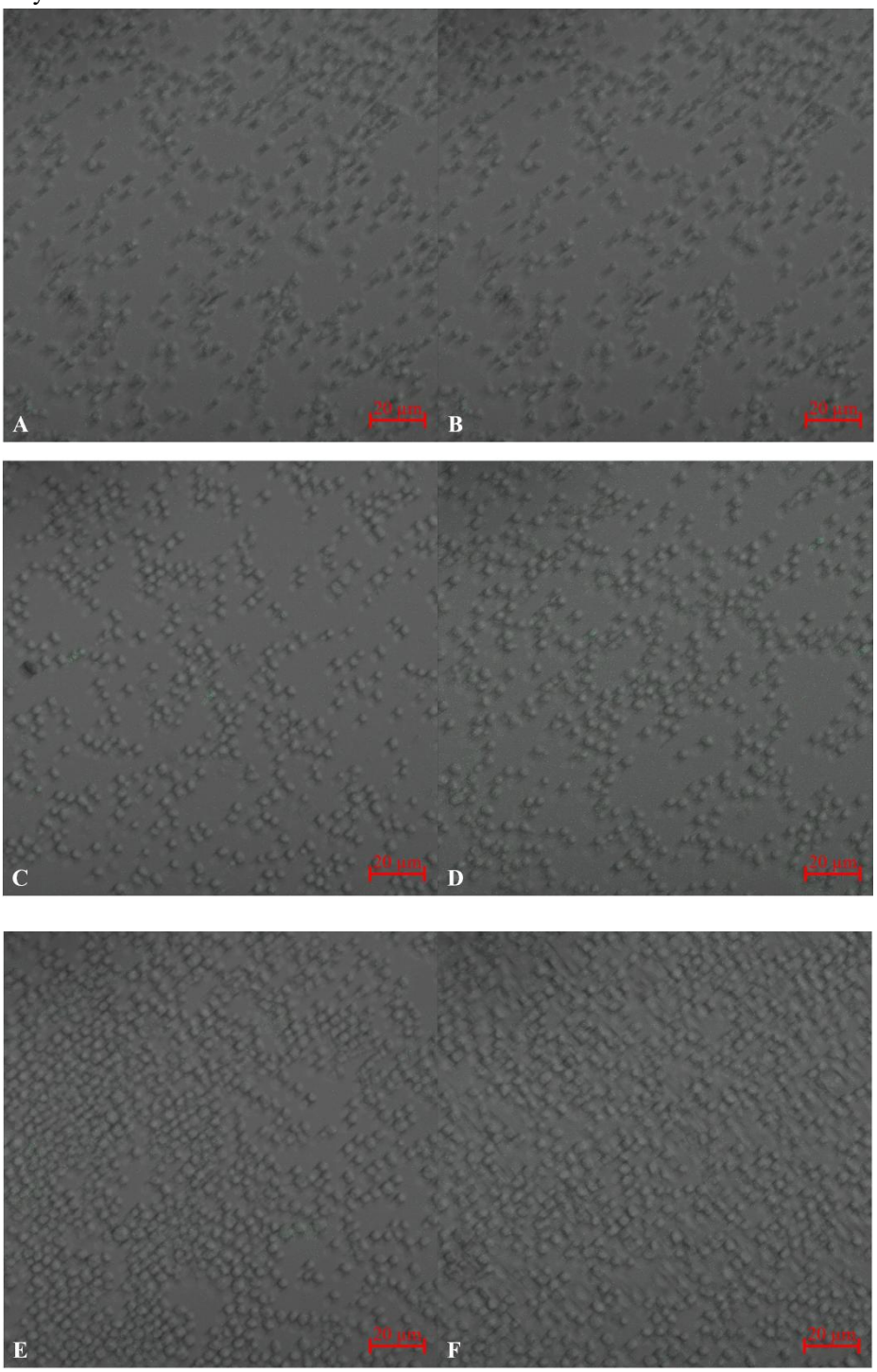


Figure 3. Repeated self-cast SDS-PAGE analysis of total protein extracted from engineered yeast strains.Sample loading order is identical to the first SDS-PAGE gel. Lane M: protein molecular weight marker; Lane 1–2: protein of first-generation modified yeast; Lane 3–4: protein of second-generation modified yeast (redundant sequences between promoter and target gene removed); Lane 5–6: wild-type BY4741 negative control.
Consistent with prior results, no specific band different from negative control was found around the expected molecular weight of 70 kDa in lanes 1–4.

August 4 to 5

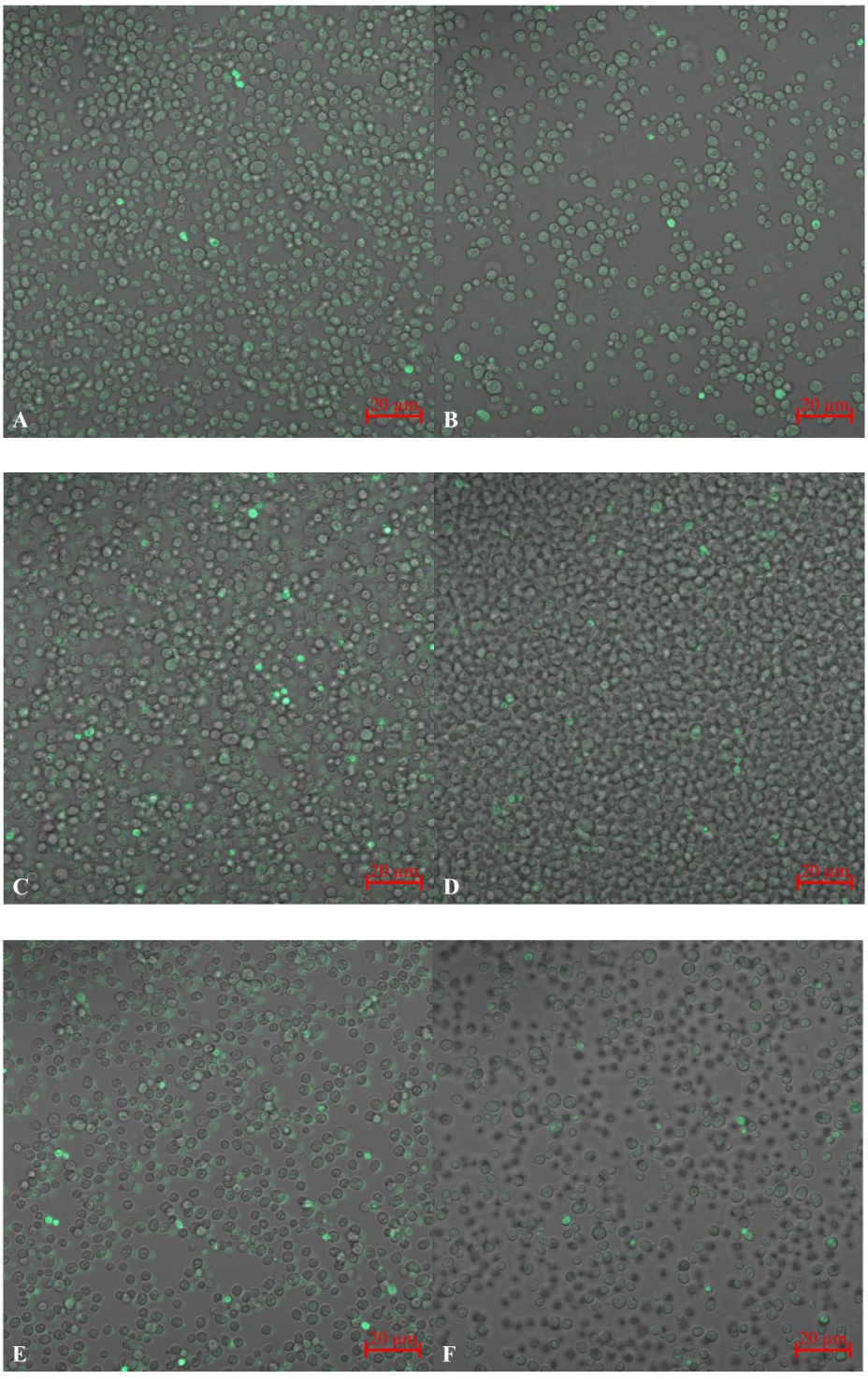
	 Figure 4. Repeated Western blot detection of total protein extracted from engineered yeast strains.Sample loading order is identical to the prior SDS-PAGE and WB gel. Lane M: protein molecular weight marker; Lane 1–2: protein of first-generation modified yeast; Lane 3–4: protein of second-generation modified yeast (redundant sequences between promoter and target gene removed); Lane 5–6: wild-type BY4741 negative control. Consistent with the first WB result, no specific band different from the negative control was found around the expected molecular weight of 70 kDa in lanes 1–4.
Explanation	Two independent replicate experiments including self-cast SDS-PAGE and Western blot were carried out to characterize target protein expression of the second-generation reconstructed yeast, where redundant sequences between the promoter and target gene were removed. In both replicates, neither the first-generation nor second-generation engineered yeast exhibited specific bands at the expected 70 kDa position when compared with the wild-type BY4741 negative control. The deletion of the redundant intervening sequence failed to restore expression of the full-length target protein. The consistent negative results from repeated experiments rule out random operational error. It is speculated that the fusion protein linker sequence may lead to protein misfolding, intracellular degradation or failed membrane localization, which results in the target protein being undetectable.
Next plan	Despite negative Western-blot results, trace functional protein may still localize to the membrane. We will carry out functional validation and confocal immunofluorescence assays to test HA-binding and fluorescence output. If no positive signal appears, we will optimize the fusion-protein linker, construct new plasmids and repeat transformation and protein-detection assays.

No 6 Date：July 29 to August 11 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Confocal immunofluorescence imaging of yeast strains after HA- protein incubation
Note	Protocol 17
Result	July 29  Figure 1. Confocal immunofluorescence imaging of yeast samples incubated with HA protein.(A-B) Strains generated from the first plasmid construction. (C-D) Strains generated from the second plasmid construction. (E-F) Wild-type BY4741 negative control.

No obvious difference in fluorescence signal was found between experimental groups and the negative-control group. Subsequent review revealed that the fluorophore-conjugated secondary-antibody incubation step was missed during sample processing.

August 7



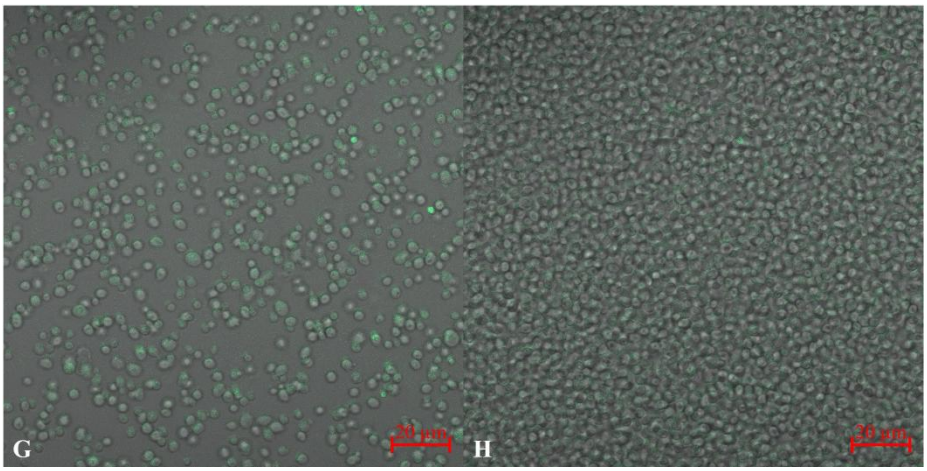
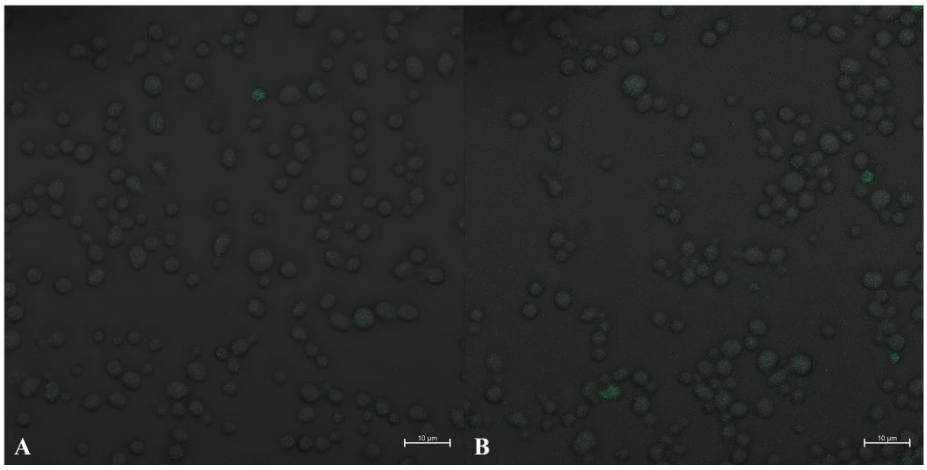
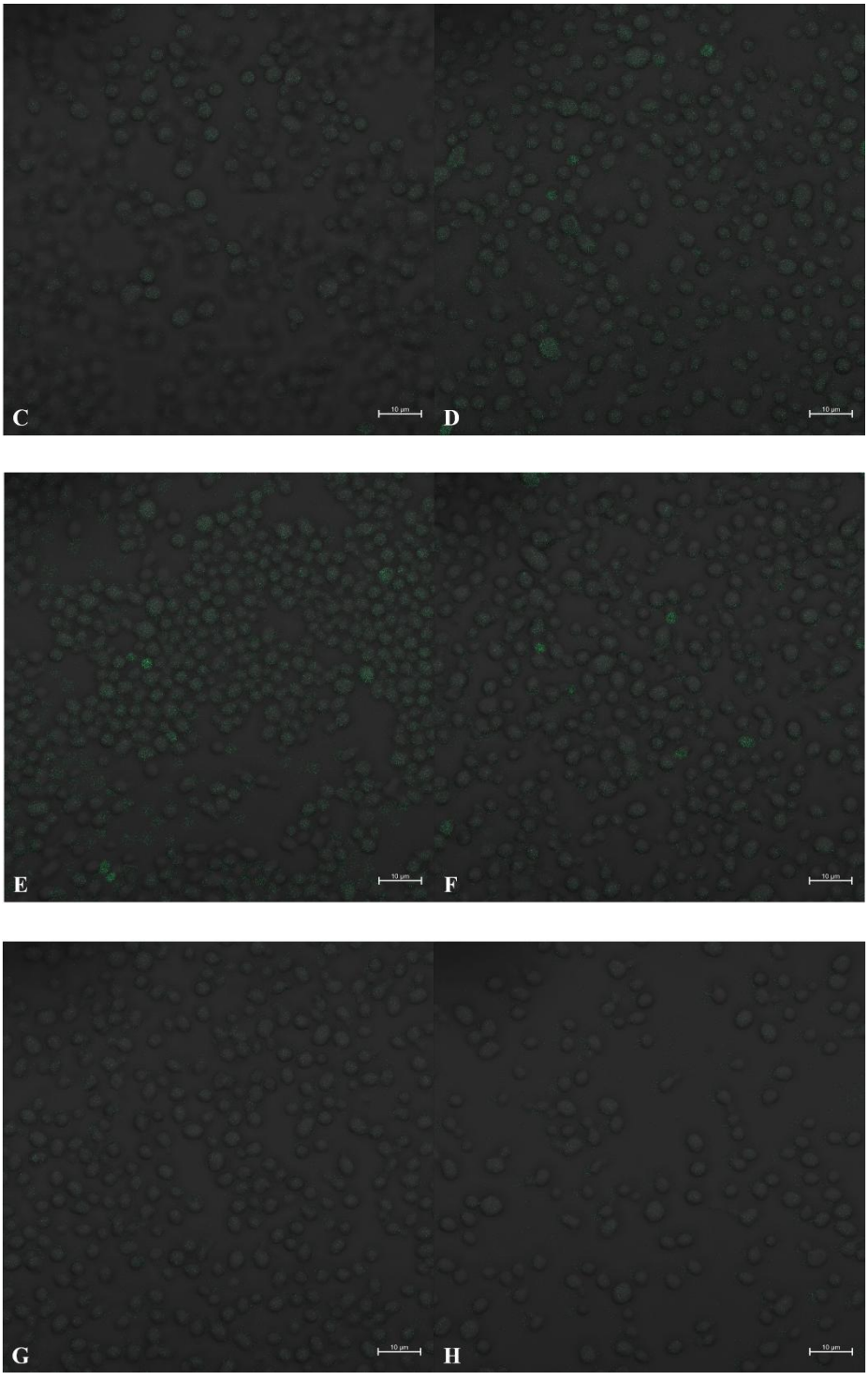


Figure 2. Confocal immunofluorescence imaging of yeast samples with different incubation treatments.(A) First-generation engineered yeast incubated with HA protein and fluorescent antibody. (B, C) Second-generation engineered yeast incubated with HA protein and fluorescent antibody. (D, E) Second-generation engineered yeast incubated only with HA protein. (F, G) Second-generation engineered yeast incubated only with fluorescent antibody. (H) Wild-type BY4741 incubated with HA protein and fluorescent antibody.
Laser power was set high, resulting in strong background fluorescence. No obvious membrane-specific fluorescent signal was observed in experimental groups.

August 9




C
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D
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E
10 µm
F
10 µm
G
10 µm
H
10 µm

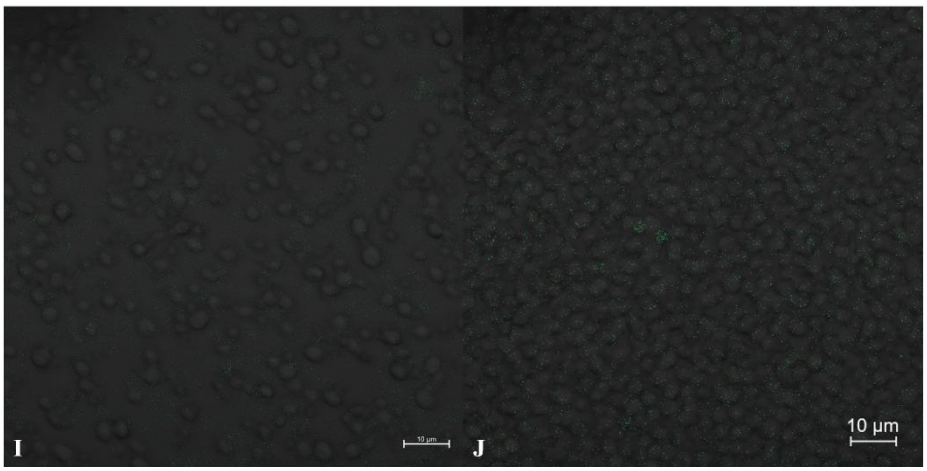
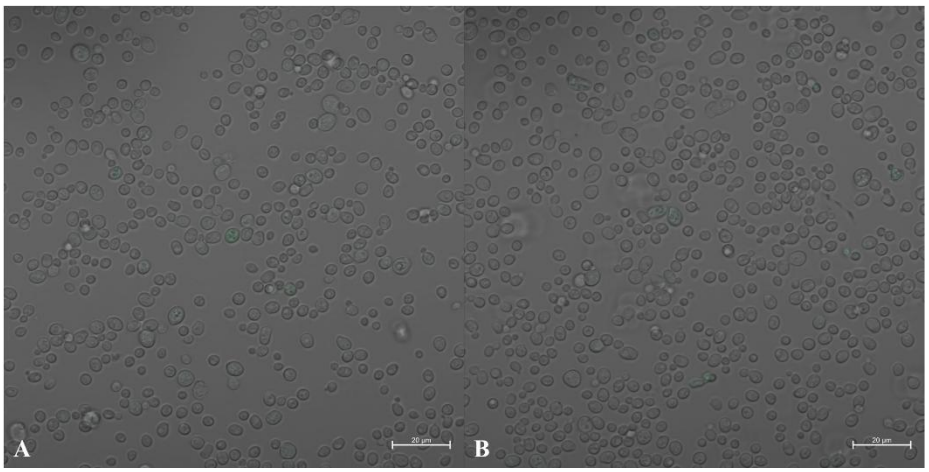
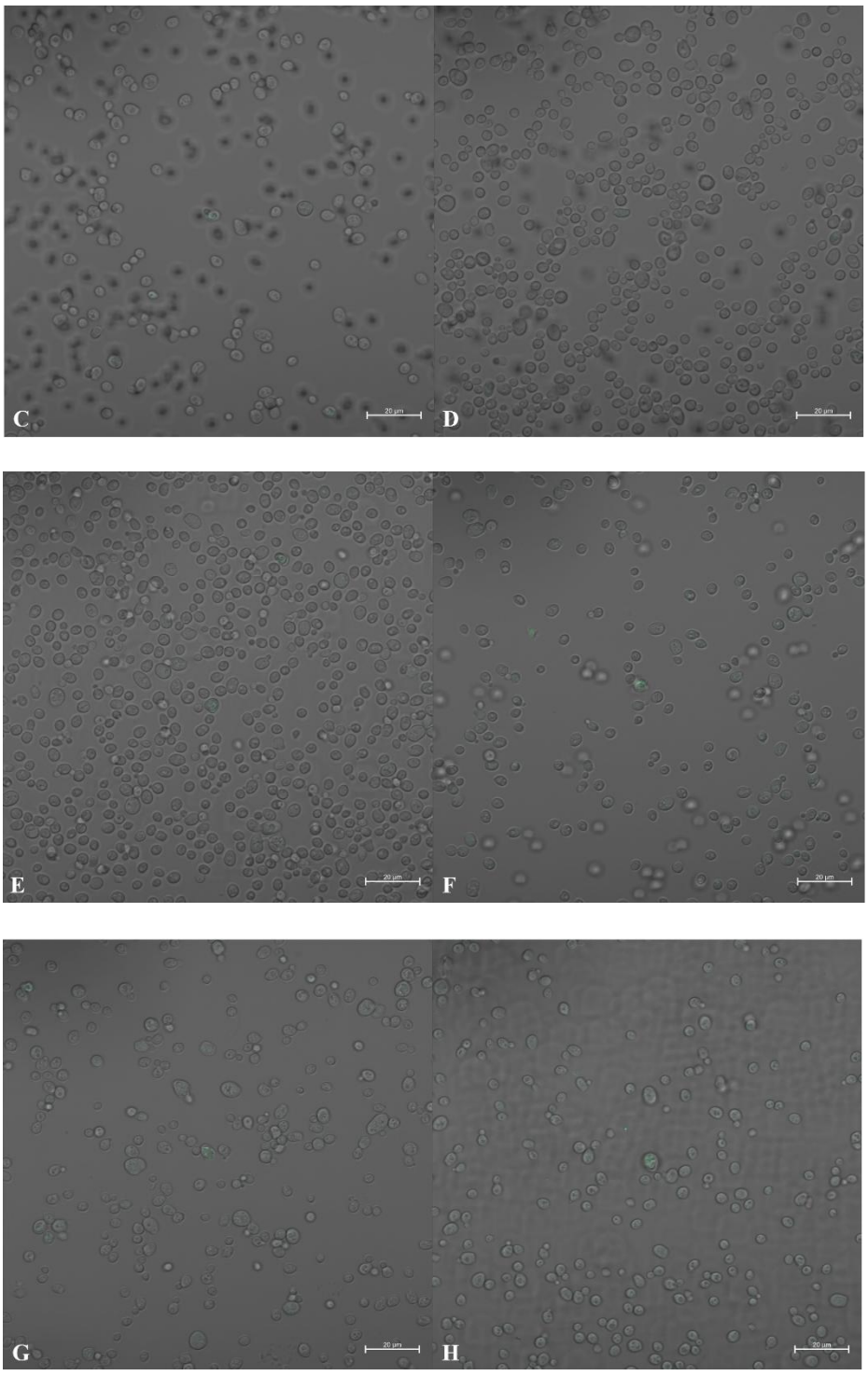


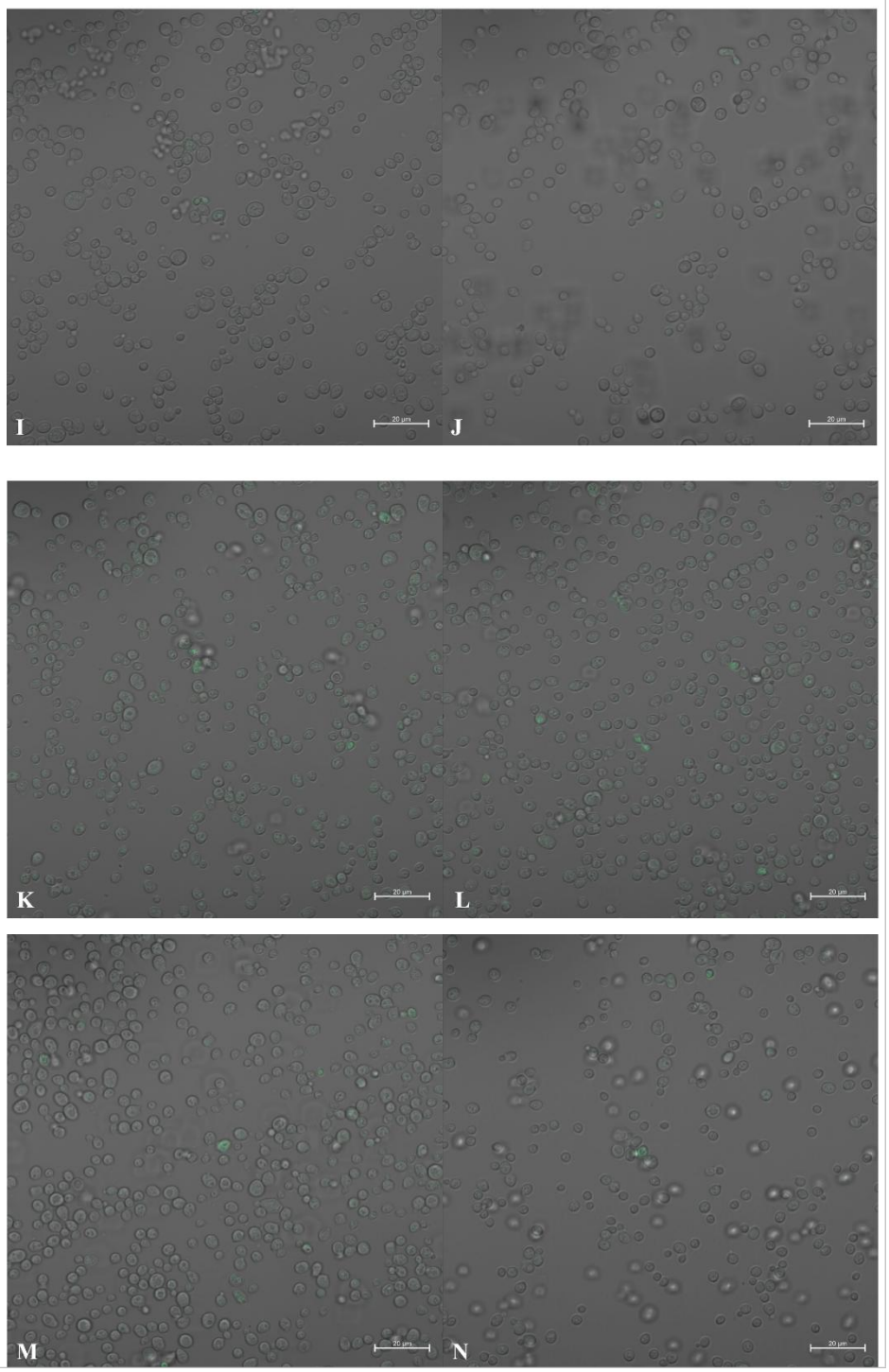
Figure 3. Confocal immunofluorescence observation of yeast samples under different incubation treatments.(A) First-generation engineered yeast incubated with HA protein and fluorescent antibody. (B, C) Second-generation engineered yeast incubated with HA protein and fluorescent antibody. (D) Second-generation engineered yeast incubated only with HA protein. (E) Second-generation engineered yeast incubated only with fluorescent antibody. (F) Second-generation engineered yeast with neither HA protein nor fluorescent antibody. (G) Wild-type BY4741 incubated with HA protein and fluorescent antibody. (H) Wild-type BY4741 incubated only with HA protein. (I) Wild-type BY4741 incubated only with fluorescent antibody. (J) Wild-type BY4741 with neither HA protein nor fluorescent antibody.

No obvious membrane-specific fluorescent signal was detected in experimental groups.

August 11




C
20 µm
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20 µm
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I
20 µm
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20 µm
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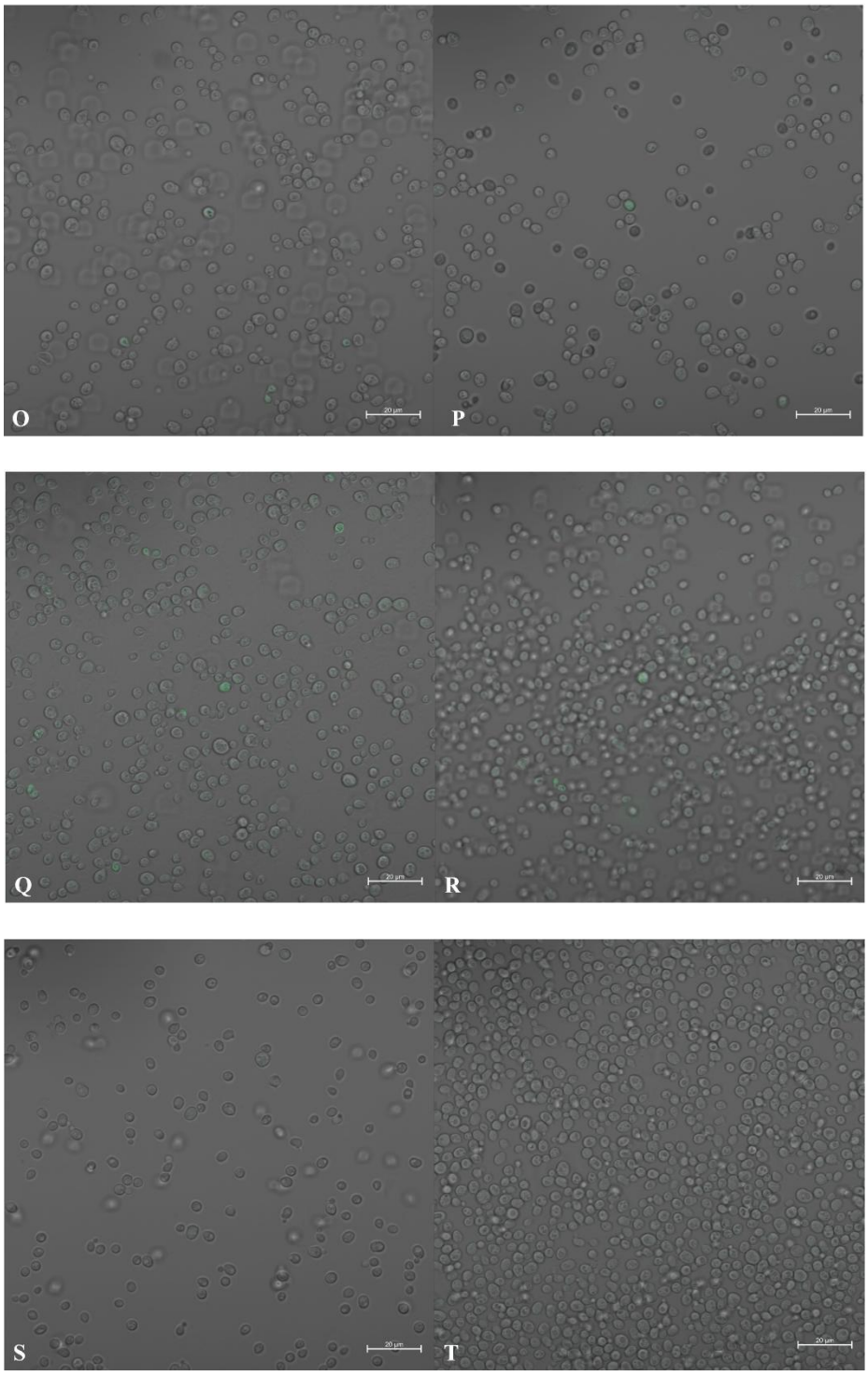
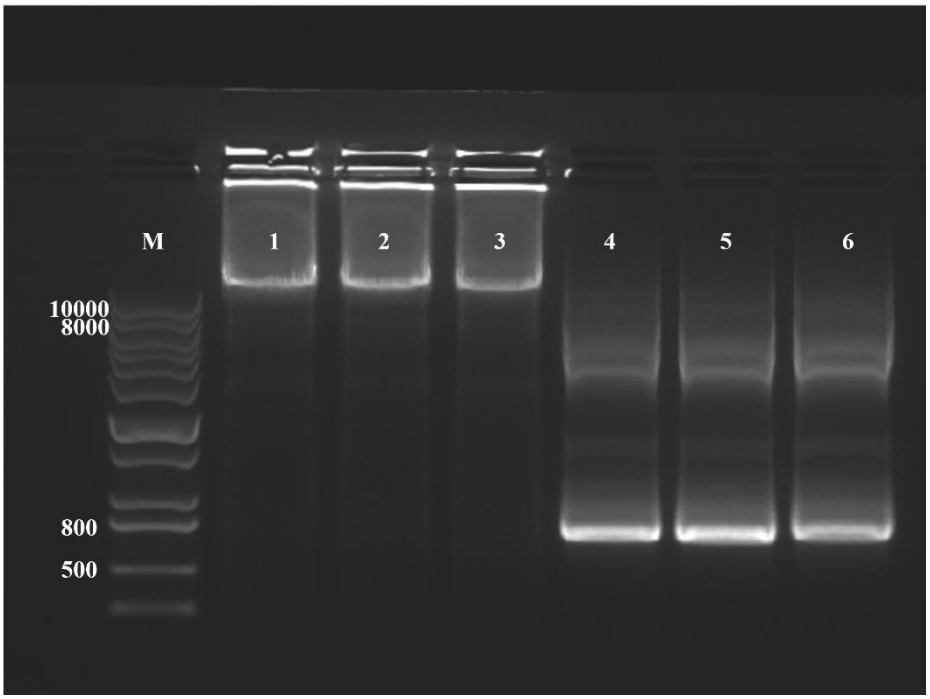


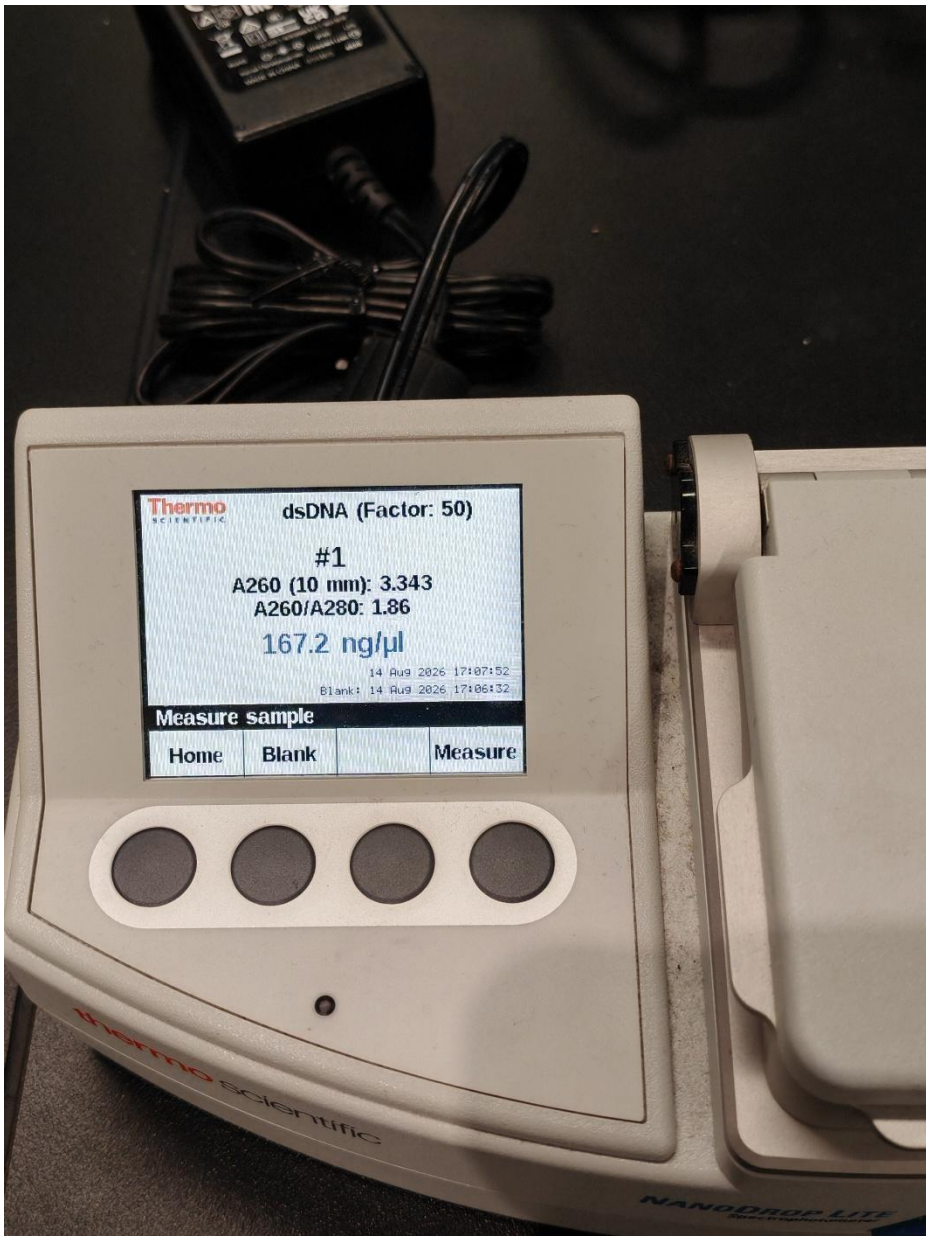
Figure 4. Confocal immunofluorescence screening with gradient HA protein and fluorescent antibody concentrations.(A-I) First-generation engineered yeast treated with HA (5, 15, 25 μg/mL) and fluorescent antibody at dilutions of 1:200, 1:100, 1:50. (J-R) Second-generation engineered yeast treated with the same gradient HA and antibody combinations. (S) Wild-type BY4741 incubated with HA protein and fluorescent antibody. (T) Wild-type BY4741 without HA protein or fluorescent antibody.

No obvious membrane-specific fluorescent signal was observed in all

	experimental groups.
Explanation	Multiple groups of confocal fluorescence detection experiments with gradient HA concentrations and different secondary antibody dilutions were performed on the first and second constructed yeast strains, alongside BY4741 negative controls. No obvious membrane-localized fluorescence signal was observed in all experimental groups. It is speculated that the linker sequence may affect the correct folding or membrane localization of the target fusion protein, leading to the failure of epitope exposure and no detectable fluorescence signal.
Next plan	Modify the linker sequence of the fusion protein, construct new plasmids, and repeat yeast transformation and confocal fluorescence detection to verify whether protein membrane localization can be restored.

No 7 Date：August 13th to 20th Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Vector reconstruction by modifying fusion-protein linker sequence: backbone linearization and insert assembly via one-step cloning
Note	Protocol 5、Protocol 7、Protocol 11
Result	August 14  Figure 1. Agarose gel electrophoresis of PCR products for backbone and insert fragments.Lane M: DNA molecular weight marker. Lane 1–3: PCR products for linearized backbone (expected 8844 bp). Lane 4–6: PCR products of target insert carrying redesigned linker (expected 762 bp). Bright bands of correct size were observed for the insert, while no clear target band was obtained for the large backbone fragment. August 14



Extracted plasmid Sample	Concentration	Purity (A260/A280)
insert fragment	167.2 ng/µL	1.86

Figure 2. Nanodrop quantification of gel-purified target insert fragment. The recovered 762 bp insert DNA was measured. Concentration = 167.2 ng/µL; A260/A280 = 1.86, demonstrating satisfactory DNA purity for subsequent one-step cloning.

August 14

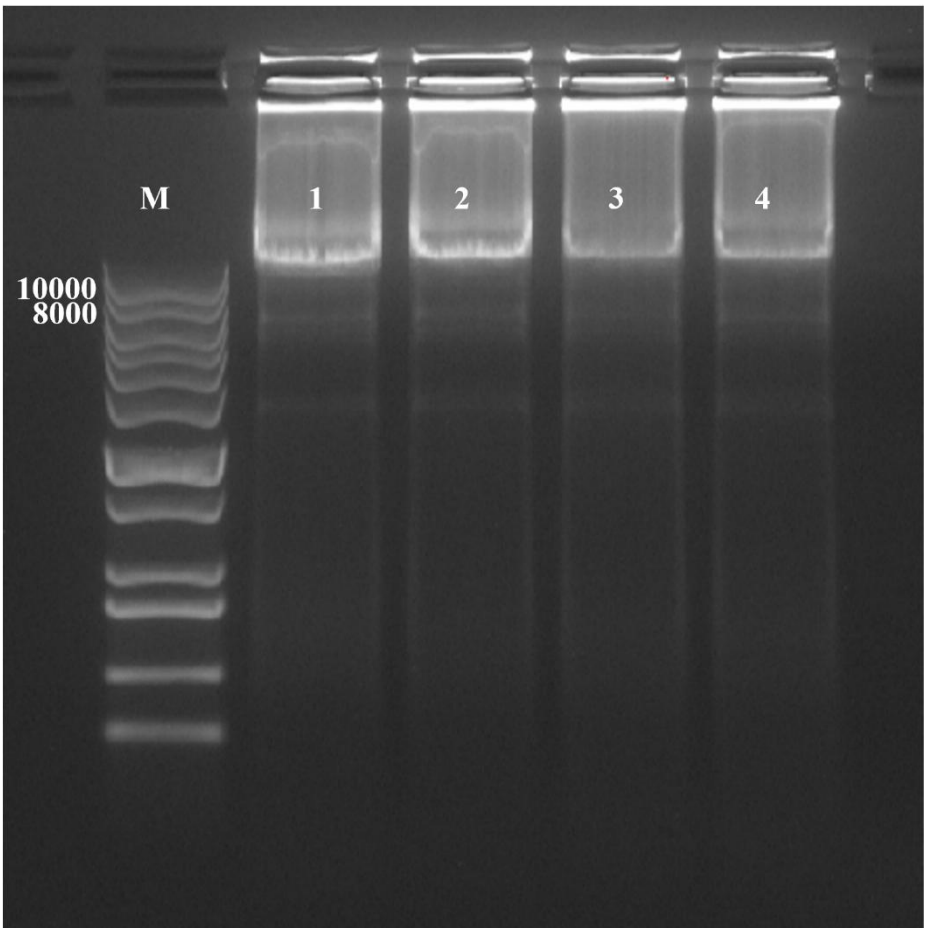


Figure 3. Agarose gel electrophoresis of the second PCR attempt for linearized backbone vector.Lane M: DNA marker. Lane 1–4: PCR products for 8844 bp linearized backbone.

No bright specific band corresponding to the expected backbone size was observed.

August 15



Extracted plasmid Sample	Concentration	Purity (A260/A280)
linearized backbone vector	17.9 ng/µL	1.72

Figure 4. Nanodrop quantification of gel-purified linearized backbone vector. Recovered 8844 bp backbone fragment was measured. Concentration =17.9 ng/µL, A260/A280 =1.72. Both concentration and purity were poor and inadequate for one-step cloning.

August 17

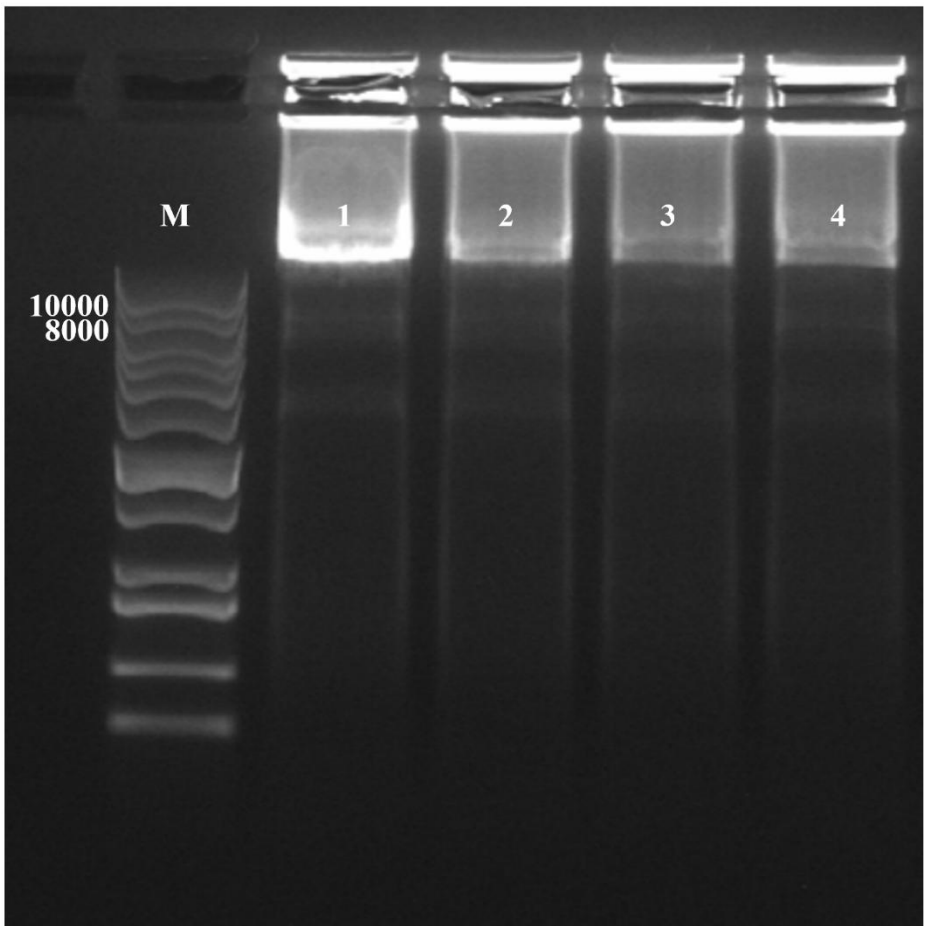


Figure 5. Agarose gel electrophoresis of the third PCR trial for linearized backbone vector using newly synthesized primers.Lane M: DNA marker. Lanes 1–4: PCR products for 8844 bp linearized backbone.

No obvious target band corresponding to the expected backbone size was detected after primer replacement.

August 18

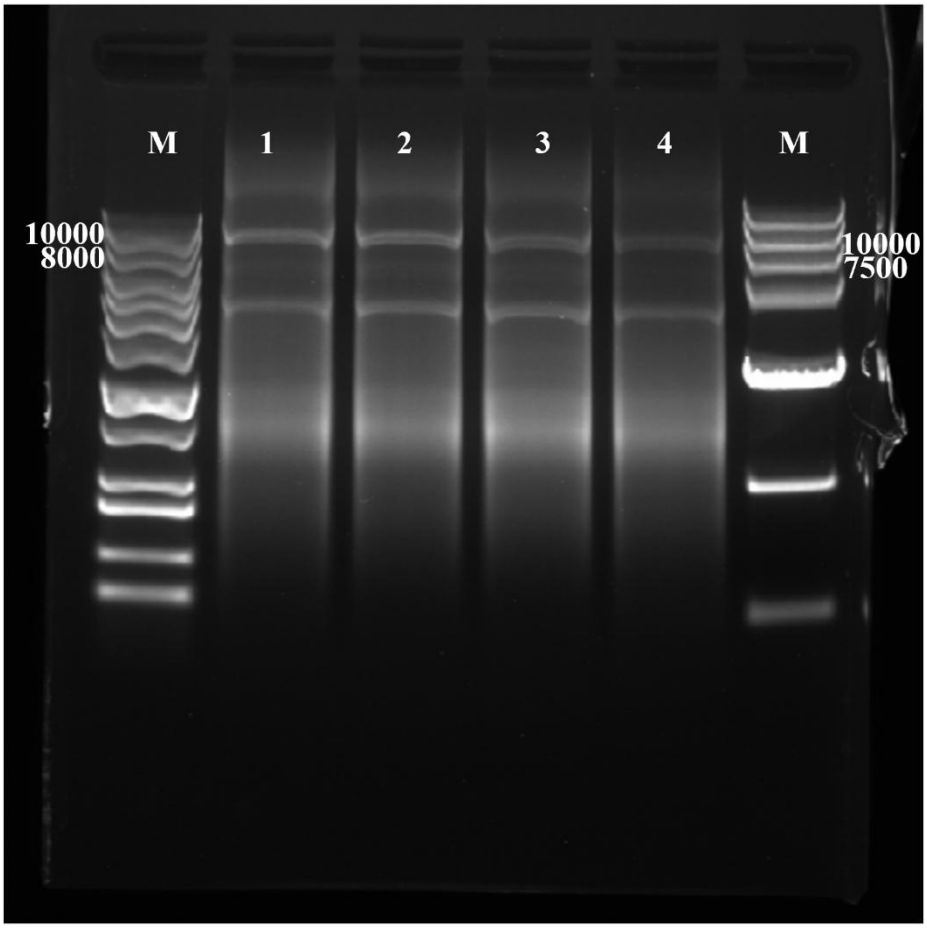


Figure 6. Agarose gel electrophoresis of the fourth PCR trial for linearized backbone vector.Dual markers were used. Left lane M: 1 kb plus DNA marker; right lane M: D15000 DNA marker. Lane 1–4: PCR products for 8844 bp linearized backbone.

Faint bands near the expected molecular-weight position could be observed, yet no distinct bright target band was obtained.

August 18



Extracted plasmid Sample	Concentration	Purity (A260/A280)
linearized backbone vector	14.1 ng/µL	1.81

Figure7.Nanodrop quantification of gel-extracted linearized backbone fragment from the previous PCR. Concentration = 14.1 ng/µL, A260/A280 = 1.81. Although the purity was acceptable, the DNA concentration remained low and unsatisfactory for subsequent cloning.

August 19

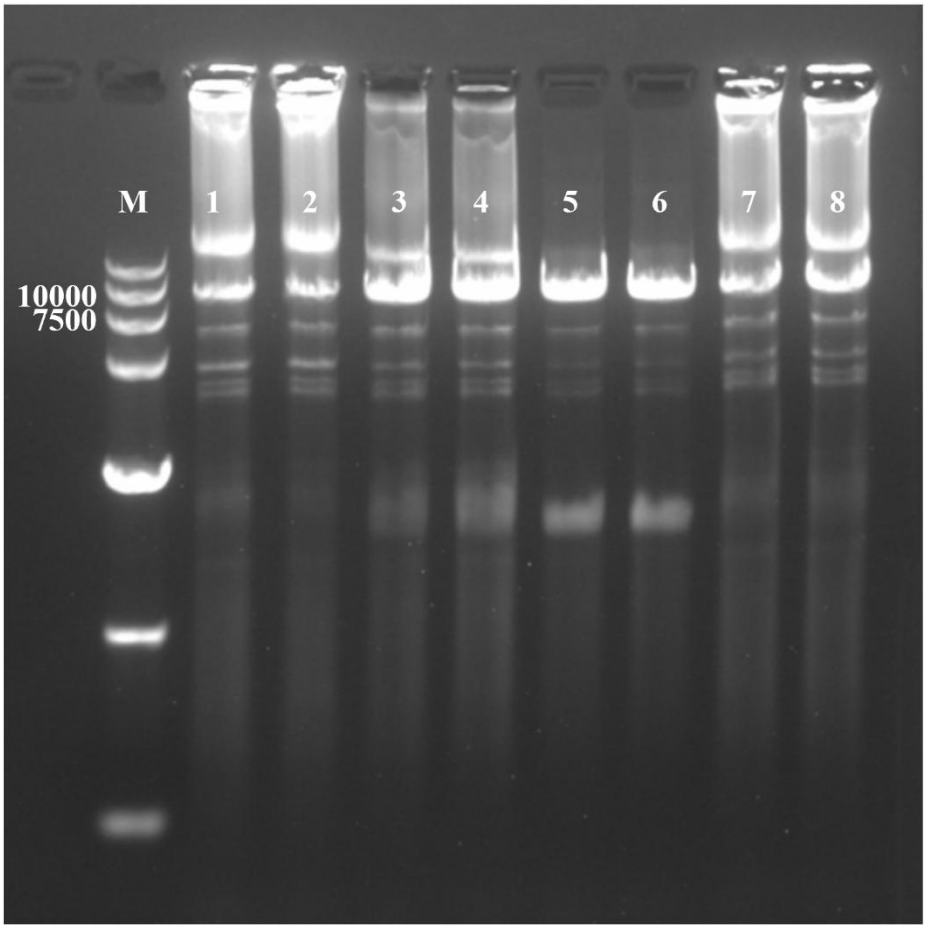


Figure 8. Agarose gel electrophoresis for optimization of backbone linearization PCR using template dilution gradient.Lane M: D15000 DNA marker. Lane 1, 2: undiluted template; Lane 3,4: 10-fold diluted template; Lane5,6: 100-fold diluted template; Lane7,8: original lyophilized plasmid (20 ng/μL).

The 8844 bp target band was clearest and brightest under 100-fold template dilution.

August 19

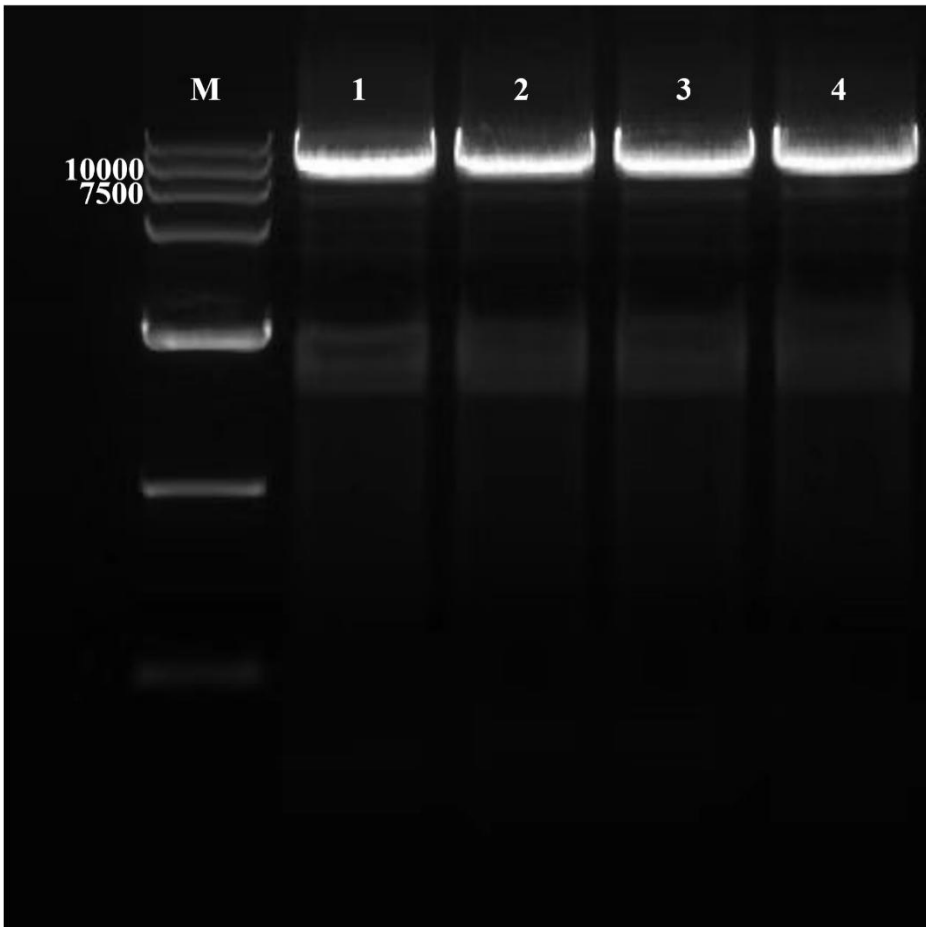


Figure 9. Agarose gel electrophoresis of backbone linearization PCR with 100-fold diluted template.Lane M: D15000 DNA marker. Lane 1–4: parallel PCR replicates. Bright target bands at the expected 8844 bp position were successfully generated.

August 19

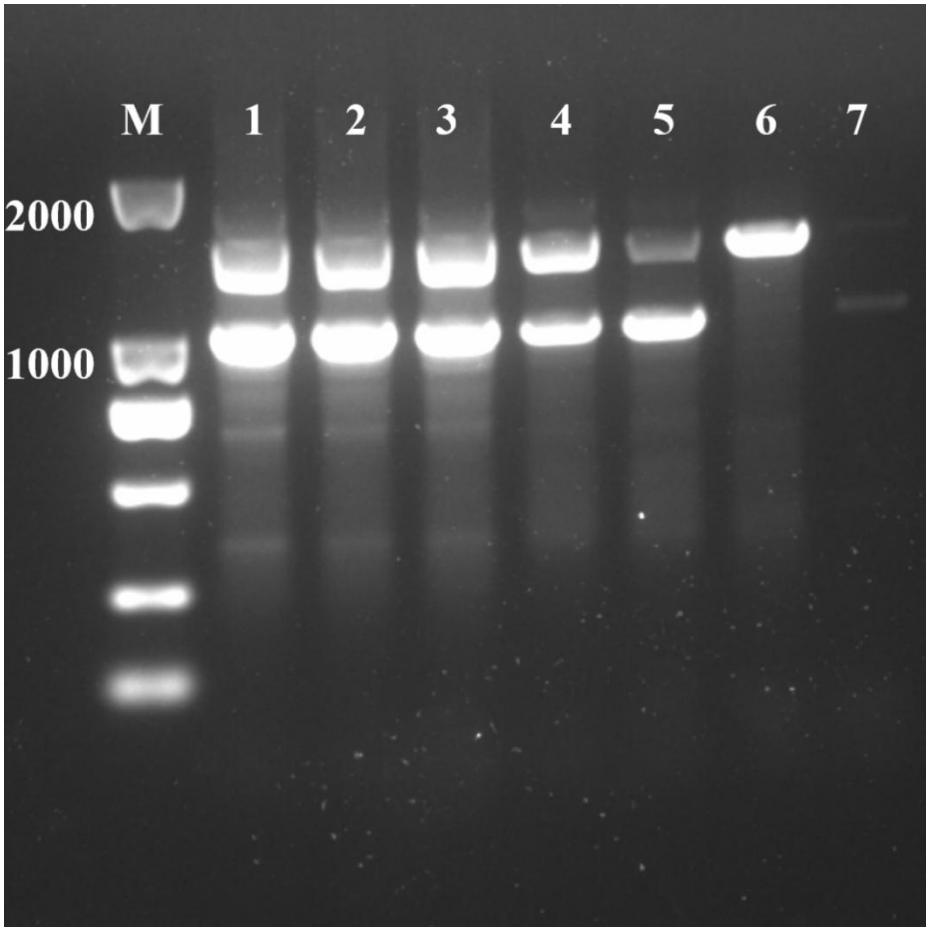


Extracted plasmid Sample	Concentration	Purity (A260/A280)
linearized backbone vector	106.3 ng/µL	1.89

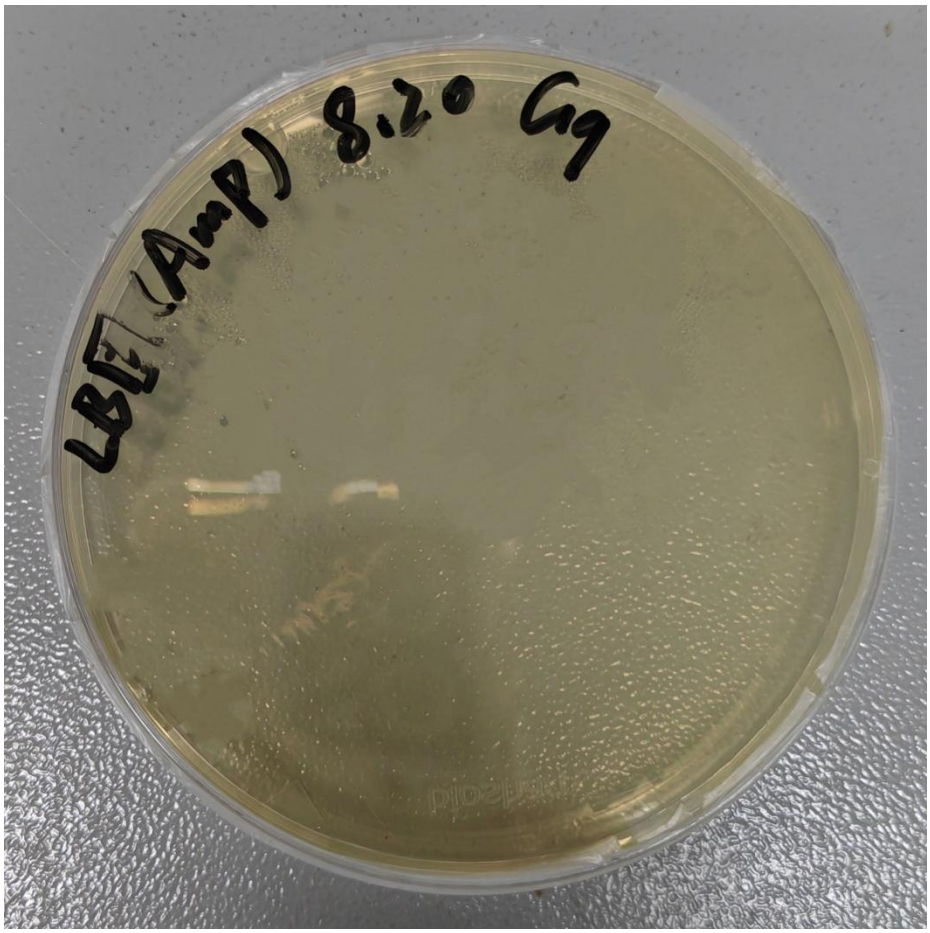
Figure 10. Nanodrop measurement of gel-purified linearized backbone fragment. The recovered 8844 bp backbone sample reached a concentration of 106.3 ng/µL with A260/A280 = 1.89.

Both concentration and purity were high and adequate for the following one-step cloning.

August 20

	 Figure 11. Agarose gel electrophoresis for PCR verification of one-step cloning product.Lane M: DL2000 DNA marker. Lanes 1–5: one-step cloning assembly products; Lane 6: backbone-only negative control; Lane 7: insert-only negative control. The expected band for successful assembly is 1051 bp. Extra non-specific bands were present in lanes 1–5. These false-positive bands matched the amplicons from both negative controls, demonstrating residual unassembled backbone and insert fragments in the cloning mixture.
Explanation	The 762 bp insert was successfully amplified and purified. After multiple failed long-fragment PCR attempts for the 8844 bp backbone, template-dilution optimization yielded high-quality linearized backbone suitable for assembly. PCR verification of the cloning product showed the expected 1051 bp band, together with extra false-positive bands. These contaminating bands matched amplicons from backbone-only and insert-only negative controls, which originated from residual unassembled DNA in the assembly reaction. The assembled one-step cloning product is intended for subsequent transformation into E. coli and yeast.
Next plan	Directly subject the crude one- step cloning reaction product to transformation without further purification.

No 8 Date：August 21 to 26 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Transformation of crude one-step cloning assembly product into Top10 E. coli competent cells for recombinant plasmid screening
Note	Protocol 1、Protocol 3、Protocol 5、Protocol 6
Result	August 21  Figure 1. LB-Amp plate after transformation of crude one-step cloning product. No colonies were obtained on this selective plate, showing that this transformation attempt failed. August 23

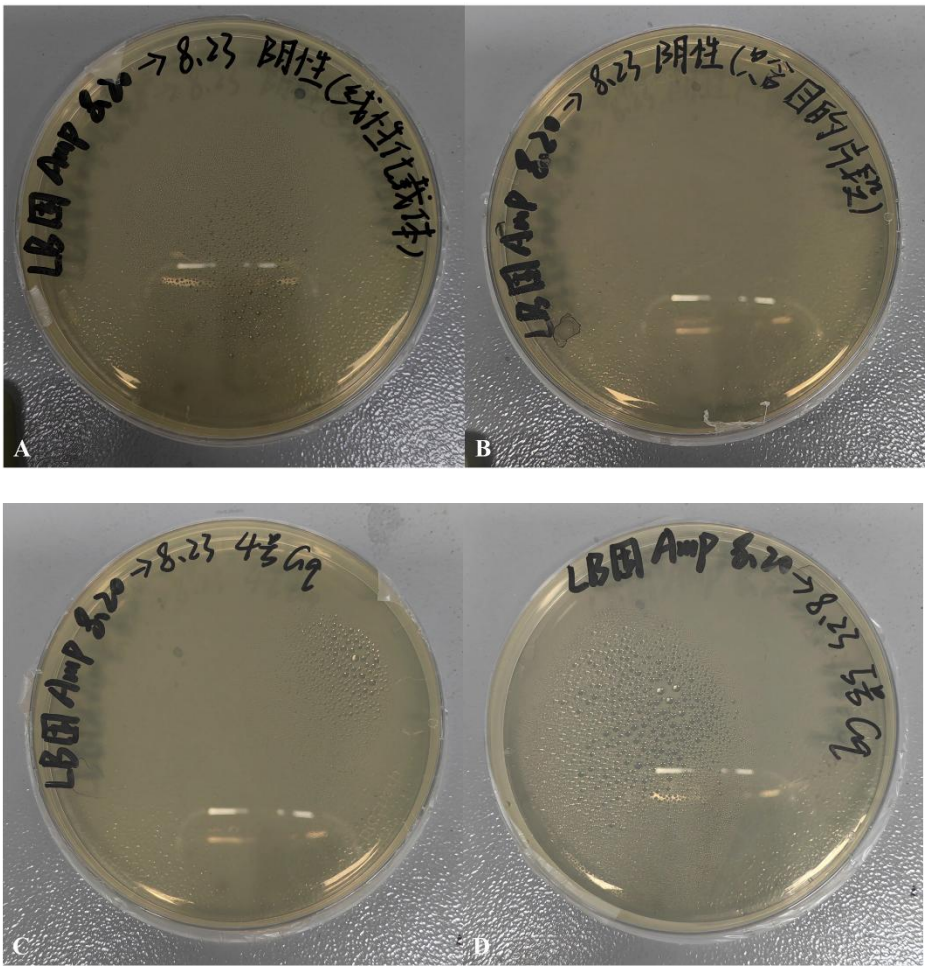


Figure 2. LB-Amp selective plates after transformation.A: Negative control, linearized backbone only; B: Negative control, insert fragment only; C and D: experimental replicates transformed with crude one-step cloning product. No colonies were observed on all plates, showing this transformation attempt failed.

August 24

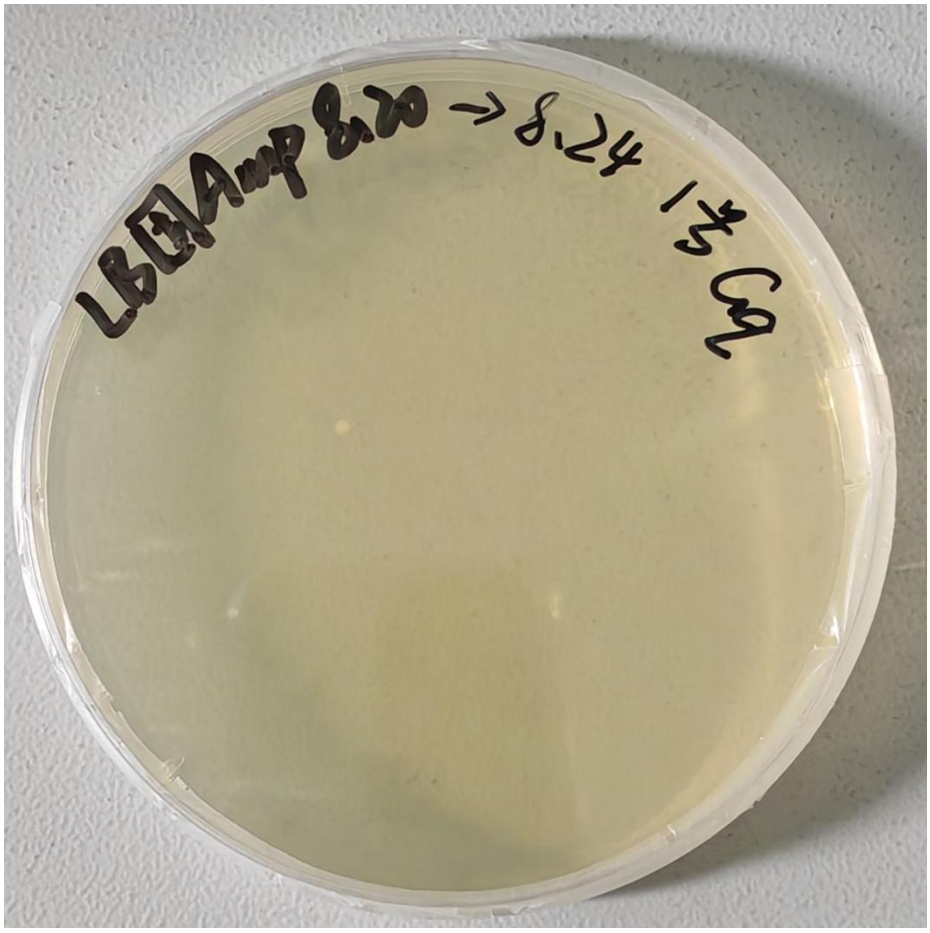


Figure 3. LB-Amp selective plate from repeated transformation. One single colony was observed on the plate, indicating successful transformation of the one-step cloning product.

August 25

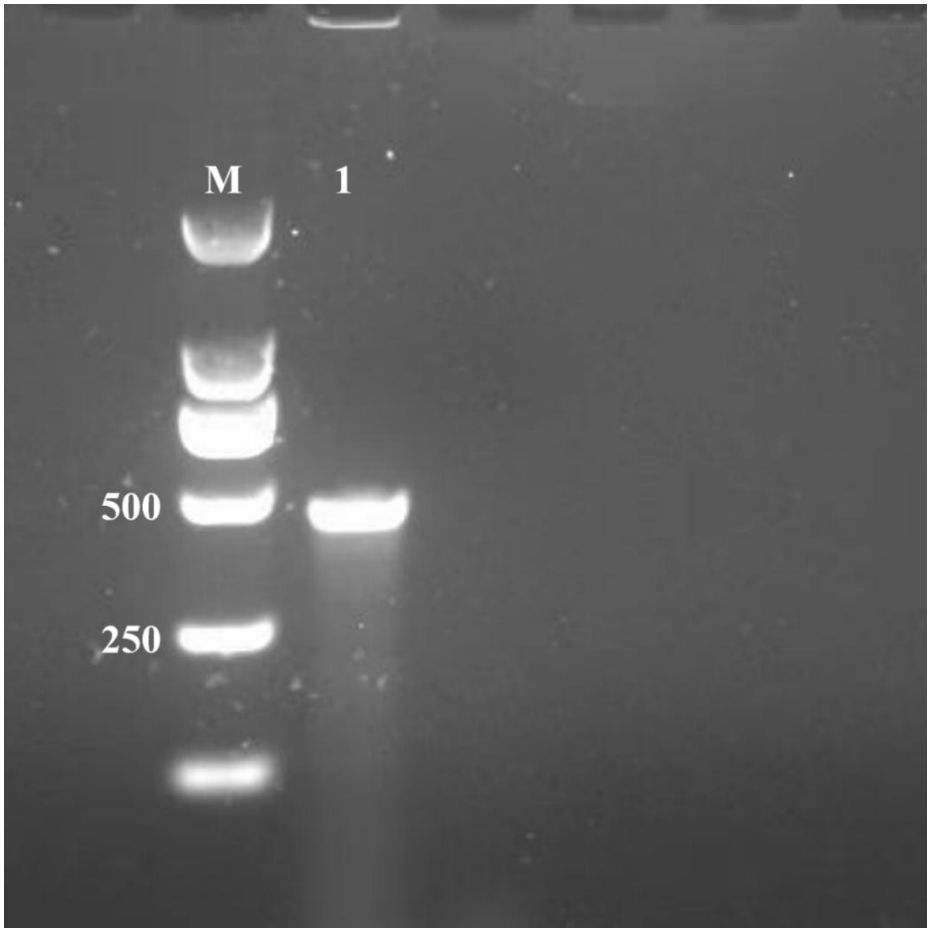
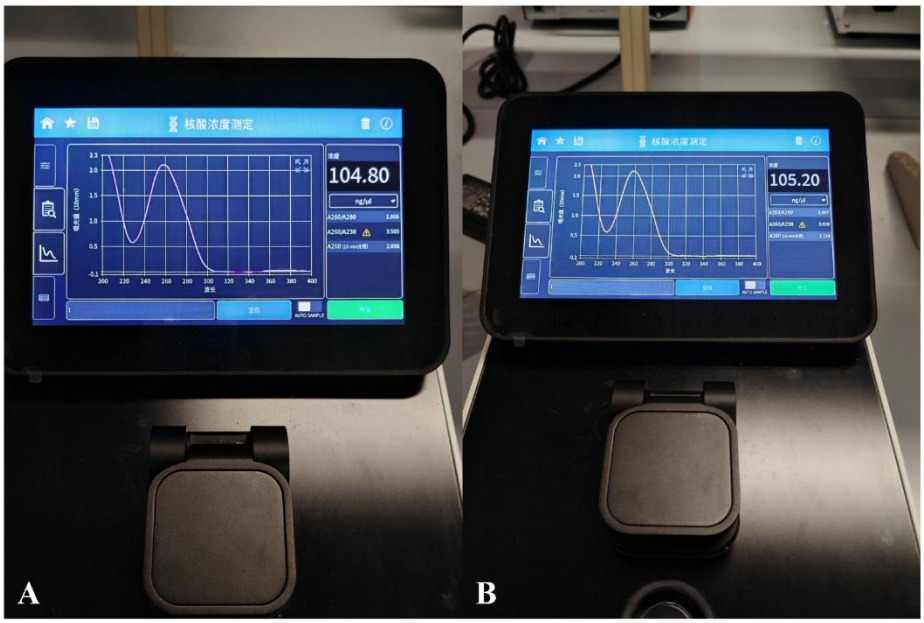


Figure 4. Agarose gel electrophoresis of E. coli colony PCR.M: DNA marker; Lane 1: PCR product from the picked single colony.

The target band at 472 bp was observed, confirming positive recombinant clone.

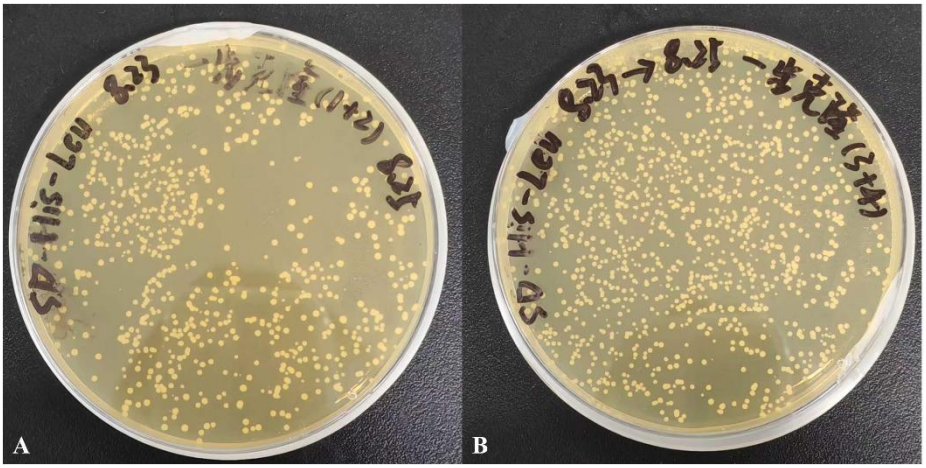
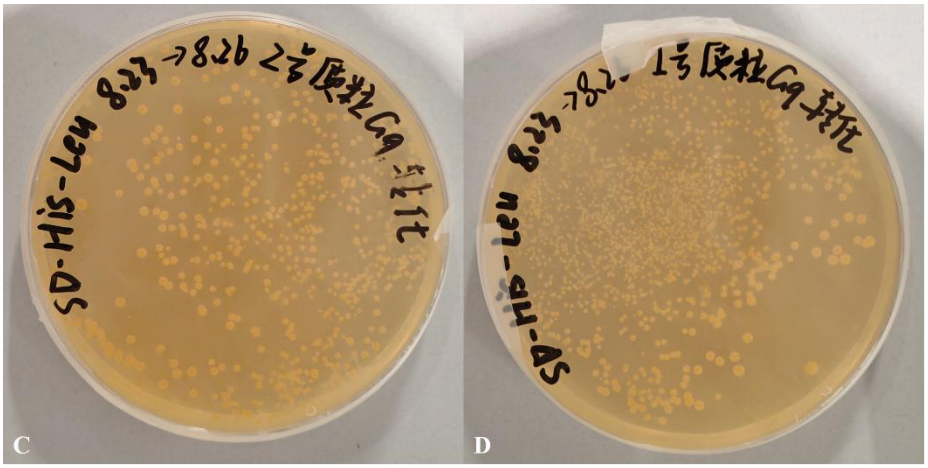
August 26



Extracted plasmid Sample	Concentration	Purity (A260/A280)

	A	104.80 ng/μL	1.866
	B	105.20 ng/μL	1.867
	Figure 5. Nanodrop quantification of extracted recombinant plasmid.A: Sample A: 104.80 ng/μL, A260/A280 = 1.866; B: Sample B:105.20 ng/μL, A260/A280 =1.867. The extracted plasmid exhibited sufficient concentration and acceptable purity.		
Explanation	The first transformation attempt including two negative controls and two experimental replicates yielded no colonies, which indicated insufficient correctly circularized recombinant plasmid in the assembly product. After redoing the transformation, one single colony grew on the selective plate. Colony PCR produced a band consistent with the expected 472 bp fragment, confirming that this colony contains the target recombinant plasmid and the transformation was successful. Plasmid extraction was performed on this positive clone. Nanodrop results showed high plasmid concentration and acceptable purity, which meets the requirements for subsequent sequencing and yeast transformation.		
Next plan	Use the extracted plasmid directly for yeast transformation, and submit the plasmid for sequencing during the yeast transformation process.		

No 9 Date：August 25 to September 10 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Transformation of recombinant plasmid into yeast competent cells and positive clone screening via colony PCR
Note	Protocol 12、Protocol 13
Result	August 25   Figure 1. SD-His-Leu selective plates after yeast transformation.A, B: transformed with one-step cloning product; C, D: transformed with recombinant plasmid extracted from positive E. coli. Abundant single colonies were obtained on all plates, indicating successful transformation. August 27

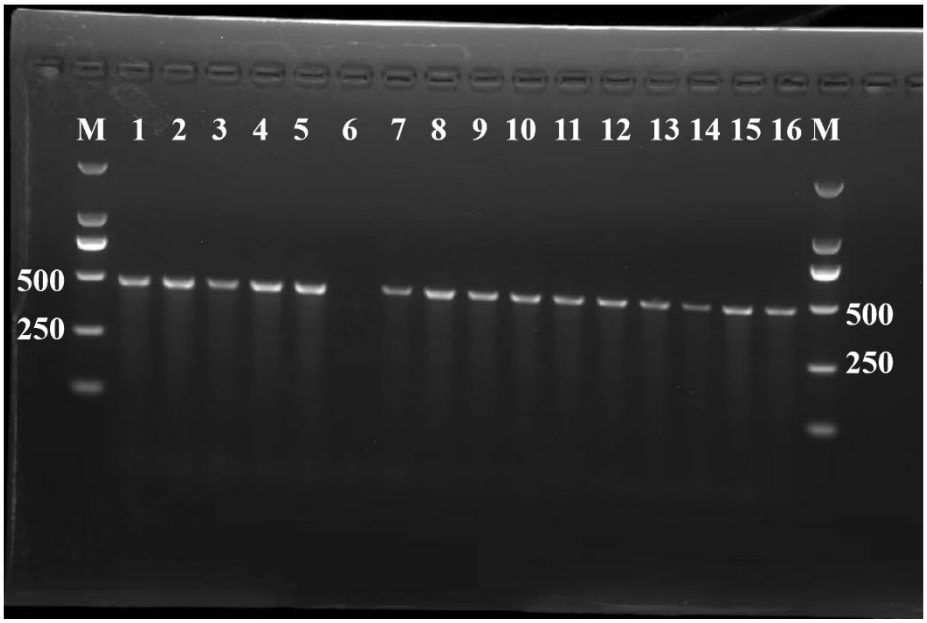
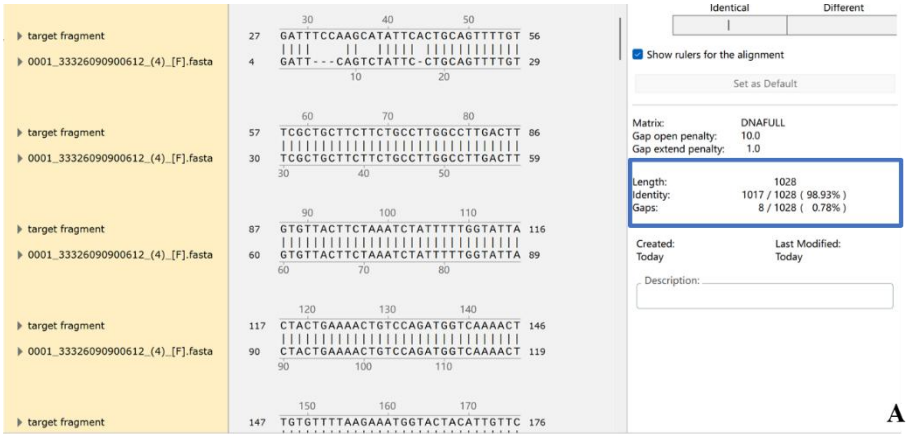
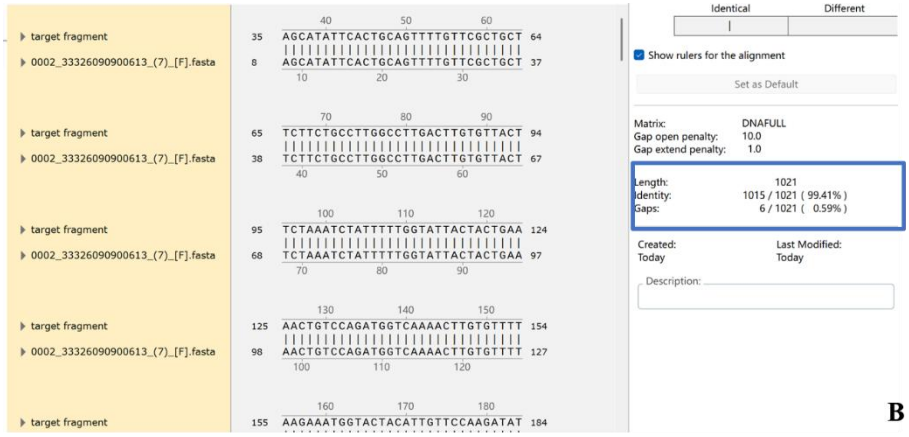
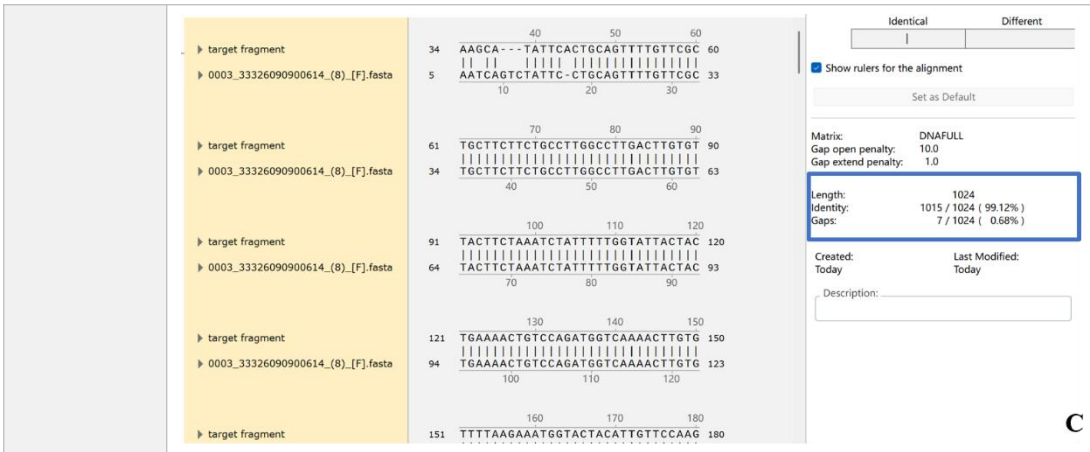


Figure 2. Agarose gel electrophoresis of yeast colony PCR.
M: DNA marker; Lane 1–16: PCR products from 16 picked yeast colonies. The target band of 472 bp was observed in all lanes except lane 6.

September 10






target fragment
0003_33326090900614_(8)_[F].fasta
Identical
Different
Show rulers for the alignment
Set as Default
Matrix: DNAFULL
Gap open penalty: 10.0
Gap extend penalty: 1.0
Length: 1024
Identity: 1015 / 1024 (99.12%)
Gaps: 7 / 1024 (0.68%)
Created: Today
Last Modified: Today
Description:
C


target fragment
0004_33326090900615_(9)_[F].fasta
Identical
Different
Show rulers for the alignment
Set as Default
Matrix: DNAFULL
Gap open penalty: 10.0
Gap extend penalty: 1.0
Length: 1030
Identity: 1018 / 1030 (98.83%)
Gaps: 11 / 1030 (1.07%)
Created: Today
Last Modified: Today
Description:
D


target fragment
0005_3332609090..._(11)_[F].fasta
Identical
Different
Show rulers for the alignment
Set as Default
Matrix: DNAFULL
Gap open penalty: 10.0
Gap extend penalty: 1.0
Length: 1034
Identity: 1019 / 1034 (98.55%)
Gaps: 13 / 1034 (1.26%)
Created: Today
Last Modified: Today
Description:
E


target fragment
0006_3332609090..._(12)_[F].fasta
Show rulers for the alignment
Set as Default
Matrix: DNAFULL
Gap open penalty: 10.0
Gap extend penalty: 1.0
Length: 1019
Identity: 1013 / 1019 (99.41%)
Gaps: 5 / 1019 (0.49%)
Created: Today
Last Modified: Today
Description:
F



Figure 3. Sequence alignment of 8 randomly selected yeast transformants.A–H: pairwise sequence alignment between each clone and the designed target fragment. I:Agarose gel electrophoresis of PCR products for sequencing. M: DNA Marker 2000; Lane 1–8: PCR products amplified from yeast transformants. The expected band size for successful transformation is 1051 bp, and the

	observed bands are at the correct position. Identity values of all samples were between 98.55% and 99.41%. Minor gaps are located at the 5' terminus of sequencing reads.t.
Explanation	Two types of samples were applied for yeast transformation: one-step cloning assembly product and recombinant plasmid extracted from positive E. coli. Plenty of colonies appeared on SD-His-Leu auxotrophic selective plates for both experimental groups, demonstrating that the constructs were successfully introduced into yeast cells and complemented the nutritional deficiency. 16 colonies were selected for colony PCR verification. 15 out of 16 colonies produced the 472 bp target fragment, while only colony 6 failed to amplify the target sequence. 8 positive yeast colonies were randomly selected for sequencing. Sequence alignment showed high identity between sequenced samples and the target fragment. The small gaps mainly existed at the 5' end, which was caused by sequencing read quality at the start of reads, and the core coding region was consistent with the design. Sequencing results confirmed that the target gene was successfully integrated into yeast.
Next plan	Perform WB assay to further verify protein expression

No 10 Date：August 30 to September 18 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Total protein extraction from verified positive yeast strains, SDS-PAGE separation and Western blot detection.
Note	Protocol 18
Result	August 30 to 31

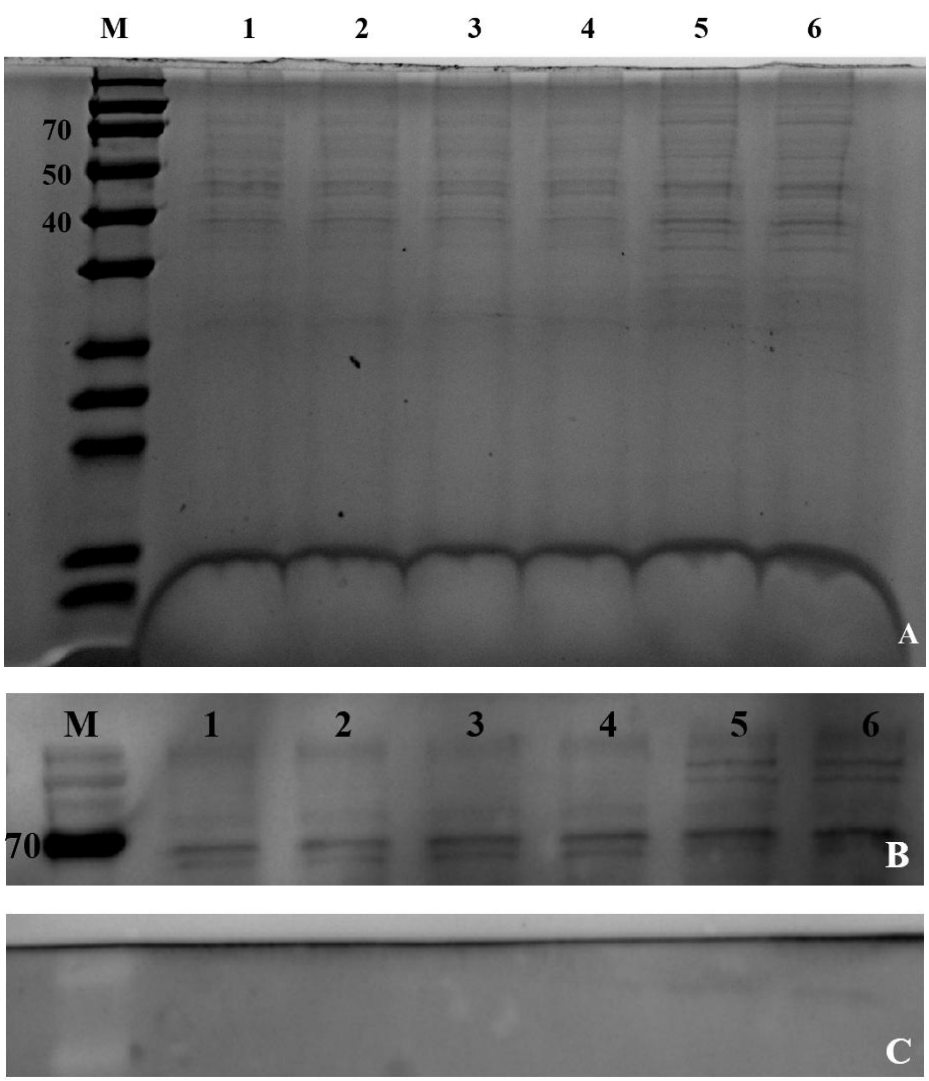
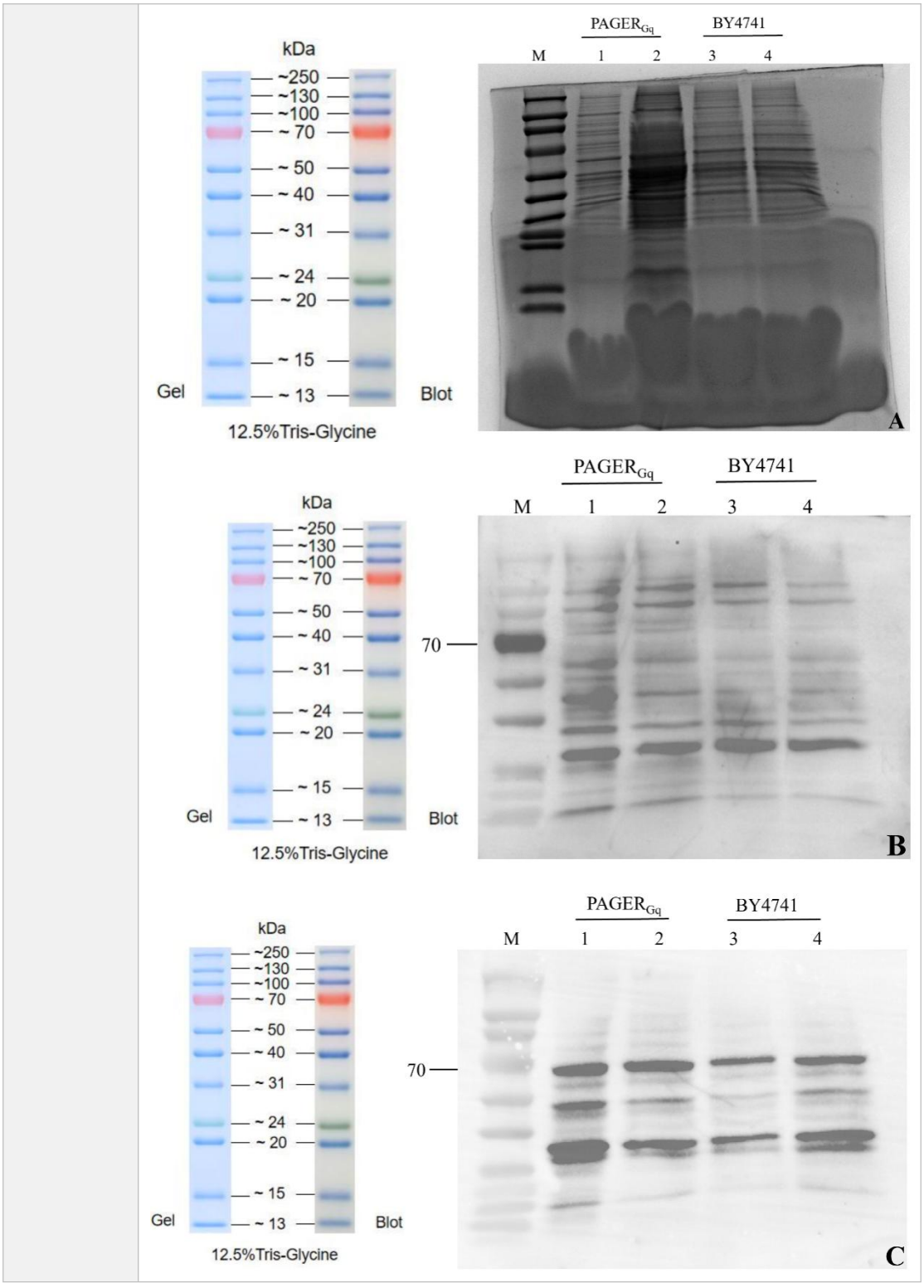


Figure 1.SDS-PAGE and Western blot results of yeast total protein. (A) Coomassie-stained SDS-PAGE gel. Lane 1–4: engineered yeast strains; Lane 5–6: wild-type BY4741. (B) Western blot probed with muscarinic Acetylcholine Receptor 1 primary antibody. (C) GAPDH loading control western blot.No specific band different from negative control was found around the expected molecular weight of 70 kDa in lanes 1–4.

September 17 to 19


kDa
~250
~130
~100
~ 70
~ 50
~ 40
~ 31
~ 24
~ 20
~ 15
~ 13
Gel
Blot
12.5%Tris-Glycine
PAGER$_{Gq}$
BY4741
M
1
2
3
4
A
70
B
C

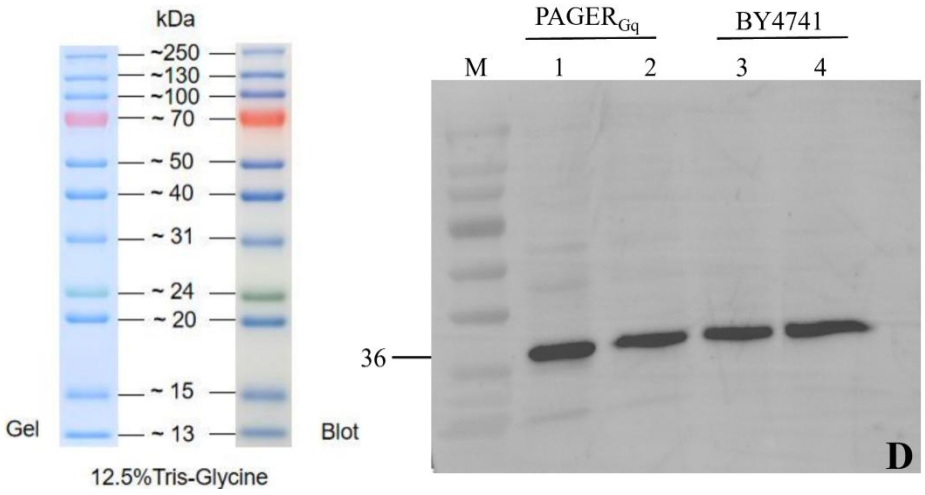


Figure 2. SDS-PAGE and WB results of yeast sensor recognition protein(A) Coomassie-stained SDS-PAGE gel of total yeast protein; (B) Western-blot probed with muscarinic Acetylcholine Receptor 1 Rabbit Polyclonal Antibody; (C) Western-blot probed with TEV Cleavage Site polyclonal Antibody; (D) GAPDH Western-blot loading control. M: protein molecular weight marker; lanes 1-2: engineered sensor-protein-expressing yeast; lanes 3-4: wild-type BY4741.

Total cellular protein was extracted from engineered yeast expressing the sensor recognition protein and wild-type BY4741 control strain, and subjected to SDS-PAGE and Western-blot detection. On Coomassie-stained SDS-PAGE gel, no distinct differential band near 70 kDa could be observed between engineered yeast and BY4741. Two independent primary antibodies, muscarinic Acetylcholine Receptor 1 Rabbit Polyclonal Antibody and TEV Cleavage Site polyclonal Antibody, were used for subsequent Western-blot detection; however, neither blot yielded specific target bands around 70 kDa that differentiated the engineered strain from wild-type BY4741. The GAPDH loading-control Western-blot showed clear and consistent bands across all sample lanes, confirming equal protein loading for each sample. Taken together, although sample loading quality was reliable, we failed to detect the target sensor recognition protein at the expected molecular weight in our test, suggesting possible low-level or undetectable protein expression under current experimental conditions.

Explanation	Total cellular protein was extracted from engineered yeast strains expressing the sensor recognition protein and wild-type BY4741 control for SDS-PAGE and Western blot verification. No differential protein bands near 70 kDa were observed between the experimental group and the control on Coomassie-stained gels. Two different primary antibodies were applied for target protein detection, yet no specific target bands at approximately 70 kDa were obtained in either trial. The GAPDH internal control displayed clear and consistent bands in all lanes, verifying equal protein loading and valid experimental operation. The results indicated that the target sensor recognition protein was undetectable under current conditions, possibly due to extremely low protein expression levels.

Next plan	Carry out fluorescence functional verification to further characterize the sensor performance of engineered yeast.

No 11 Date：September 2 to 13 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Fluorescence detection after incubating engineered yeast with HA or TEV protease.
Note	Protocol 16
Result	September 2

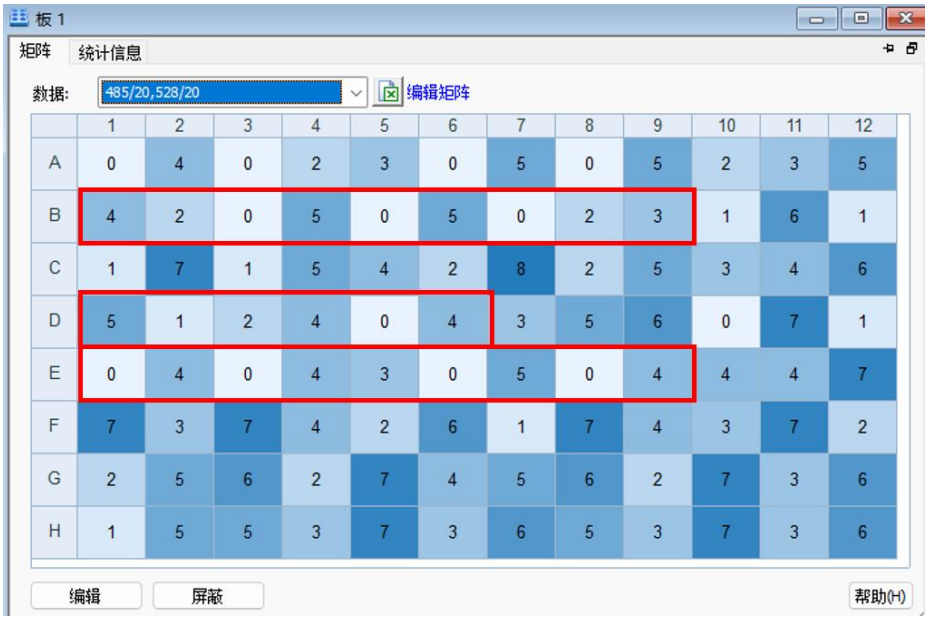


Figure 1. Fluorescence intensity heatmap of the plate reader assay. Row B: 3rd-round engineered yeast treated with TEV and 25 nM / 50 nM / 100 nM DCZ. Row D: 3rd-round engineered yeast control groups, including DCZ-only group and the TEV-only group. Row E: Wild-type BY4741 groups: wells 1–3: BY4741 + TEV + 100 nM DCZ; wells 4–6: BY4741 + 50 nM DCZ only; wells 7–9: BY4741 + TEV only.

Yeast pellets were collected by centrifugation and resuspended in sorbitol buffer before detection. No obvious difference in fluorescence signal was observed between experimental groups and negative controls.

September 5

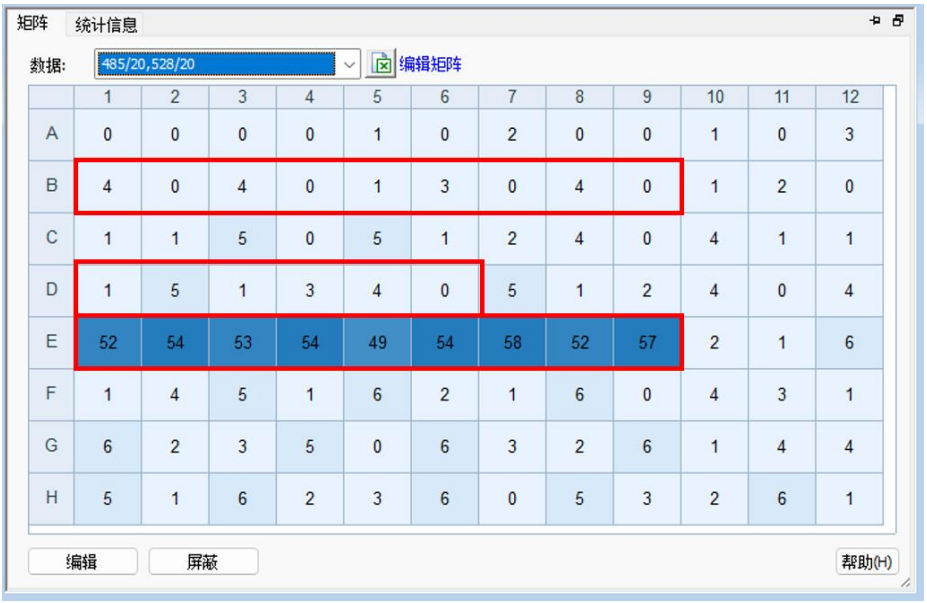
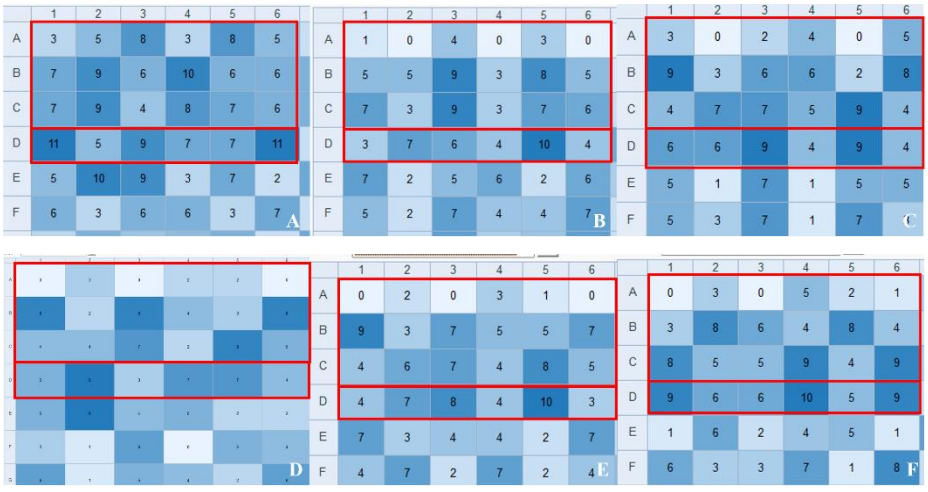
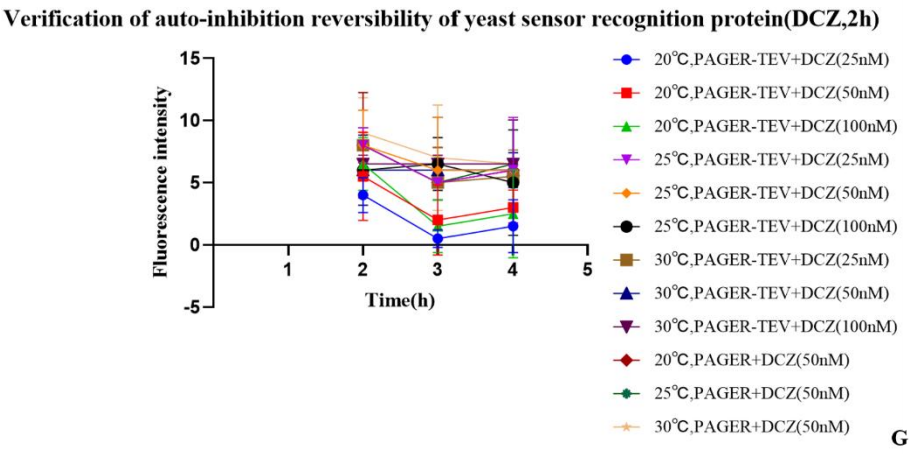


Figure 2. Fluorescence intensity heatmap when cells were resuspended in YPD medium. Row B: 3rd-round engineered yeast treated with TEV and 25 nM / 50 nM / 100 nM DCZ. Row D: 3rd-round engineered yeast control groups, including DCZ-only group and TEV-only group. Row E: Wild-type BY4741 groups: wells 1–3: BY4741 + TEV + 100 nM DCZ; wells 4–6: BY4741 + 50 nM DCZ only; wells 7–9: BY4741 + TEV only.

High fluorescence values were observed in Row E due to the dark color of the YPD medium. No obvious fluorescence difference was detected among subgroups of BY4741. The 3rd-round engineered yeast did not produce obvious target fluorescence signal.

September 7 to 8





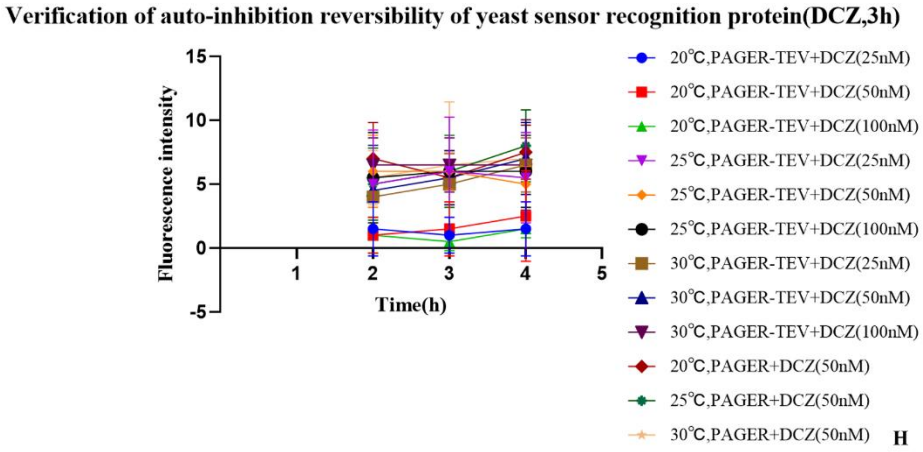


Figure 3. Fluorescence intensity heatmap of the plate reader assay. A-C: 2 h, 3 h, 4 h lyticase treatment at 20 °C, 25 °C, 30 °C, followed by 2 h DCZ reaction. Row A: 20 °C; Row B:25 °C; Row C:30 °C. Wells 1–2: 25 nM DCZ + TEV; wells 3–4: 50 nM DCZ + TEV; wells 5–6:100 nM DCZ + TEV. Row D: control groups without TEV, treated with 25/50/100 nM DCZ at corresponding temperatures. D-F: 2 h, 3 h, 4 h lyticase treatment followed by 3 h DCZ reaction. Figure G and H: Time-course fluorescence intensity curves of 2 h and 3 h DCZ reaction respectively, showing fluorescence signals under all temperature and concentration combinations. No obvious fluorescence signal was obtained.

September 11 to 13

	1	2	3	4	5	6	7	8	9
A	2	4	0	4	1	2	5	0	5
B	7	7	11	4	10	8	6	11	5
C	7	1	5	4	1	7	2	4	7
D	11	5	11	9	9	13	1	5	5
E	38	35	40	41	31	30	35	39	37
F	7	3	5	8	3	8	5	3	8
G	2	7	5	3	8	3	5	7	2
H	3	8	3	6	7	3	8	3	4

A

	1	2	3	4	5	6	7	8	9
A	2	5	0	6	0	4	4	0	7
B	12	5	9	10	4	12	4	8	10
C	0	5	5	0	8	1	5	5	0
D	8	12	4	12	12	6	8	2	4
E	34	38	31	43	31	23	39	30	36
F	1	8	5	2	8	1	6	6	1
G	8	4	3	8	2	6	5	2	7
H	7	2	8	3	5	7	1	7	4

B

	1	2	3	4	5	6	7	8	9
A	5	0	3	3	0	6	0	3	5
B	11	5	9	8	5	11	7	8	9
C	3	3	6	0	6	5	2	7	2
D	13	7	8	14	9	13	4	2	7
E	31	36	35	37	31	25	34	36	32
F	5	5	2	8	4	4	7	1	7
G	2	7	2	5	6	2	7	2	4
H	6	2	8	2	6	6	3	7	1

C

	1	2	3	4	5	6	7	8	9
A	0	5	1	3	3	1	6	0	5
B	12	7	9	8	6	11	5	10	9
C	1	5	3	3	6	1	6	2	4
D	14	6	11	12	11	13	1	7	3
E	32	36	34	39	31	23	36	32	35
F	5	3	8	2	6	5	2	7	1
G	6	3	7	3	5	6	2	7	3
H	6	5	2	7	3	6	6	2	7

D

	1	2	3	4	5	6	7	8	9
A	0	3	3	0	6	0	1	6	0
B	11	9	5	10	5	7	8	3	9
C	0	7	1	2	6	0	5	3	0
D	3	10	5	9	11	4	6	4	1
E	37	37	32	41	30	36	37	31	39
F	8	0	5	6	1	7	1	5	5
G	1	5	5	1	7	2	5	5	0
H	0	7	2	2	7	0	7	3	2

E

	1	2	3	4	5	6	7	8	9
A	4	0	3	1	0	5	0	4	0
B	8	9	11	5	11	4	8	6	6
C	6	0	5	2	2	6	0	6	0
D	4	9	8	8	12	5	7	3	3
E	36	32	34	36	31	36	31	35	34
F	1	7	1	5	5	2	7	1	5
G	5	4	0	6	0	5	5	1	6
H	3	2	6	1	4	5	1	7	3

F

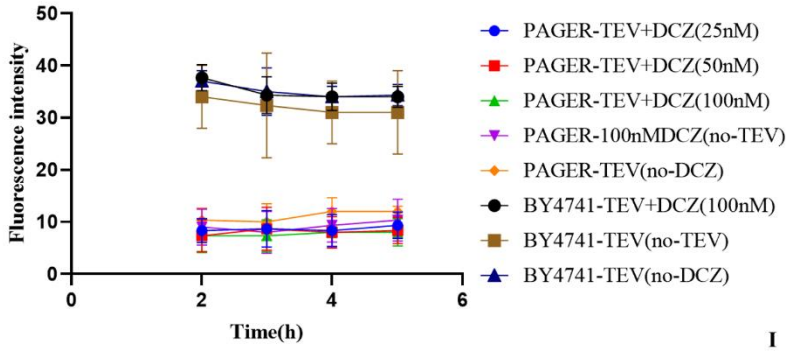
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B	11	7	12	5	9	6	6	8	4
C	3	0	5	0	4	3	1	5	1
D	5	8	8	8	13	7	5	4	1
E	36	32	33	36	30	35	32	32	37
F	3	2	6	2	3	5	1	6	4
G	5	1	5	4	2	6	1	4	5
H	6	1	6	3	3	6	1	5	4

G

	1	2	3	4	5	6	7	8	9
A	0	1	3	0	4	0	4	2	1
B	11	6	13	6	10	8	5	9	4
C	3	1	6	0	5	3	1	6	0
D	4	9	8	10	13	6	5	3	1
E	36	32	32	36	29	36	31	33	35
F	6	3	2	7	0	5	4	2	7
G	0	7	2	4	6	0	6	3	3
H	2	2	6	1	6	3	2	6	0

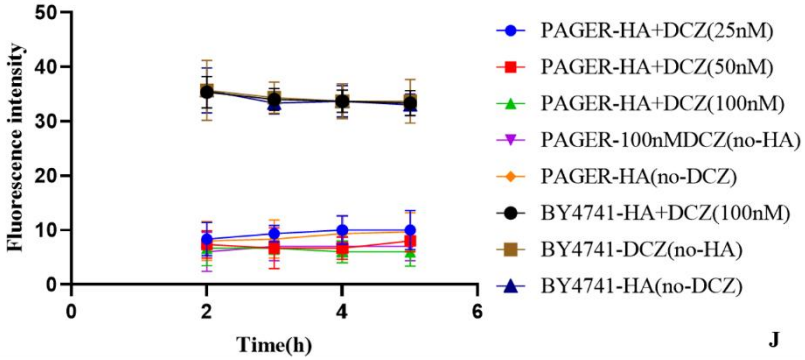
H

Verification of auto-inhibition reversibility of yeast sensor recognition protein(TEV)



I

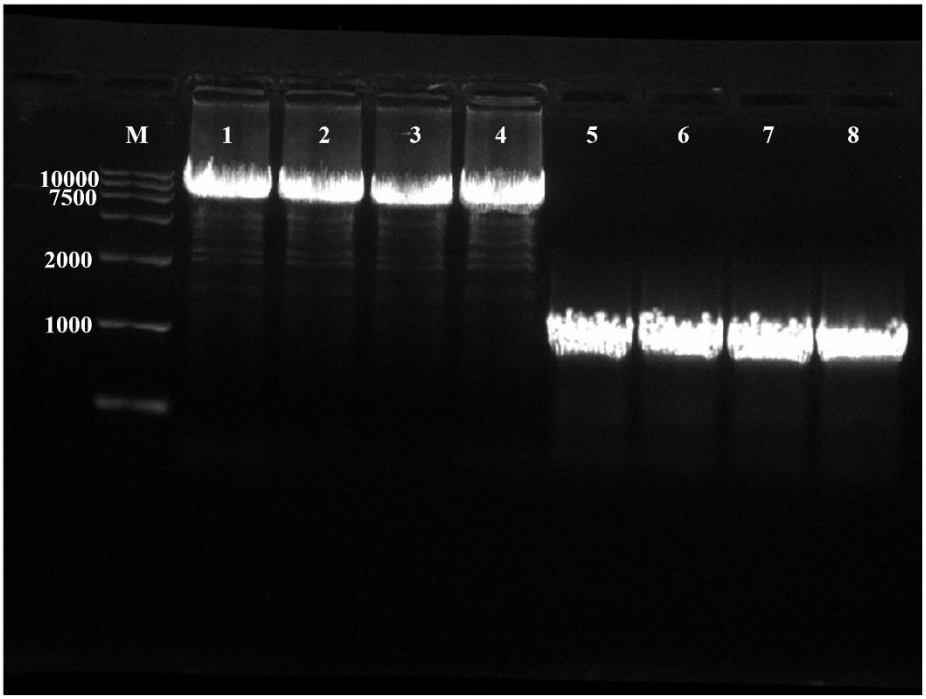
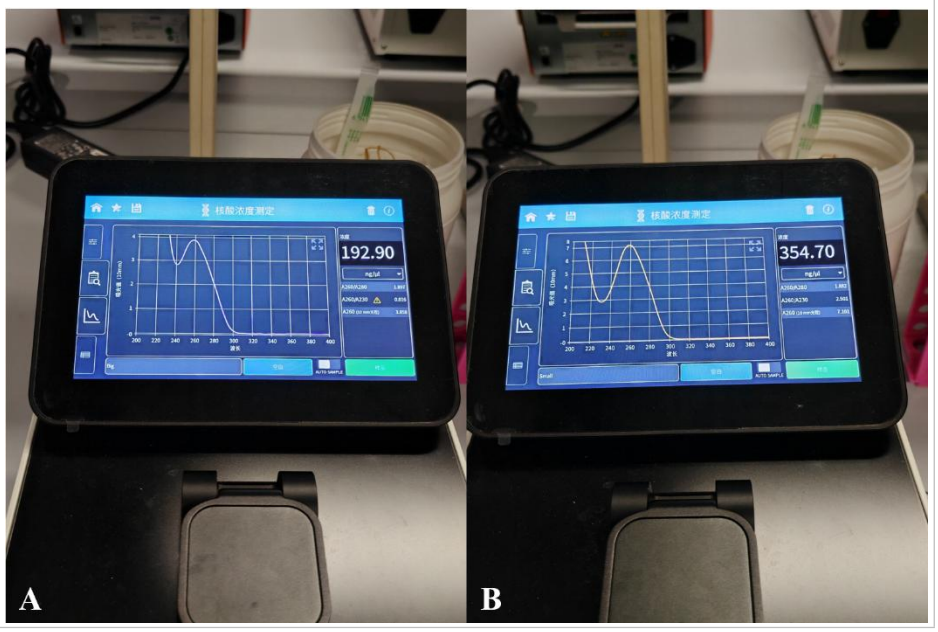
Verification of auto-inhibition reversibility of yeast sensor recognition protein(HA)



J

	Figure 4. Fluorescence intensity heatmap of the plate reader assay. Yeast samples were pre-treated with lyticase at low temperature and low speed (20 °C, 100 rpm, overnight for 16 h) before the addition of TEV protease / HA protein and DCZ. A–D: TEV protease treated groups followed by DCZ incubation for 2 h, 3 h, 4 h, 5 h respectively. E–H: HA protein treated groups followed by DCZ incubation for 2 h, 3 h, 4 h, 5 h respectively. I and J: Quantitative plots of corresponding fluorescence data. High background signal was observed due to the dark color of YPD medium. No obvious fluorescence difference was detected within BY4741 groups, and no significant fluorescence signal was observed in experimental groups compared with negative controls.
Explanation	Consistent fluorescence assay results from multiple plate-reader tests showed no distinguishable fluorescence signal differences between experimental groups and negative control groups under all optimized conditions, including varied lyticase treatment temperature, duration, DCZ concentration, reaction time, and overnight low-speed permeabilization. These negative results indicate that the engineered sensor protein failed to exert normal functional response in yeast cells. The absence of specific fluorescence activation suggests defects in protein expression, folding, or structural switching, which may originate from unreasonable design of the current gene sequence.
Next plan	Further optimize and modify the target gene sequence to correct potential structural or expression defects, and repeat the fluorescence functional verification to validate sensor performance.

No 12 Date：September 20 to 22 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Reconstruction of sensor gene fragments. The signal peptide and ICL3 sequence of hM1D were deleted.
Note	Protocol 5、Protocol 7、Protocol 11
Result	September 20  Figure 1. Agarose gel electrophoresis of linearized vector backbone and target gene fragment. M: DNA marker. Lane 1–4: linearized vector backbone (expected size: 7815 bp). Lane 5–8: linearized target fragment (expected size: 1370 bp). Both fragments were observed at the expected positions. September 20 

Extracted plasmid sample	Concentration	Purity (A260/A280)
linearized backbone vector	192.0 ng/μL	1.897
insert fragment	354.70 ng/μL	1.882

Figure 2. Nucleic acid spectrophotometer was used to detect the concentration of recovered DNA samples. Figure 2A shows the detection result of gel-extracted linearized vector backbone with concentration of 192.90 ng/μL. Figure 2B shows the detection result of gel-extracted target insert fragment with concentration of 354.70 ng/μL.

Both fragments have adequate concentration and good purity.

September 20

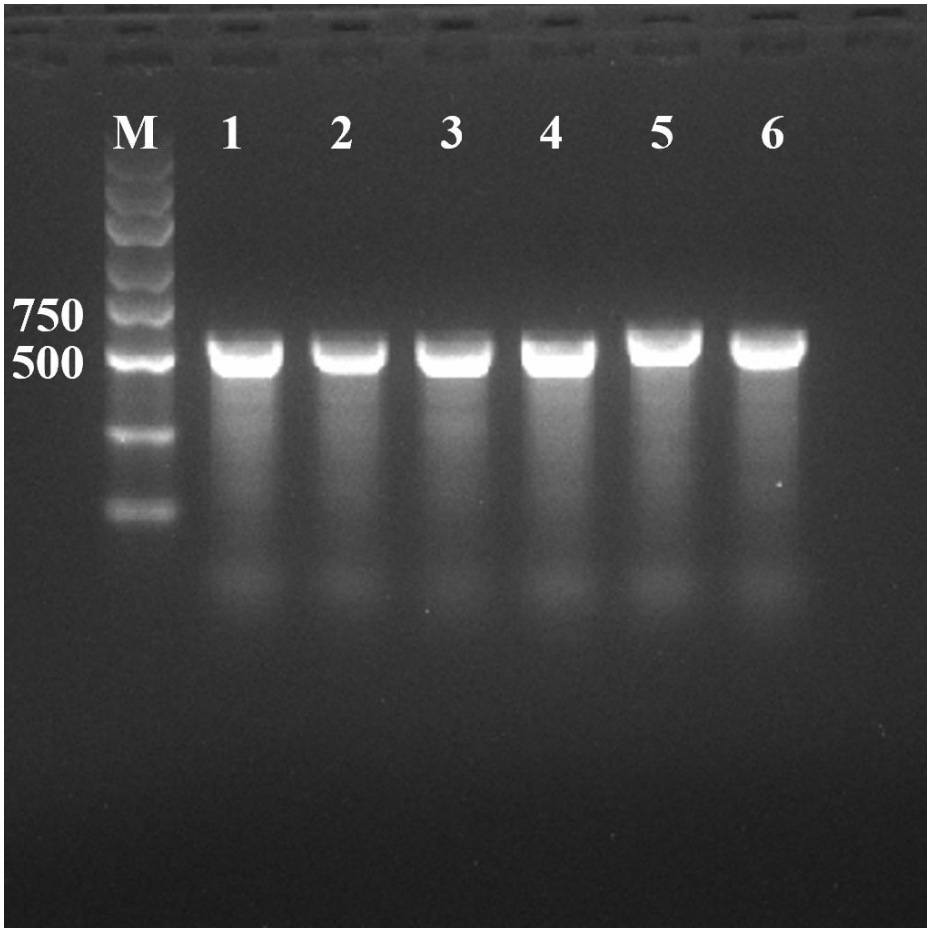
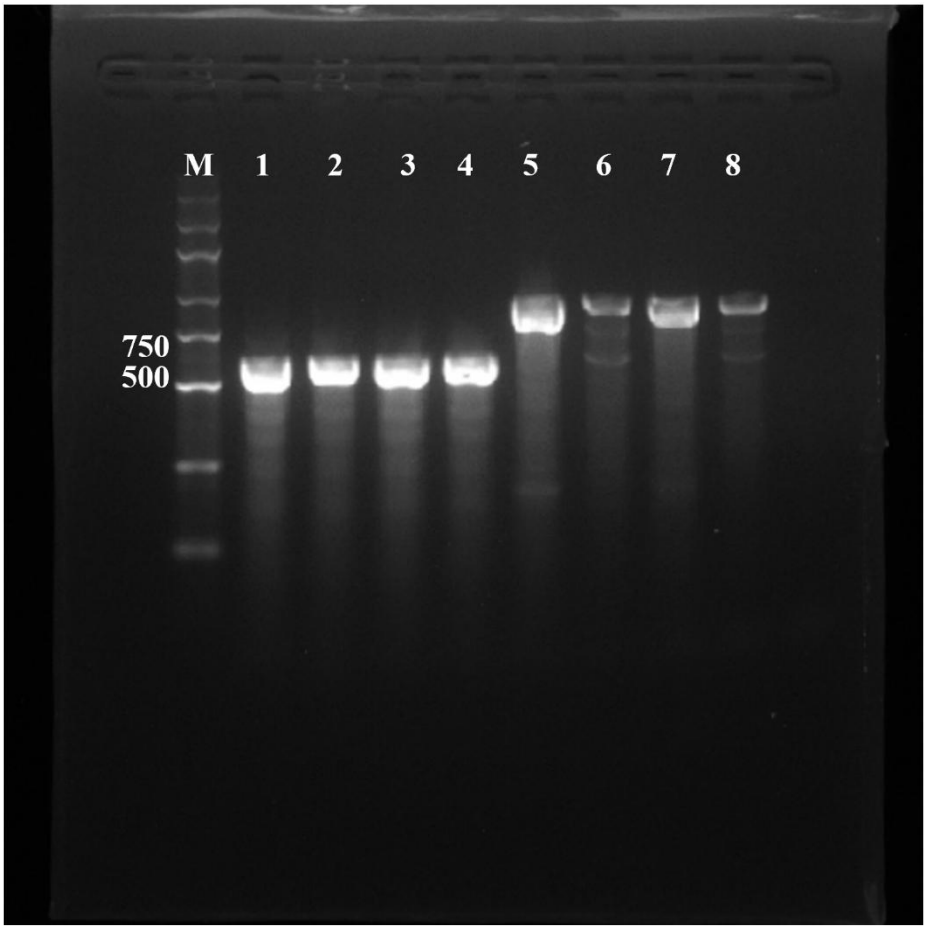


Figure 3. Agarose gel electrophoresis was performed for PCR verification of one-step cloning products. M: DNA marker. Lane 1–4: PCR products of one-step cloning samples, expected band size 556 bp. Lane 5: vector-only negative control. Lane 6: insert-only negative control.

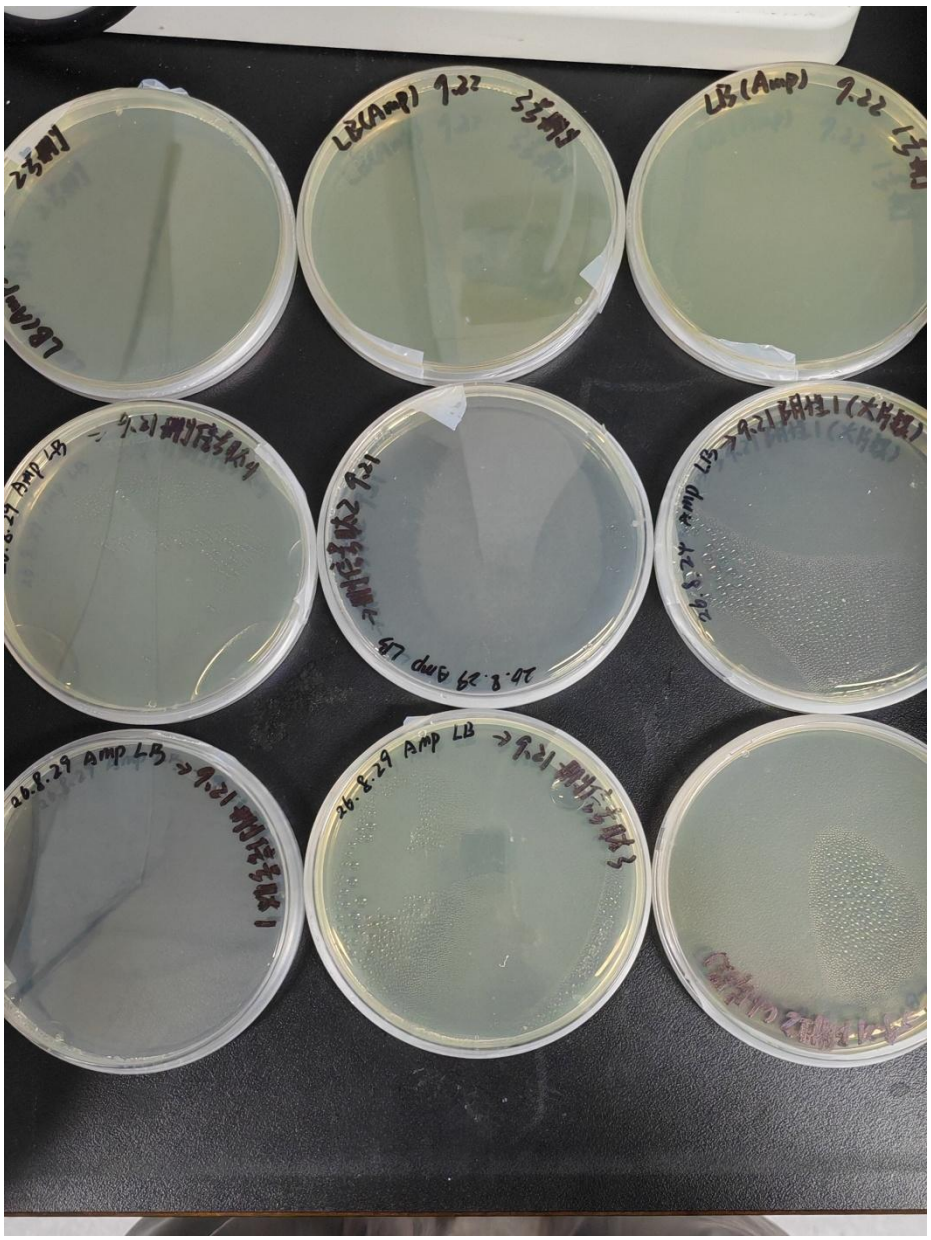
Bands of similar size were observed in both negative control groups, which was caused by inappropriate primer design.

September 21

	 Figure 4.Agarose gel electrophoresis was performed for PCR verification using newly designed primers, with an expected target size of 601 bp. M: DNA marker. Lane 1–4: one-step cloning products. Lane 5: gel-purified vector fragment negative control. Lane 6: gel-purified insert fragment negative control. Lane 7: vector-only one-step cloning negative control. Lane 8: insert-only one-step cloning negative control. Lanes 1–4 produced bands at the expected 601 bp position. Bands were amplified in negative control groups, but their sizes differed from the target fragment.
Explanation	To resolve the previous failure of functional fluorescence detection caused by abnormal protein expression, we modified the target gene by removing the signal peptide and ICL3 sequence of hM1D. The linearized vector backbone and truncated target insert were obtained via PCR amplification, gel extraction and nucleic acid purification. After optimizing the primer design, the PCR verification showed that the one-step cloning samples yielded the target band of expected size. Although non-specific bands appeared in negative controls, their fragment sizes were inconsistent with the target, which verified the successful assembly of vector and insert by homologous recombination. The newly constructed recombinant plasmid provides the foundation for subsequent transformation and protein functional assay.

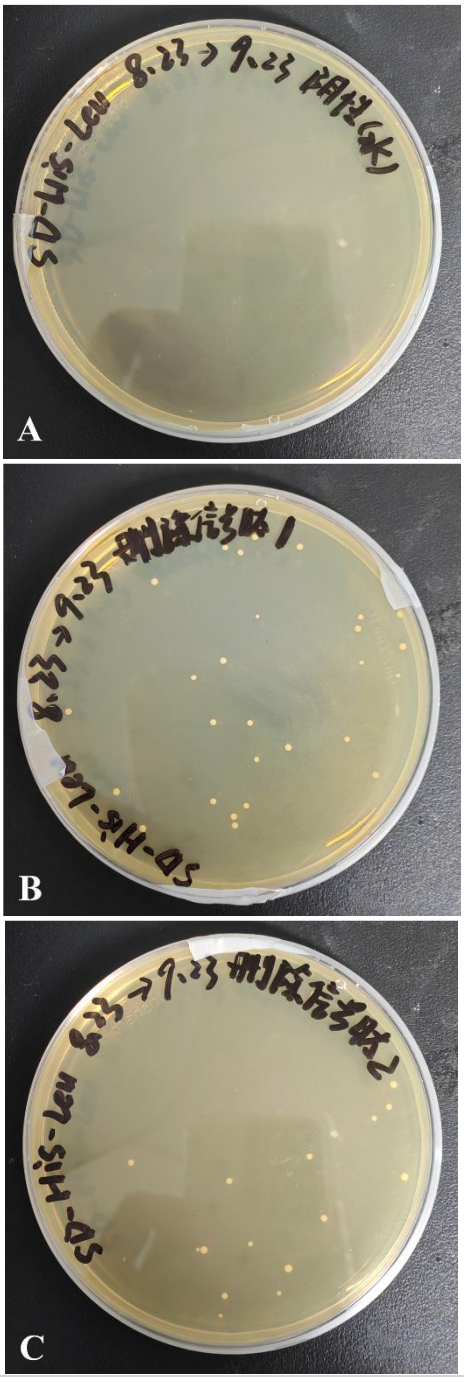
Next plan	Transform the one-step cloning product into E. coli competent cells, perform colony PCR screening to obtain positive single colonies, extract plasmids and carry out subsequent yeast transformation and fluorescence functional verification.

No 13 Date：September 22 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Transformation of *E. coli* competent cells using the one-step cloning product
Note	Protocol 1
Result	September 22  Figure 1. Figure 1: LB agar plates with ampicillin after transformation of crude one-step cloning product. No transformant colonies were observed on all plates.
Explanation	No colonies grew on the ampicillin selective medium. Possible reasons include low concentration or poor quality of assembled recombinant DNA, inhibitory

	components existing in the crude one-step cloning reaction system, low transformation efficiency of competent cells, or inappropriate transformation operation. The assembled recombinant plasmid was not successfully delivered into *E. coli* cells to confer ampicillin resistance.
Next plan	Skip *E. coli* amplification step. Directly use the one-step cloning product to perform yeast transformation.

No 14 Date：September 23 to 30 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Transformation of one-step cloning product into yeast competent cells and positive clone screening via colony PCR
Note	Protocol 12、Protocol 13
Result	September 23 

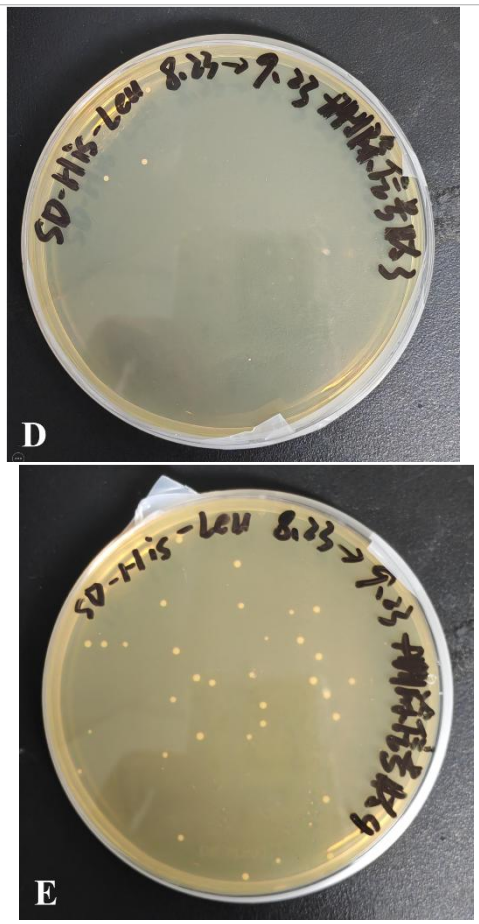


Figure 1. Yeast transformation and colony PCR verification of the yeast sensor recognition protein membrane-expression vector. A: Negative control plate; B-E: SD-His-Leu selective plates with yeast transformants.

The recognition protein sequence was cloned into the yeast membrane-expression vector and transformed into *Saccharomyces cerevisiae*, then plated on SD-His-Leu medium. No colonies grew on the negative control plate (A), confirming effective selection. Single colonies formed on plates B-E.

September 26

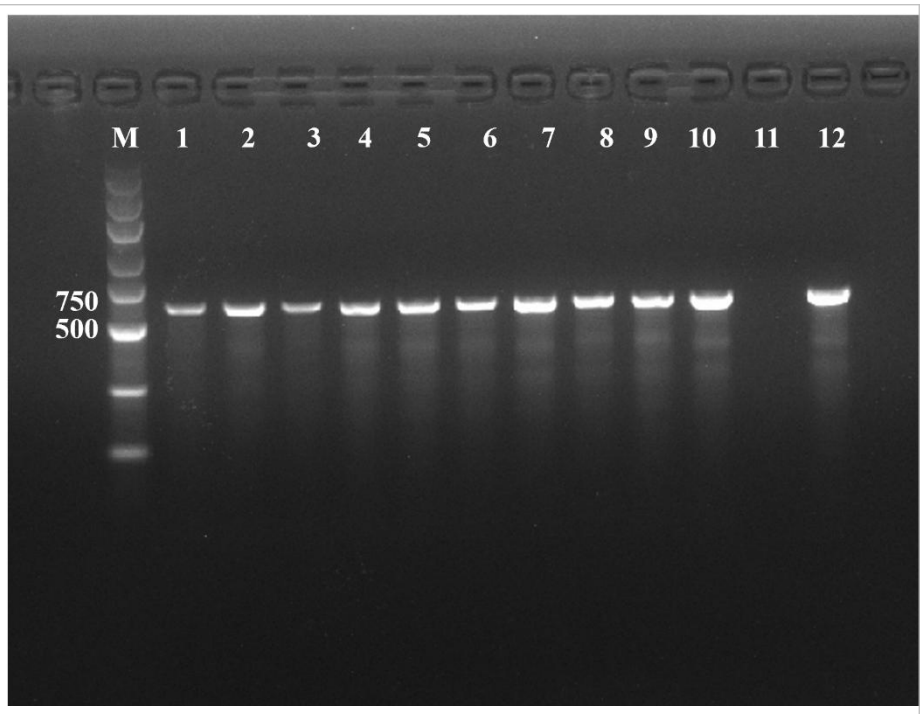
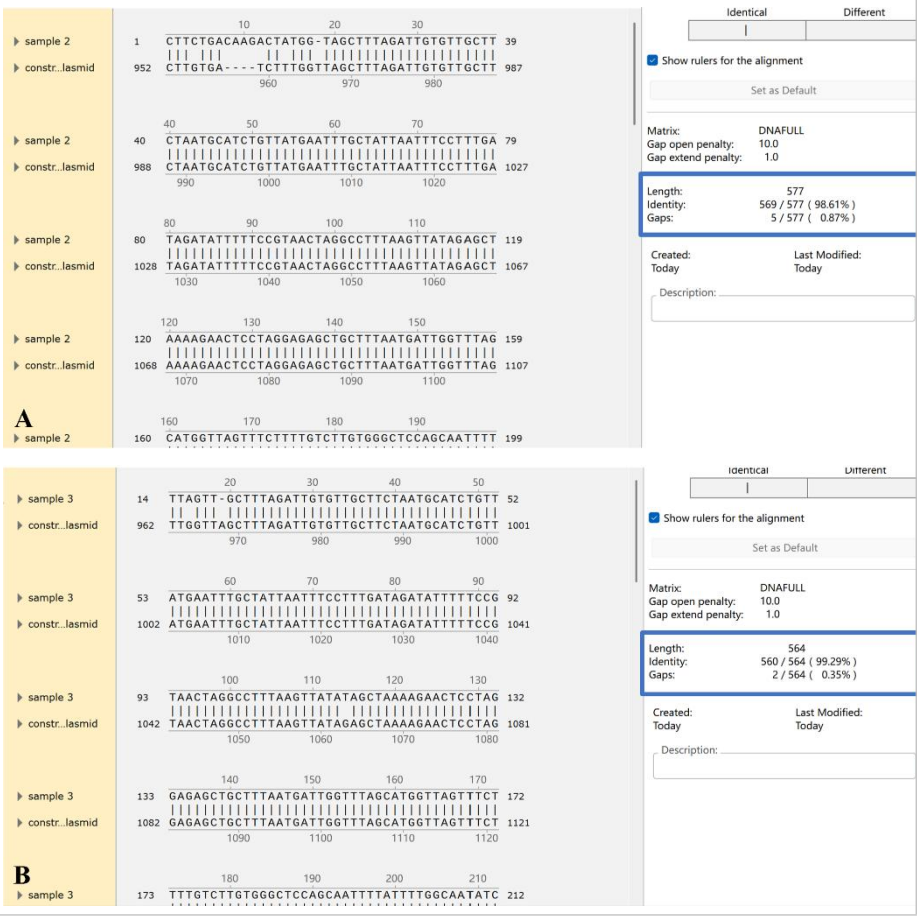
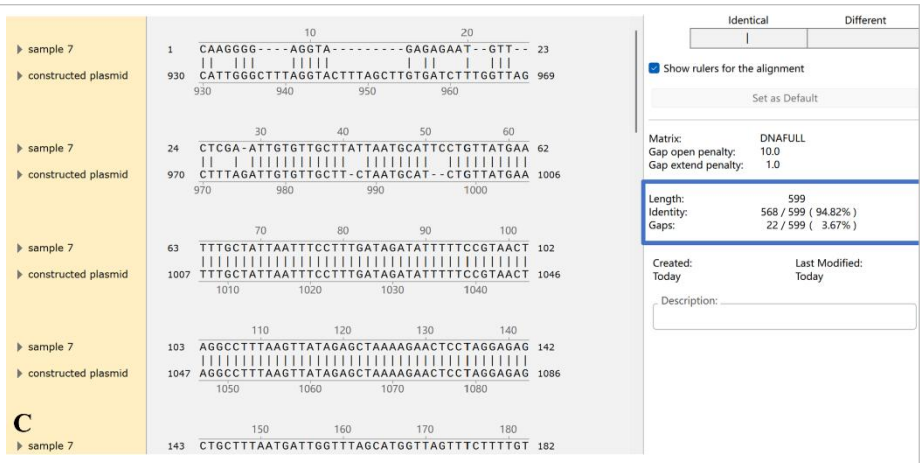


Figure 2. Colony PCR agarose gel. M: DNA marker; lanes 1–12: picked colonies (expected 601 bp) Colony PCR of 12 isolates showed clear 601 bp bands in lanes 1–10 and 12; lane 11 had no product, with minimal non-specific bands. Positive yeast transformants were obtained for sequencing.

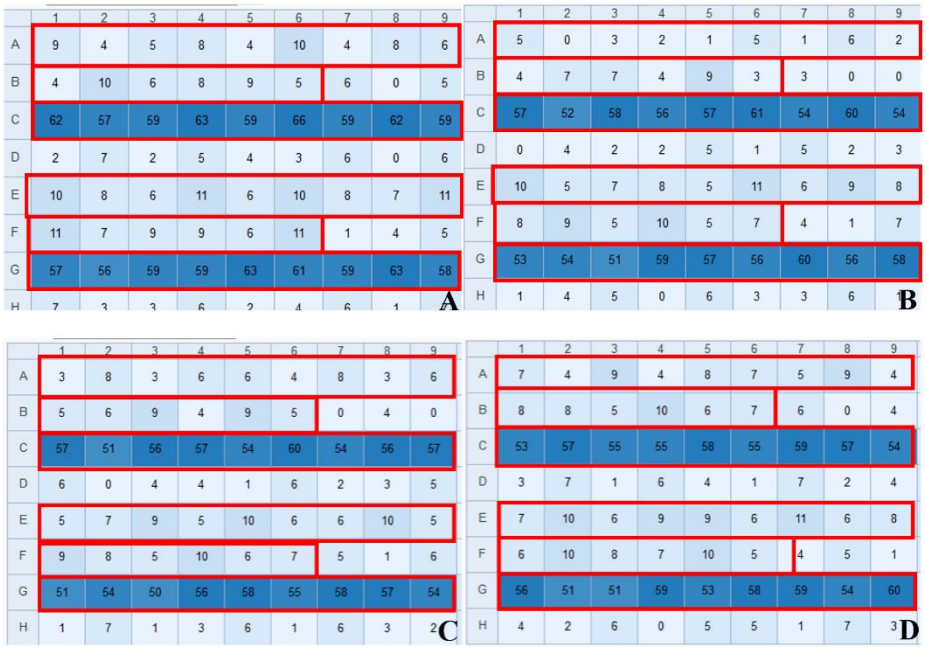
September 30



	 Figure 3. Sanger sequencing alignment verification of yeast sensor recognition protein membrane-expression vector. A: Sequence alignment of sample 2; B: Sequence alignment of sample 3; C: Sequence alignment of sample 7. The right panel shows alignment statistics for each sample, including total aligned length, sequence identity and gap proportion. Plasmids from PCR-positive yeast colonies were sequenced. Samples 2 and 3 matched the designed sequence with no damaging mutations and were selected for follow-up assays. Sample 7 had poor alignment and was discarded.
Explanation	The one-step cloning product was directly transformed into *Saccharomyces cerevisiae*. No colonies grew on the negative control plate, confirming valid SD-His-Leu selection pressure. Colony PCR results showed most picked colonies carried the target 601 bp fragment. Subsequent Sanger sequencing verified samples 2 and 3 had high sequence identity to the designed construct with no critical mutations in the recognition protein coding region, while sample 7 contained mismatches and was discarded. Positive clones with intact open reading frame were successfully obtained.
Next plan	Use verified yeast clones to perform protein expression detection and fluorescence functional verification of the membrane sensor protein.

No 15 Date：September 29 Experimenters: Aishi Zeng Recorder: Aishi Zeng

Experiment	Fluorescence detection after incubating engineered yeast with HA or TEV protease.
Note	Protocol 16
Result	September 29



	1	2	3	4	5	6	7	8	9
A	9	4	5	8	4	10	4	8	6
B	4	10	6	8	9	5	6	0	5
C	62	57	59	63	59	66	59	62	59
D	2	7	2	5	4	3	6	0	6
E	10	8	6	11	6	10	8	7	11
F	11	7	9	9	6	11	1	4	5
G	57	56	59	59	63	61	59	63	58
H	7	3	3	6	2	4	6	1	2

A

	1	2	3	4	5	6	7	8	9
A	5	0	3	2	1	5	1	6	2
B	4	7	7	4	9	3	3	0	0
C	57	52	58	56	57	61	54	60	54
D	0	4	2	2	5	1	5	2	3
E	10	5	7	8	5	11	6	9	8
F	8	9	5	10	5	7	4	1	7
G	53	54	51	59	57	56	60	56	58
H	1	4	5	0	6	3	3	6	1

B

	1	2	3	4	5	6	7	8	9
A	3	8	3	6	6	4	8	3	6
B	5	6	9	4	9	5	0	4	0
C	57	51	56	57	54	60	54	56	57
D	6	0	4	4	1	6	2	3	5
E	5	7	9	5	10	6	6	10	5
F	9	8	5	10	6	7	5	1	6
G	51	54	50	56	58	55	58	57	54
H	1	7	1	3	6	1	6	3	2

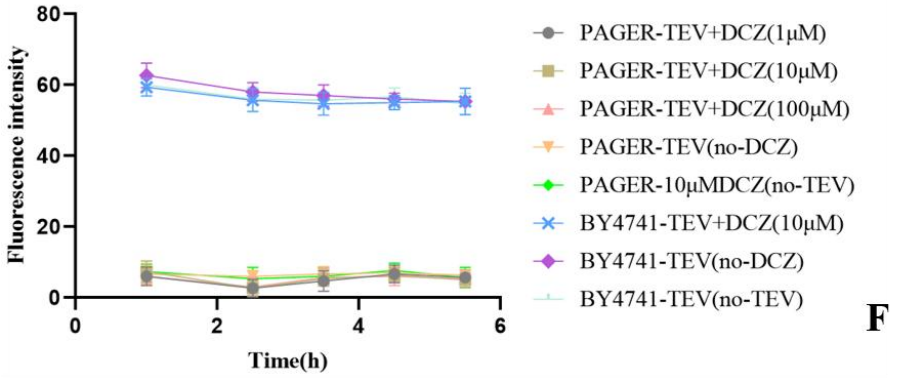
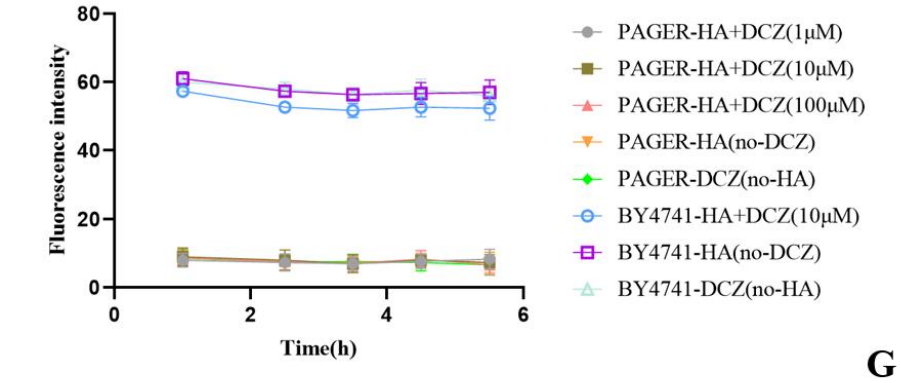
C

	1	2	3	4	5	6	7	8	9
A	7	4	9	4	8	7	5	9	4
B	8	8	5	10	6	7	6	0	4
C	53	57	55	55	58	55	59	57	54
D	3	7	1	6	4	1	7	2	4
E	7	10	6	9	9	6	11	6	8
F	6	10	8	7	10	5	4	5	1
G	56	51	51	59	53	58	59	54	60
H	4	2	6	0	5	5	1	7	3

D

	1	2	3	4	5	6	7	8	9
A	5	5	7	3	7	5	5	8	3
B	5	6	8	4	9	4	3	2	1
C	57	51	58	54	55	57	54	58	52
D	2	3	5	0	6	1	4	4	2
E	10	5	10	6	7	9	5	10	5
F	4	10	8	6	10	4	5	3	3
G	56	49	52	58	53	60	55	56	57
H	6	2	2	6	1	6	6	2	4

E

	Verification of auto-inhibition reversibility of yeast sensor recognition protein  **Functional validation of yeast sensor recognition protein**  Figure1. Time-course fluorescence measurement of yeast biosensor after DCZ treatment. Panels A–E show fluorescence intensity detected at 1 h, 2.5 h, 3.5 h, 4.5 h and 5.5 h after DCZ addition, respectively. Row A: 4th engineered yeast clone incubated with TEV protease and gradient concentrations of DCZ (1 μM, 10 μM, 100 μM). Row B: control groups of 4th engineered yeast clone, including TEV-only group (no DCZ) and DCZ-only group (no TEV protease). Row C: wild-type BY4741 control groups treated with TEV protease and DCZ. Rows E, F and G follow the identical sample layout of Row A, B and C, respectively, with TEV protease replaced by HA protein. Panels F and G represent processed time-course quantitative fluorescence curves. No obvious difference in fluorescence signal was observed between experimental groups and negative controls. fluorescence signal.
Explanation	Under current experimental conditions, the engineered yeast failed to produce obvious target fluorescence. High background signal comes from YPD medium. There is no distinguishable fluorescence difference between engineered strains and wild-type BY4741 controls, demonstrating the sensor cannot respond to TEV protease or HA protein to induce fluorescence output under this setup.
Next plan	Conduct Western blot to confirm target protein expression. If expression is absent, modify the gene's signal peptide sequence.